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Covariation & salience in linguistic contact: a sociophonetic study of liquid variation in Boston Spanish

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BOSTON UNIVERSITY
GRADUATE SCHOOL OF ARTS & SCIENCES

Dissertation

**COVARIATION & SALIENCE IN LINGUISTIC CONTACT:
A SOCIOPHONETIC STUDY OF LIQUID VARIATION IN BOSTON SPANISH**

by

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*Nature does not hurry, yet
everything is accomplished.*

Lao Tzu

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**COVARIATION & SALIENCE IN LINGUISTIC CONTACT:
A SOCIOPHONETIC STUDY OF LIQUID VARIATION IN BOSTON SPANISH**

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ABSTRACT

This dissertation examines how Spanish-speaking Bostonians of Puerto Rican and Dominican origin navigate linguistic variation and lectal coherence, focusing on the role of salience in shaping the outcomes of linguistic contact. Speakers do not adopt, resist, or modify linguistic features at random—social and cognitive factors shape how variation unfolds in contact settings. While prior research has shown that some features vary in their susceptibility to change in bilingual environments, the extent to which salience influences this process remains underexplored. By analyzing both low-salience features (e.g., *subject pronoun expression*, *subject placement*, and *filled pauses*) and high-salience features (e.g., *coda /s/* and *liquid variation*), this study explores how speakers engage with variation in response to social and linguistic pressures in a dynamic contact setting.

Using sociolinguistic interviews from 22 Spanish speakers in Boston, the study employs quantitative variationist methods to analyze the distribution and interaction of multiple variables. The results show that low-salience features exhibit modest, yet, measurable and systematic structural convergence with English grammatical norms, reflecting bilingual optimization strategies. In contrast, high-salience features provide speakers with greater social and stylistic flexibility, allowing them to deploy these features in ways that align

with their identity, ideology, and interactional goals. In some cases, such as with rhotics, speakers maintain a broader sense of Caribbean linguistic identity—even after extended residence in the mainland U.S.—using these features as shibboleths of national identity. For example, the lateralization of the tap continues to differentiate Puerto Rican and Dominican speech styles, serving as a salient marker of regional affiliation. However, other high-salience features, such as laterals, appear to carry less social weight, making them more susceptible to contact-induced restructuring or stylistic reinterpretation.

To examine how these patterns of variation intersect, the study incorporates a covariation analysis that investigates whether and how the five variables pattern together—particularly in relation to speakers’ exposure to English and broader contact-induced pressures. This analysis reveals a fundamental asymmetry: lower-salience features tend to shift in tandem, reflecting a broader structural realignment shaped by bilingual optimization. Higher-salience features, by contrast, do not show the same kind of coordinated movement. Instead, they remain sites of sociolinguistic agency, affording speakers the flexibility to reinforce, neutralize, or reinterpret these forms in ways that serve social and identity-related goals. These findings contribute to sociolinguistic theory by demonstrating that salience is not merely an inherent property of linguistic variables but a dynamic, ideologically mediated phenomenon—one that shapes how and why speakers retain, modify, or reconfigure particular linguistic features in language- and dialect-contact settings.

Moreover, the current research advances understanding of linguistic agency, identity, and change in multilingual communities by conceptualizing salience as a continuum rather than a dichotomy. By framing linguistic features along a salience continuum, the study reveals how different variables interact with social forces in contact settings, offering in-

sights into the ways speakers navigate variation to manage group affiliations and respond to sociolinguistic pressures. Additionally, by incorporating liquid variation into the study of salience and covariation, this dissertation broadens the empirical and theoretical scope of the study of Spanish in the United States, shedding new light on the ways bilingual speakers actively engage with variation.

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List of Abbreviations

AIC		Akaike Information Criterion
AOA		Age of Arrival
CI		Confidence Interval
FPs		Filled Pauses
ICC		Intraclass Correlation Coefficient
PLUS		Percent Life in the US
SPPs		Subject Personal Pronouns
SE		Standard Error
SV		Subject Verb
VS		Verb Subject

1 Introduction

On a typical day, a Spanish-speaking Bostonian of Puerto Rican or Dominican origin might seamlessly navigate multiple linguistic environments. At home, they speak in rapid, Caribbean-inflected Spanish, dropping final /s/ and occasionally lateralizing a tap, just like their parents and grandparents do. At work or school, their Spanish shifts—perhaps they hyperarticulate their rhotics or pronounce syllable-final /s/ more frequently, subtly adjusting to perceptions of prestige or clarity. When chatting with bilingual friends, they code-switch fluidly, peppering their Spanish with English borrowings and sometimes even altering the structure of their sentences to align more closely with English syntax.

Yet, not all linguistic features are treated the same. Some, like pronoun presence or subject placement, tend to escape speakers' conscious awareness and rarely become the subject of overt commentary or concern. Others—like the realization of rhotics or coda /s/—are highly noticeable and frequently referenced as markers of group identity or dialectal difference. A Puerto Rican speaker might velarize their trill to distinguish themselves from a Dominican peer, while a Dominican speaker might lean into their own dialectal patterns as an assertion of identity. These choices are not random. Rather, they reflect asymmetries in linguistic awareness and the social meanings attached to particular forms.

This dissertation investigates these dynamics, focusing on how Spanish-speaking Bostonians navigate linguistic variation in a bilingual setting. Specifically, it examines how differential salience—the degree to which linguistic features are noticed, socially meaning-

ful, and available for speaker manipulation—shapes patterns of variation and covariation. While some variables may serve as markers of identity and be deployed strategically, others may shift more subtly in response to broader structural pressures, such as those that arise in contexts of bilingualism. Recent work in sociolinguistics has shown that linguistic variables often do not vary in isolation: instead, they exhibit patterned relationships with one another—a phenomenon known as covariation. These patterns may reflect shared social meanings, stylistic coherence, or cognitive pressures associated with bilingual speech.

By analyzing variables at the lower end of the salience spectrum, which tend to shift in response to bilingual optimization, alongside variables at the higher end of the salience spectrum, which offer speakers greater opportunities for social signaling and stylistic agency, this study sheds light on the mechanisms through which linguistic features co-vary, converge, or resist change in contact settings.

1.1 Research Questions and Objectives

This study builds on prior research on Spanish in the United States by incorporating variation in the production of liquids into the discussion of the themes of salience and covariation, expanding the range of variables examined at the higher end of the salience spectrum.

The dissertation sets out to answer the following research questions:

1. How do linguistic and social factors shape variation in the production of liquids in Spanish?
2. Do the data support the hypothesis that features at the lower end of the salience continuum, such as *subject personal pronouns*, *subject placement*, and *filled pauses*, pattern

differently than those at the higher end, such as *coda /s/* and *liquid variation*?

- (a) In this contact situation, do speakers' treatment of the features at the lower end of the salience continuum follow a pattern of change?
- (b) Do speakers treat coda /s/ and liquid variation differently than features at the lower end of the salience continuum?

These questions inform the methodological and theoretical approach of the dissertation, linking individual linguistic variables to broader patterns of potential language change, identity construction, and contact-induced restructuring.

1.2 Context and Scope

Boston's Puerto Rican and Dominican communities provide an ideal setting for exploring these questions. As bilingual speakers immersed in an English-dominant environment, they are constantly navigating multiple linguistic boundaries—not only between Spanish and English, but also between distinct varieties of Spanish. These speakers make choices about which features to retain, adapt, or downplay, often in response to shifting social contexts. For some, linguistic behavior reflects an orientation toward maintaining dialectal distinctiveness; for others, accommodation or leveling may be preferred, whether consciously or not. These choices are shaped by a range of factors, including migration history, community networks, and ideological perceptions of language and identity.

Data for this study come from the *Spanish in Boston Project*, a large-scale variationist survey designed to provide insight into the linguistic practices and attitudes of Spanish-speaking Bostonians, with an emphasis on the outcomes of language and dialect contact. This broader project has investigated a set of four sociolinguistic variables: (i) subject

pronoun expression, (ii) subject placement, (iii) filled pauses, and (iv) coda /s/. Findings suggest modest but measurable intergenerational change for the lower-salience variables—such as subject pronoun expression, filled pauses, and subject placement—while higher-salience variables like coda /s/ exhibit patterns of relative stability. These results suggest that features lower in salience may shift more predictably in response to bilingual pressure, while more salient features are shaped by local social pressures and speaker ideologies.

The present dissertation deepens this line of inquiry by focusing on *liquid variation*—the realization of rhotics (/r/, /r/) and the lateral /l/—a feature that is dialectally distinctive, perceptually salient, and prominent in folk ideologies of Caribbean Spanish. Crucially, speakers in this study also directly comment on liquid variation in metalinguistic discourse, underscoring its salience not only in broader ideological narratives but also in local speaker awareness. While liquid variation has been extensively studied in Spanish sociolinguistics, its relationship to salience in contact settings remains underexplored. By situating this variable alongside both high- and low-salience variables, this study probes whether liquid variation patterns more like coda /s/ or more like lower-salience variables. The goal is not to argue that highly salient variables resist contact-induced change, but rather that speakers are more likely to treat them independently, in ways shaped by local social pressures, ideological stances, and communicative goals. Unlike lower-salience variables—which often show convergence across generations in ways that reflect broad cognitive pressures—salient variables are less subject to such uniform pressures. Their social visibility makes them more available for stylistic use, but also renders their trajectories more variable across settings. By drawing together existing insights into salience and

organizing them within a continuum-based framework, this dissertation offers an analytic approach to understanding how different types of variables are implicated in linguistic change, stability, and identity negotiation in contact settings.

1.3 Hypotheses and Contributions

Guided by the research questions, this dissertation hypothesizes the following patterns of linguistic variation:

LOWER-SALIENCE features (e.g., pronoun presence, subject placement, filled pauses) will exhibit significant convergence with English norms. Their lack of social visibility makes them more susceptible to so-called bilingual optimization strategies, allowing for structural realignment with minimal social cost.

HIGHER-SALIENCE features (e.g., coda /s/, liquid variation) will serve as resources for social meaning, with their realization shaped by speaker ideology and communicative goals. While these features may maintain alignment with national and regional dialectal norms, they are also subject to reconfiguration depending on whether speakers aim to reinforce dialectal differences, signal group membership, or accommodate external norms.

This research makes several contributions to sociolinguistic theory and methodology:

BROADENING THE SALIENCE FRAMEWORK- By incorporating liquids into salience studies, this dissertation refines the theoretical understanding of salience as a continuum rather than a binary property.

EXPANDING THE STUDY OF CO-VARIATION- By analyzing both low- and high-salience variables together, this research adds to the growing research focusing of how linguistic features interact and shift in bilingual contexts.

CONTEXTUAL INSIGHTS- Findings are situated within the unique sociolinguistic dynamics of the Boston community, emphasizing the role of social meaning, language ideologies, and identity in shaping linguistic behavior.

1.4 Structure of the Dissertation

Following the introduction, this dissertation is organized into six chapters, each addressing different aspects of the study:

CHAPTER 2- This chapter provides a comprehensive review of the literature, exploring salience, covariation, and the variables central to this study.

CHAPTER 3- This chapter details the methodology, including data collection procedures, acoustic analysis, and the role of social variables in the analysis. It also describes the *Spanish in Boston Corpus* and the sociolinguistic background of the community.

CHAPTER 4 - This chapter examines speaker profiles, highlighting their demographic backgrounds, patterns of language use, and attitudes toward dialectal variation. It lays the groundwork for understanding the sociolinguistic context of the participants' speech and provides critical insights into how their language use and attitudes interact with broader patterns of variation.

CHAPTER 5- This chapter presents an in-depth examination of liquid variation, focusing on the linguistic predictors and social factors that influence the realization of /r/,

/r/, and /l/.

CHAPTER 6- This chapter explores potential covariation between liquids and other variables, situating the relationships between variables within a framework of salience and contact-sensitive patterns of variation.

CHAPTER 7- This chapter synthesizes findings from all previous chapters, discussing their implications for sociolinguistic theory, language contact, and the role of salience in bilingual communities. It also highlights the study's contributions and offers suggestions for future research directions.

1.5 Significance of the Study

By examining the interplay between salience, covariation, and language- and dialect-contact, this dissertation contributes to a more nuanced understanding of how bilingual speakers navigate complex sociolinguistic terrain. The findings show that not all variables respond similarly to contact pressures: some exhibit measurable shifts across generations, while others remain relatively stable, or display variability that defies straightforward patterns. This variability underscores an important point—contact-induced change is not inevitable, nor is it uniformly distributed across the linguistic system.

Rather than dividing variables into those that “converge” and those that are “maintained for social indexing,” this study highlights how speaker awareness and social meaning shape the treatment of different variables. Variables that are lower in salience tend to pattern together, showing modest convergence consistent with bilingual optimization strategies. In contrast, more salient variables—those that are perceptually prominent, ide-

ologically loaded, or explicitly referenced by speakers—are treated more independently. Their outcomes are more sensitive to local social dynamics and less constrained by cognitive pressures. In some communities, this may lead to maintenance or even amplification; in others, salient variables may undergo leveling, depending on how speakers orient to them. In doing so, the study builds on foundational insights from Trudgill (1986) and Labov (1972), offering further evidence that the degree to which speakers are aware of a variable—its salience—plays a key role in its behavior in contact settings. But importantly, this awareness does not make outcomes more predictable; rather, it opens the door to greater variability, since socially salient variables are more likely to be taken up as stylistic resources, sites of differentiation, or targets of reconfiguration.

By placing liquid variation within this broader framework, this dissertation offers a structured approach to understanding how salience and speaker agency shape the trajectories of linguistic variables in contact settings—showing that what changes, what remains stable, and why, depends not just on structure, but on social context, speaker goals, and community ideologies.

2 Previous Literature & Motivation for the Current Study

In this chapter, I provide an overview of the existing literature that informs and motivates the current study. I delve into key themes in sociolinguistics, focusing on the concepts of *salience* and *awareness*, and their impact on linguistic variation. Following this, I examine the ideas of *covariation* and *coherence*, exploring how these concepts have been treated in previous research. Lastly, I discuss the variables central to the study. I review the literature on liquid variation, the variable for which I developed the methodology for this dissertation. Additionally, I introduce four other variables: coda /s/, filled pauses, subject personal pronouns, and subject placement. By studying these variables, I aim to highlight gaps in the current literature and demonstrate the need for further investigation. This chapter sets the stage for the methodology and analysis presented in subsequent chapters.

2.1 Hispanic Linguistics in Contact: A Comparative Variationist Framework

The evolution of Spanish has been profoundly shaped by its ongoing interaction with diverse linguistic communities. The historical forces that birthed Castilian also paved the way for continuous exchanges with other languages. Penny (2000) illustrates that modern Spanish has been molded not only by the reciprocal influences among speakers of various languages but also by the impact of mutually intelligible dialects. Additionally, its role as the language of Spanish colonizers positioned its speakers in enduring contact both within

Spain and around the world.

Extensive research has examined the linguistic effects of interactions between Spanish speakers and those of indigenous and African languages—among the earliest contacts in the Hispanic Americas. Scholars have primarily focused on three domains shaped by the Conquest: first, pre-Columbian native languages such as Nahuatl (Lipski 1994), Taíno (Lipski 1996b), and Mayan (Michnowicz 2015); second, the emergence of Caribbean creoles from the integration of enslaved populations during colonial times (Lipski 1996a); and third, the influence of neighboring Romance languages like Catalán (de Prada Pérez 2009) and Portuguese (Carvalho & Child 2011). Moreover, beyond Europe and the Americas, studies have explored Spanish’s interactions in Asia and Africa—focusing on regions such as the Philippines (Quilis 1992; Lipski 2001; Lipski 2005b), Equatorial Guinea (Granda 1991; Lipski 1985; Lipski 2005b; Quilis 1988), and, Morocco (Sayahi 2004). For this investigation, however, the primary emphasis is on Spanish language contact within the United States.

Few regions within the Spanish-speaking world have attracted as much scholarly attention as the United States, and for good reason. Rooted in historical and political ties that trace back to the arrival of the conquistadors in the 16th century (López Morales 2009), Spanish-speaking communities have long been woven into the American demographic fabric. By 2005, Hispanics had become the largest minority group in the country, surpassing African Americans (Lipski 2003). Moreover, continuous migration from diverse Spanish-speaking regions has helped preserve distinctive dialectal features even as these communities interact with English speakers and with one another.

Researchers have recently adopted a broadly applicable method for evaluating con-

tact outcomes known as the *comparative variationist framework* (Poplack & Levey 2010). Of particular relevance to the current study, this approach has been profitably employed to assess the effects of language and dialect contact across a wide range of linguistic settings—not limited to Spanish-speaking communities. Grounded in the foundational principles of variationist sociolinguistics, this framework examines whether patterns of linguistic variation remain stable or shift under the pressures of contact, while explicitly cautioning against presuming that contact necessarily results in change. As Poplack & Levey (2010: 412) argue, “contact-induced change is not an inevitable, nor possibly even a common, outcome of language contact.” A core strength of this approach lies in its reliance on systematic comparison—often using apparent-time or real-time data—to trace continuity or change in variation patterns over time. Without such comparison to a relevant precursor, “any claims about what has changed must remain speculative” (Poplack & Levey 2010: 394). By foregrounding structured linguistic variation and its interaction with social and cognitive factors, the comparative variationist framework offers a powerful tool for distinguishing genuine contact effects from natural language-internal variation. This makes it especially well-suited for investigating how bilingual speakers navigate pressures to converge, diverge, or maintain features across generations. With such a comparison in place, it becomes feasible to diagnose language changes that emerge as a direct result of contact.

Building on this methodological foundation, drawing robust conclusions about linguistic change also demands data from sufficiently representative samples. In this respect, the two largest spoken Spanish corpora in the United States—the *Spanish in Boston Corpus* and the *Otheguy and Zentella Corpus of Spanish in New York City*—have been instrumental in advancing our understanding of both language and dialect contact dynamics (Otheguy &

Zentella 2012; Erker & Otheguy 2016; Erker 2017).

Central to this approach is the recognition that the linguistic behavior of individuals mirrors broader community patterns. Applying this approach, research on Spanish in the United States has identified multiple outcomes of language contact. For instance, some studies have documented intergenerational dialectal stability alongside signs of dialect leveling (Otheguy & Zentella 2012; Erker & Otheguy 2020), while others have observed structural convergence occurring in tandem with the preservation of distinct crosslinguistic differences (Silva-Corvalán 2008; Torres Cacoullos & Travis 2016; Benevento & Dietrich 2015).

Silva-Corvalán (Silva-Corvalán 2008) proposes a plausible mechanism for some contact-driven changes observed among Spanish-speaking communities in the United States. She argues that the cognitive demands of managing two languages exceed those of operating in a single language, prompting bilinguals to adopt strategies that ease this cognitive load. One such strategy is *linguistic convergence*—a process whereby two languages gradually become more alike structurally, as features from one language are incorporated into the other (Benevento & Dietrich 2015: 407). An illustrative case of this is the purported *simplification* and *overgeneralization* of regular verbal morphology among U.S.-born Spanish speakers, as noted by Silva-Corvalán.

However, while her explanation is compelling, it is important to situate it within a broader framework of bilingual processing. Muysken (Muysken 2013) argues that bilingual outcomes reflect optimization strategies shaped by cognitive constraints, social context, and language distance. Similarly, Hawkins and Filipović (Hawkins & Filipović 2024) propose that cognitive pressures such as working memory limitations and processing effi-

ciency can drive structural convergence in contact settings. These perspectives underscore the need to view convergence as a product not only of individual cognitive economizing but also of broader systemic dynamics. Indeed, there is reason to believe that linguistic features are differentially susceptible to convergence. Specifically, features that lack strong social salience or prescriptive norms may shift more readily in contact settings than those that are sociolinguistically prominent (Erker & Otheguy 2016; Erker 2017).

2.2 Salience in Sociolinguistics

Although salience has not always been an explicit focus in sociolinguistic theory, it has long been an implicit factor in the study of language variation. Different sociolinguistic patterns—such as *change from above and below*, *age-grading*, *style-shifting*, and *social stratification*—engage with salience to varying degrees. While earlier variationist research primarily examined language change occurring below the level of conscious awareness (see Labov 1964; Labov 1966), recent work has increasingly emphasized the role of salience in shaping linguistic variation (see Rácz 2013; Babel 2016a; Erker 2022, and others). However, the differential salience of linguistic variables—the extent to which certain features are noticed or attended to by speakers and listeners—has often remained an implicit rather than explicitly theorized factor in sociolinguistic inquiry, despite its longstanding relevance to the field.

Investigating the topic of salience in sociolinguistics begins by considering whether and to what extent speakers are aware of their treatment of sociolinguistic variables. Some variables are overtly stigmatized, subject to style-shifting, and referenced in meta-commentary, indicating speaker awareness. Other variables, however, appear to go unno-

ticed. Attempts at characterizing speakers' varying levels of awareness begin with Labov's (1972) taxonomy of phases through which variables can go through in the context of change. Labov classifies variables involved in linguistic change into three categories: *indicators*, *markers*, and *stereotypes*. *Indicators*, which fall below the level of conscious awareness, exhibit regular distribution in terms of linguistic and social factors, without social indexation, stylistic variation, or attached judgments. Awareness emerges with the latter two categories. *Markers* also show regular distribution and social indexation with stylistic variation, though speakers are not necessarily explicitly aware of their usage. *Stereotypes* exhibit the highest level of awareness, with their explicit awareness accessible through meta-commentary.

While Labov (1972) engages the question of salience primarily in the context of language change, Trudgill (1986) explores it through the lens of dialect contact. Rather than analyzing how features become salient, Trudgill is concerned with how differential awareness of linguistic features shapes speakers' behavior in contact situations. According to him, speakers adjust the features of their own varieties that they are most aware of, with salience being influenced by overt stigmatization, involvement in a change in progress, phonological contrast maintenance, and phonetic differences between variants (Trudgill 1986: 11).

The pioneering work of both Labov and Trudgill indicate that rather than viewing salience as a discrete property of linguistic features, it is more useful to conceptualize it as a continuum. Labov's taxonomy of indicators, markers, and stereotypes, along with Trudgill's criteria for heightened salience, provides a framework for positioning linguistic variables along a gradient of linguistic awareness. Variables at the lower end of the con-

tinum remain below the threshold of conscious awareness, while those at the higher end are overtly recognized, stigmatized, and subject to hypercorrection or meta-commentary. Figure 2.1 synthesizes these insights, illustrating how features may vary along the salience spectrum.

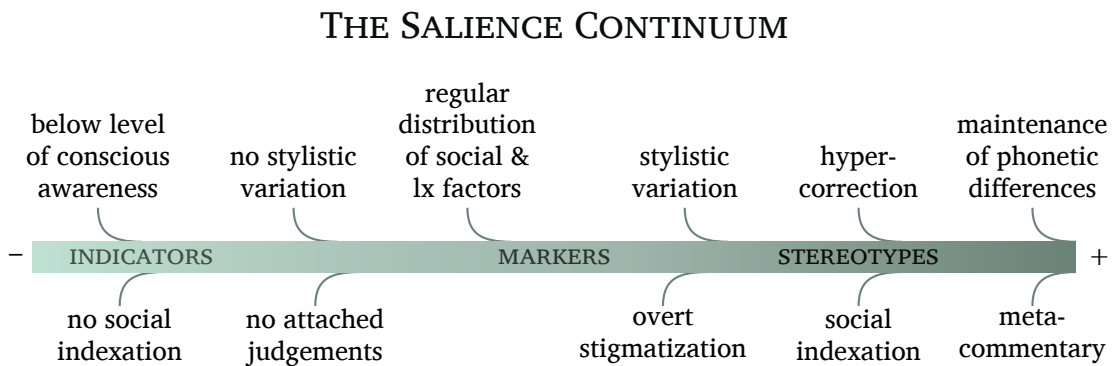


Figure 2.1: The salience continuum. This continuum illustrates the gradience of salience in sociolinguistic variation, synthesizing Labov's (1972) taxonomy with Trudgill's (1986) criteria for heightened salience. Variables at the lower end exhibit minimal awareness and social indexation, while those at the higher end are overtly recognized, stigmatized, or subject to hypercorrection and meta-commentary. This model highlights how salience is not a binary property but a dynamic spectrum shaped by linguistic, cognitive, and social factors.

Building on this foundation, sociolinguistic studies have increasingly considered the role of *speaker agency*—the capacity of individuals to make choices and exert control over their linguistic practices—in shaping variation. More recent work has refined these insights by adopting a dynamic perspective on the social meaning of variation—emphasizing how linguistic forms can actively construct, rather than merely reflect, social identities. This orientation aligns with what has been called the third wave of sociolinguistics, which moves beyond macro-sociodemographic categorizations to more nuanced understandings of locally relevant social categories and the social meanings embedded in linguistic variation. Researchers in this tradition emphasize the role of social meaning in variation, introducing concepts such as *indexicality* and *positionality*. *Indexicality* refers to the context-

based social meanings of linguistic forms, from phonetic variants to dialect choices, linked to social ideologies and stances (Bucholtz & Hall 2005: 594). *Positionality* suggests that identity includes not just broad social categories, but also local ethnographic and temporary interactional roles (Bucholtz & Hall 2005: 592). These studies highlight that linguistic features do not merely reflect social categories but actively construct and negotiate social identities.

In their work, Eckert (1988) and Mendoza-Denton (2008) exemplify the shift towards more in-depth, ethnographically informed methodologies. For instance, Eckert (1988) conducted extensive ethnographic research to explore how social networks within a high school influenced the use of linguistic variables. Similarly, Mendoza-Denton (2008) used detailed ethnographic methods to examine the linguistic practices of Latina gang girls in California, providing a rich account of how their language use is intertwined with social identity and group dynamics. These studies underscore the value of capturing the dynamic and complex nature of social interactions, revealing how linguistic variation is deeply embedded in the fabric of everyday social life.

Despite the many insights gained from context-dependent approaches, Schilling (2013) cautions that linguists should not lose sight of the broader structures and norms that constrain speaker agency and creativity. While these studies celebrate the agentive and creative aspects of language use, it is crucial to remember that speakers operate within pre-existing social orders and ideologies. As researchers acknowledged the limitations of purely social explanations for linguistic variation, they began to consider the cognitive aspects of salience. Rácz (2013) proposed a dual model that distinguishes between cognitive and social salience. Cognitive salience pertains to the inherent properties of linguistic

forms that make them noticeable, while social salience involves the social awareness and significance attributed to these forms. This distinction underscores that linguistic variation is not only socially meaningful but also subject to cognitive constraints.

The concepts of *awareness* and *control* are central to understanding speaker agency in sociolinguistic variation. *Awareness* refers to a speaker's ability to recognize and reflect on linguistic variation, while *control* pertains to the extent to which speakers can consciously regulate their linguistic behavior in response to social, stylistic, or ideological factors. Both exist on a continuum—from unconscious perception to deliberate manipulation—and, while often linked, they are not synonymous. Speakers can be aware of a linguistic feature without having control over it, and vice versa (Nycz 2016; Babel 2016b). Moreover, scholars distinguish between *implicit* and *explicit* knowledge of variation: implicit knowledge involves perceiving differences without conscious awareness, whereas explicit knowledge entails active recognition and reflection (Nycz 2016: 64; Preston 2016: 186; Squires 2016: 83).

Preston (2016) includes control as one of four scales of what he terms *folk linguistic awareness*¹, alongside availability (encompassing Silverstein's (2001) limitations of awareness), accuracy, and detail. Building on these definitions, researchers have examined control and awareness across various domains of language: production (Babel 2016b; McGowan 2016; Zimman 2016), acquisition (Nycz 2016), and processing (Beck 2016; Carmichael 2016; Squires 2016). They have also explored how varying degrees of awareness enable speakers to assert in-group or out-group membership through diverse mech-

¹Like Preston (2016: 177), I use 'folk' to refer to language speakers who are not professionally trained in linguistics and are naive to existing academic research on the variables they reference. This term makes no assumptions about their social characteristics.

anisms—ranging from silence (Babel 2016b), to orthography (Choksi & Meek 2016), to even clothing choices (Mendoza-Denton 2008).

Control can manifest in various ways, from explicit modulation of phonetic or syntactic forms to more abstract sociolinguistic positioning. For instance, speakers may exert control over their linguistic identity through active choices, such as shifting their speech style in different contexts Eckert (2008), or through passive mechanisms, such as strategic silence to resist hegemonic norms (Babel 2016b). At the same time, external constraints—such as ideology, prestige, power structures, stigma, and folk linguistic beliefs—can shape the degree of awareness and control that speakers have (Preston 2016; Choksi & Meek 2016; Carmichael 2016; Zimman 2016). Moreover, Babel (2016b) and Nycz (2016) contend that in situations of contact, the level of awareness is not necessarily correlated with accommodation to or divergence from the dialect/language of contact. They explain that high awareness does not guarantee change, and absence of awareness does not preclude it.

An alternative but related perspective could be that certain linguistic variables, which users are not explicitly aware of, tend to exhibit natural covariation and coherence. In contrast, variables that users are more explicitly aware of allow them to exert agency, controlling these variables according to their language ideologies, context, and level of control. This suggests that explicit awareness may facilitate intentional manipulation of linguistic features, while lack of awareness may lead to more unconscious, accommodative adjustments influenced by the surrounding dialect or language. This brings us to the themes of *coherence* and *covariation*. In the next section, I will delve deeper into these concepts, examining how they interact with salience and awareness to shape linguistic variation and change.

2.3 Covariation and Coherence

When we think of the most famous scholarship in sociolinguistics, it tends to be the case that we link the study with a particular variable: Labov's (1963) Martha's Vineyard study investigated diphthong centralization; his 1966 NYC Department Store study examined rhoticity, etc. Interestingly, while Labov's Department Store study is primarily known for its insights into rhoticity, Labov does consider two other variables in that study: height of vowels [æ] and [ɔ:] and word-initial /ð/ (dh-stopping). So, while the simultaneous description of multiple variables in a given setting has long been a feature of sociolinguistic research, the question of the relationships between variables has only more recently become a topic of specific theoretical interest.

Recent research has taken up this question explicitly, arguing that investigating how variables pattern together is essential for understanding sociolinguistic systems more holistically (see Guy & Hinskens 2016; Erker & Otheguy 2016; Erker 2022, and others). This research has moved the dial away from exploring variables individually towards examining the relationship between different variables within the community, i.e., towards the topics of *covariation* and *coherence*. *Covariation* refers to the statistical correlation between the occurrences of the variants of two or more specific linguistic variables. It's a measure of how frequently distributions of the variants of different variables are found together in the speech of individuals or groups within a speech community, e.g. a potential correlation (or lack thereof) between rates of t/d-deletion, r-lessness, and alveolar realizations of /r/. *Coherence*, on the other hand, is broader; it refers to the idea that linguistic variables systematically align with each other across a language system or within a speech commu-

nity in a non-random manner, reflecting underlying social, cultural, or linguistic principles that govern their use. The distinction between these concepts lies in their scope and focus: covariation is a measurable phenomenon examining statistical relationships between specific variables, whereas coherence offers a broader lens on how variables interact to reflect and construct social realities, e.g. evidence of correlative covariation between rates of t/d deletion, r-lessness and alveolar realizations of ING could undergird the coherence of an informal speech style .

Labov's concept of *orderly heterogeneity* introduced the notion that language variation is structured and meaningful within a speech community. Guy's (2013) idea of (lectal-) coherence extends this to consider how these variations align or correlate with each other across the linguistic system of a community, suggesting that:

If sociolects are indeed socially and cognitively coherent varieties, we should expect some degree of correlation among the different variables present in a community... If social varieties have a similar coherence, the variables that characterize them should exhibit some kind of binding force, a pattern of correlation or clustered usage. Guy (2013: 64)

Both concepts, since their inception, have been primarily explored in relation to variation in non-contact settings (see Labov 1966; Guy 2013; Oushiro & Guy 2015; Oushiro 2016; Guy & Hinskens 2016). More recent studies have begun examining coherence in communities characterized by potential change as a result of language and dialect contact. This shift is crucial for grasping the complex dynamics of language variation and change. For instance, Erker & Otheguy (2016) studied the speech of Spanish speakers in the largest Spanish-speaking urban center in the United States, New York City, where speakers are subject to contact with not only English, and many other languages, but also

various dialects of Spanish. The authors found structural convergence with English in four of the five variables studied: pronoun use, subjunctive morphology, pronoun placement, and word order. In another example, Erker (2017) observed covariation among four of the five variables in a study of Spanish in another major U.S. urban center, Boston.

Tamminga & Wade (2022) further explore the concept of coherence, emphasizing the need for continued theoretical and methodological refinement in defining the appropriate linguistic and social units of analysis to fully capture its complexity across social and temporal scales. They argue for a broad definition of coherence as the systematic covariation of linguistic variables, highlighting the need for flexible analytical approaches that account for the dynamic nature of linguistic variability (Tamminga & Wade 2022: 34). Additionally, Guy et al. (2022) explore how linguistic features in Brazilian Portuguese covary based on their social meanings, suggesting that variables closely aligned in their social indexicality are more likely to exhibit coherence in usage. This begs the question: if coherence is expected from variables that define a speech community, what can we learn from those that deviate from expected patterns of coherence? The answer may lie in speakers' agency and language awareness.

2.4 The Present Study

The present study investigates the interaction between salience, speaker agency, and covariation in the linguistic behavior of Spanish speakers in Boston. By examining both highly salient variables as well as those at the lower end of the salience continuum, this study seeks to uncover how different features are treated within a contact setting—whether they covary systematically in a largely unconscious or uncontrolled way or are manipu-

lated independently by speakers and show potentially divergent patterns of usage. Prior research has demonstrated that salience plays a crucial role in shaping linguistic variation, influencing the extent to which speakers exert agency over their speech and engage in identity construction. This study builds on these insights by exploring how speakers navigate the relationship between socially prominent variables, such as coda /s/ and liquid variation, and less salient variables.

To bring the various topics discussed so far—salience, control, agency, covariation, coherence, and linguistic contact—into a cohesive analytical frame, it is useful to consider the concept of *bricolage*. *Bricolage*, or the idea that “individual resources (in this case, variables) can be interpreted and combined with other resources to construct a more complex meaningful entity”, offers a lens through which to view the strategic use of salient linguistic features (Eckert 2008: 456). Through bricolage, speakers can leverage their awareness of certain features to index group membership, express patriotic pride, distinguish themselves from others, or signal their educational level, among other identity markers. This process not only underscores the agency speakers have over their linguistic choices but also highlights the role of salience in facilitating these choices. Expanding on these ideas, Sharma (2022) examines coherence across different scales—from wholesale dialects to individual linguistic variables—arguing for a more nuanced understanding that accounts for both routinized and agentive aspects of speech. She suggests moving away from a binary opposition between structure and agency towards recognizing degrees of coherence that includes both large-scale dialectal variation and smaller-scale bricolage (Sharma 2022: 21).

Erker (2022) provides a case study of the Spanish-speaking community in Boston,

demonstrating how social salience influences linguistic behavior in a contact setting. He finds that less socially salient features, like filled pauses and subject positions, show significant differences between adult and child arrivals, indicating a shift towards English structural norms among the latter. Conversely, the highly salient feature of coda /s/ shows similar patterns of deletion across both groups, underscoring its robustness in marking linguistic identity in the context of contact and reflecting the own-group dialect loyalty of the participants. This highlights the complex interplay between social salience and linguistic change, showing that highly salient features possess strong “social signaling potential” and are more resistant to change when maintenance of particular patterns of usage aligns with speakers’ social goals. When change in a highly salient feature does occur, Erker argues that such change is likely socially motivated rather than primarily cognitively driven (Erker 2022: 157).

In other words, more salient features are arguably treated more independently by language users, while less salient ones are more likely to cohere or covary systematically (see also Oushiro & Guy 2015; Erker & Otheguy 2016; Erker 2017; Cole 2020). Insofar as non-salient features covary, change in one among a set of covarying variables has been compared to a falling domino, where a change in one variant may trigger change in another. Guy & Hinskens describe this potential scenario as follows:

The more coherent a set of coexisting linguistic variables, the bigger the chances that a change in the variant of one of the variables will trigger a switch to another variety – like a falling domino can make a row of neighboring dominoes fall.
Guy & Hinskens (2016: 2)

The present study builds on these themes by exploring the relationship between multiple sociolinguistic variables, including two features that are prominent in folk ideologies of

Spanish dialectal variation: *coda /s/* and *liquid variation*. The low salience features in this dissertation are (i) *filled pauses*, (ii) *subject pronoun expression*, and (iii) *subject placement*. To the extent that the lower salience variables are undergoing reconfiguration via contact, the expectation is that they will “move” together. Conversely, I anticipate the potential for the salient features, (iv) *coda /s/* and (v) *liquid variation*, to be treated independently—both from each other and from the lower-salience variables—with each, in some cases, following its own distinct path shaped by speakers’ language attitudes and ideologies. *Features (ii) - (v)* are among the most extensively studied in the context of Spanish variation and change. Notably, Erker (2017) has conducted significant research on features (i)-(iv), examining these variables collectively and highlighting *coda /s/* as particularly salient to Spanish speakers. However, more work is needed on other salient features, and this study aims to fill that gap. In fact, Erker (2017) concludes with a call for the inclusion of multiple salient variables, which he reiterates in Erker (2022: 160):

Assemblages of other powerfully indexical features should be examined alongside larger sets of non-salient features, and the relative salience of each feature, already informed by folk-ideologies as well as through the express attitudes of speakers, should be more fully established empirically in the context of direct study.

One such feature is *liquid variation*, which expands the focus on salient variables in the Spanish of Boston. While *coda /s/* has been extensively studied in this community and is well-established as a socially prominent variable, *liquid variation* has not yet been examined in this context. Its inclusion allows for a more comprehensive investigation of how speakers engage with highly salient linguistic features. There is strong reason to view both *coda /s/* and *liquid variation* as sites of prominence in folk ideologies of Spanish, as

speakers frequently comment on them when discussing dialectal differences. Both features qualify as *stereotypes* in the Labovian sense, and, following Trudgill's (1986) criteria, they exhibit systematic dialectal and stylistic variation, have high-status variants, involve clear phonetic contrasts, and contribute to phonological contrast maintenance.

Furthermore, speakers' metalinguistic commentary on these variables points to their salience within the Spanish-speaking community. In fact, these two features are routinely commented on by participants in this study, which forms part of the *Spanish in Boston Project*², a large-scale variationist survey of the Spanish-speaking community in the Greater Boston Area. In contrast, the other three features included in this study are not mentioned at all by participants in that project. In terms of the liquids, Caribbean speakers seem to be particularly attuned to the nuances of this phenomenon. The focus on these features—especially considering their prominence in folk ideologies of Spanish dialectal variation—underscores the importance of salience in linguistic variation and change. By focusing on two highly salient features like coda /s/ and liquid variation and exploring their treatment alongside less salient features, this study aims to illuminate the intricate ways in which linguistic variation and identity construction intersect within the Spanish-speaking community of Boston.

Building on these foundational concepts, my dissertation expands the study of sociolinguistic variation in the Spanish-speaking community of Greater Boston by incorporating *liquid variation*—a feature not yet examined in this context. While previous research has centered on the social meaning of coda /s/, the inclusion of liquids broadens the inquiry into how speakers engage with salient variables. This approach not only deepens our

²The Spanish in Boston Project is funded by the National Science Foundation (BCS-1423840) directed by Daniel Erker (Boston University).

understanding of liquids in this community but also tests an emerging hypothesis in co-variation research: speakers tend to treat less salient features as a cohesive style, while manipulating highly salient ones more independently, in ways influenced by language attitudes and ideologies. By targeting both (1) dialectal variation and (2) cross-linguistic interference, this study contributes to research on Spanish variation and sheds light on potential contact-induced change. A key question, then, is whether *liquid variation* aligns more closely with the highly salient *coda /s/*, or with lower-salience features. Given that speakers frequently cite liquid variation as emblematic of Caribbean Spanish—highlighting its perceptual prominence—I hypothesize that, like *coda /s/*, liquids will show more independent patterning.

Crucially, this hypothesis was shaped by the fact that speakers in the *Spanish in Boston Corpus* exhibit a high degree of own-group dialectal loyalty. While studies from other Spanish-speaking communities have shown evidence of change in salient features—often driven by speakers’ motivation to assimilate dialectally (Silva-Corvalán 1994; Otheguy & Zentella 2012; Erker 2017)—many of the participants in the *Spanish in Boston Corpus* appear instead to use these features to actively assert their dialectal distinctiveness within their local context. Importantly, my hypothesis does not assume that salient variables are inherently resistant to change over generations. Rather, I argue that *coda /s/* and *liquid variation* will pattern independently not simply because they are salient, but because (a) socially prominent variables provide sites for greater speaker agency and (b) participants in the *Spanish in Boston Corpus* demonstrate a motivation to dialectally distinguish themselves. This social orientation provides a key explanatory factor for why I expect *liquids* to behave similarly to *coda /s/*, rather than following the pattern of less salient variables.

2.4.1 Liquid Variation

The term *liquid* encompasses two types of consonants: laterals (l-like sounds) and rhotics (r-like sounds). In Spanish, onset consonant clusters are only allowed when a liquid sound occurs in the second position, which is one of their most important grouping properties. Laterals are produced by obstructing airflow along the central part of the oral cavity, allowing air to flow around the sides of the tongue, while Spanish rhotics are articulated through contact between the tongue and the alveolar ridge. Spanish has three liquid phonemes: two rhotics, /ɾ/ (tap) and /r/ (trill), and one lateral, /l/. The tap and the trill contrast in intervocalic position, as seen in minimal pairs such as *pero* /pero/ ‘but’ and *perro* /pero/ ‘dog’ (Willis & Bradley 2008).

A well-documented feature of Spanish is liquid variation, especially in Caribbean locales, which results in diverse realizations of underlying rhotics and laterals, particularly in coda position (Navarro Tomás 1948; Poplack 1979; López Morales 1983; Alba 1990; Lipski 2008; Hualde 2014, *inter alia*). Several intralinguistic factors systematically constrain *liquid variation* in Spanish, including (i) *preceding and following phonological environment*, (ii) *speech style*, (iii) *position in syllable or word*, (iv) *syllable stress*, (v) *word*, and (vi) *syllable length* (Alba 1990; Poplack 1979; Ramos-Pellicia 2004; Beaton 2016; Valentín-Márquez 2019). In addition, extralinguistic factors such as *sex*, *education level*, *regional origin*, and *generation* also shape its variation (Alba 1990; Poplack 1979; Prosper-Sánchez 1995; Ramos-Pellicia 2007; Willis & Bradley 2008; Valentín-Márquez 2019).

Poplack (1979: 144) notes that rhotic variation in Spanish is highly complex and more difficult to explain than other phonetic alternations, such as those of /s/, due to the wide

range of phonetic processes affecting rhotics, among other factors. These processes include vocalization, lateralization, velarization, assimilation, aspiration, and deletion. These processes lead to diverse allophonic realizations, which are summarized in Table 2.1. The table presents the surface realizations of each process, the phonetic processes involved, and a corresponding example, offering a structured overview of rhotic variation.

Table 2.1: Summary of Phonological Processes Affecting Rhotics in Caribbean Spanish

SURFACE REALIZATION	PHONETIC PROCESS	EXAMPLE	TRANSLATION
[i], [e], [a] [o], [u]	vocalization	<i>comer</i> → [ko.'mei]	'to eat'
[l], [ɫ]	lateralization	<i>puerta</i> → ['pwel.ta]	'door'
[r], [R]	frication	<i>carro</i> → ['ka.ro]	'car'
[x]	velarization	<i>carro</i> → ['ka.xo]	'car'
(context-dependent)	assimilation	<i>cuerpo</i> → ['cuep.po]	'body'
[h]	aspiration	<i>amor</i> → [a.'moh]	'love'
[Ø]	deletion	<i>comer</i> → [ko.'meØ]	'to eat'

Rhotic variation in Caribbean Spanish exhibits regionally distinct patterns, with different phonological processes occurring across various speech communities. Lateralization, in which /r/ is realized as [l], is particularly common in Puerto Rico, the Dominican Republic, central Cuba, eastern Venezuela, the Canary Islands, and occasionally in Murcia, Spain (Lipski 2012; Campos-Astorkiza 2012). In contrast, vocalization, where /r/ surfaces as a vowel, has been documented primarily in the Cibao region of the Dominican Republic (Alba 2004: 85), though it also occurs sporadically in Cuba, Puerto Rico, and southeastern Spain (Lipski 2012: 6).

Another process, assimilation (or gemination), in which coda /r/ takes on the place features of the following consonant, has been reported in Cuba and the Caribbean coast of Colombia (Quilis 1993; Alba 2004; Lipski 2012; Martínez-Gil 2012). Similarly, aspiration of coda /r/, where it is realized as [h], is well attested across the Spanish-speaking Caribbean, particularly in Puerto Rico, Cuba, and the Dominican Republic (Hualde 2014).

It should be noted that not all phonetic realizations of /r/ are evaluated in the same way by speakers. While deletion of coda /r/ is widespread in eastern Cuba, Panama, the Caribbean coast of Colombia, much of Venezuela, and along the Pacific coast of Colombia, Ecuador, and Peru, as well as in Afro-Bolivian Spanish (Hualde 2014), its social evaluation differs from other phonetic processes. Lipski (2012: 6) notes that deletion, despite being a prominent feature of many Caribbean varieties, does not necessarily carry the same degree of stigma as other allophones, like [x]. Velarization of coda /r/, which results in realizations such as [x] and is reported as a characteristic feature of Puerto Rican Spanish (Delgado-Díaz & Galarza 2015; Lipski 2008), has been met with significant sociolinguistic awareness and even overt stigma in some contexts. Despite this, velarization has been found to persist among Puerto Rican speakers in diaspora communities, such as those in Grand Rapids, Michigan, where speakers continue to produce [x], suggesting that it remains a meaningful marker of Puerto Rican identity beyond the island (Valentín-Márquez 2019). Similarly, lateralization, which is especially salient in Puerto Rican and Dominican varieties, has been subject to prescriptive correction and metalinguistic commentary. Shenk (2021) discusses explicit campaigns against lateralization, particularly in Puerto Rico, where speakers are often encouraged to "correct" this feature in more formal or public speech contexts.

Compared to rhotics, /l/ is generally more resistant to variation (Alba 1990: 214; Lipski 2012: 6). Nonetheless, several phonological processes affecting /l/ have been documented, including vocalization, rhotacism, assimilation, aspiration, deletion, and even nasalization. These processes lead to diverse allophonic realizations, which are summarized in Table 2.2. As with the rhotics above, this table presents the documented surface

realizations of each process at a segmental level, the phonetic processes involved, and a corresponding example.

Table 2.2: Summary of Phonological Processes Affecting the Lateral in Caribbean Spanish

SURFACE REALIZATION	PHONETIC PROCESS	EXAMPLE	TRANSLATION
[i], [e], [a] [o], [u]	vocalization	<i>capital</i> → [ka.pi.'tai]	'capital'
[r], [r]	rhotacism	<i>capital</i> → [ka.pi.'tar]	'capital'
[ɬ]	velarization	<i>pulpo</i> → ['puɬ.po]	'octopus'
(context-dependent)	assimilation	<i>caldo</i> → ['kad.do]	'soup'
[h]	aspiration	<i>alto</i> → ['ah.to]	'tall, stop'
[Ø]	deletion	<i>pulpo</i> → ['puØ.po]	'octopus'

Lateral variation in Spanish exhibits regionally distinct patterns. *Vocalization*, where /l/ is realized as a vowel, is widely attested in the northern Dominican Republic, parts of Colombia, and the Canary Islands, as well as among Isleño Spanish speakers in Louisiana (Alba 1990: 134; Lipski 2008: 136; Alba 2004: 86). *Rhotacism*, where /l/ is realized as [r], is found in Andalusia, southwestern Dominican Republic, Cuba, Puerto Rico, Colombia, Venezuela, Ecuador, Peru, Chile, and Argentina, as well as in the Canary Islands (Alba 2004: 86; Quilis 1993: 325-326). *Assimilation*, where /l/ takes on the place of articulation of a following consonant, has been reported in the Dominican Republic and Cuba (Alba 1990: 134; Alba 2004: 86; Lipski 2008: 112). *Aspiration* of /l/ to [h] is documented in Cuba, Puerto Rico, the Dominican Republic, Colombia, and the Canary Islands, and is also found among Cuban Americans in the U.S. (Quilis 1993: 327; Lipski 2008: 114). Finally, *deletion* of /l/ is common in the Dominican Republic, Cuba, Puerto Rico, Panama, coastal Colombia, Chile, and the Canary Islands (Alba 1990: 134; Quilis 1993: 326-327). *Nasalization* of /l/ has been reported in Cuba (Quilis 1993: 328-329).

Phonetic contrast between variables can become reduced due to articulatory weakening, neutralization, coarticulatory overlap, or other phonetic and phonological processes.

In such cases, speakers have been reported to employ acoustic enhancement strategies to preserve distinctions, ensuring that reduced contrasts remain perceptible. These strategies include segmental lengthening, as well as differences in spectral properties and coarticulatory cues. For example, Willis & Bradley (2008) provide acoustic evidence that the contrast between taps and trills is maintained in Dominican Spanish, despite a reduction in trill occlusions. Their study found that even when the phonetic realization of the trill is weakened, speakers compensate by lengthening the duration of the segment, ensuring that the distinction between taps and trills remains perceptible (Willis & Bradley 2008: 95). Similarly, Delgado-Díaz & Galarza (2015) found that speakers maintain contrast in perception between /h/ and /r/ even when both are realized as [x]. These studies confirm that despite phonetic reduction, acoustic differences can still be observed in formant structure, allowing speakers to successfully differentiate intended contrasts.

Despite extensive research on liquid variation in Spanish, there is an asymmetry in the literature that is relevant to this dissertation—namely, the relative lack of variationist studies on Dominican Spanish compared to Puerto Rican Spanish. While research on liquid variation in the Spanish of Dominicans³ has grown, it remains less extensive than studies on Puerto Rican Spanish.⁴

2.4.2 Additional Variables: An Introduction

This section provides an introduction to four variable phenomena in Spanish for which I haven't developed an original methodology in this dissertation: *coda /s/*, *filled pauses*, *subject personal pronouns*, and *subject placement*. It begins with a general overview of the

³US Latinos of Dominican origin are the 5th largest group of Hispanic origin (U.S. Census Bureau 2019).

⁴Puerto Ricans are the 2nd largest group of Hispanic origin in the United States (U.S. Census Bureau 2019).

nature of each variable and a brief description of the internal and external factors that have been found to constrain their variation. The discussion is followed by a dedicated section that situates these variables within a continuum of salience.

2.4.2.1 Subject Pronouns Expression

The variable use of subject personal pronouns (SPPs) in Spanish refers to the phenomenon in which finite verbs can appear with an explicit subject pronoun, as in example 1a, or without one, as in example 1b. It is a feature of Spanish morphosyntax that has been a central focus of variationist sociolinguistic research. Early studies on SPP variation examined Spanish spoken in Buenos Aires, Puerto Rico, and the Mexican-American community of East Los Angeles (Barrenechea & Alonso 1973; Morales 1982; Silva Corvalán 1982), establishing a foundation for decades of subsequent research. Since then, numerous studies have analyzed both linguistic and social constraints on pronoun usage, making it one of the most well-documented sociolinguistic variables in Spanish (Bayley & Pease-Alvarez 1997; Flores-Ferrán 2004; Otheguy et al. 2007; Torres Cacoullos & Travis 2010; Vidal-Covas 2013; Orozco et al. 2014; Cerrón-Palomino 2014; Shin et al. 2015; Ortiz López 2019; Dionne 2023, *inter alia*).

- | | | | | | | |
|-----|----|-----------------|-----------|----|---------------|----------|
| (1) | a. | Yo bailo. | (PRESENT) | b. | Bailo. | (ABSENT) |
| | | I dance-1SG.PRS | | | dance-1SG.PRS | |
| | | ‘I dance.’ | | | ‘I dance.’ | |

Research has identified a series of linguistic constraints that systematically shape the distribution of SPPs. The most consistent external constraint is *region of origin*, with Caribbean and coastal South American dialects favoring higher overt pronoun rates, compared to Andean and Peninsular varieties, which more frequently omit pronouns (Cameron

1993; Orozco & Hurtado 2021b; Otheguy & Zentella 2012; Escobar & Potowski 2015). Spanish speakers with more years spent in an *English-speaking environment* (e.g., U.S. Latinos) may exhibit higher pronoun rates, likely due to cross-linguistic influence (Otheguy et al. 2007; Otheguy & Zentella 2012; Erker 2022). Other social factors, such as *age*, *gender*, and *education level*, have produced mixed results across studies (Bentivoglio 1987; Cameron 1992; Bayley & Pease-Alvarez 1997; Flores-Ferrán 2002; Otheguy & Zentella 2012; Shin & Erker 2015; Orozco 2015).

One of the strongest linguistic predictors is *person and number*: first-person singular pronouns (*yo*) and third-person pronouns (*él*, *ella*) are likely to occur at a higher rate than plural subjects (Otheguy et al. 2007; Otheguy & Zentella 2012; de Prada Pérez 2009; Carvalho & Child 2011; Erker & Guy 2012; Orozco & Hurtado 2021a). Another key constraint is *switch reference*, which refers to whether the subject remains the same or changes from the previous clause. Speakers are more likely to use an overt SPP when there is a switch in referent (e.g., “*Yo fui al mercado y ella se quedó en casa.*”) because pronouns help clarify subject continuity and reduce ambiguity (Cameron 1995; Erker & Guy 2012; Posio 2014; Torres Cacoullos & Travis 2018). Relatedly, *preceding pronoun presence* also conditions variation—SPPs are more likely to occur when the previous clause contained an overt pronoun, a pattern attributed to *priming effects* (Flores-Ferrán 2004; Torres Cacoullos & Travis 2019; de Prada Pérez 2020; Orozco & Hurtado 2021a).

Verb frequency has been found to amplify the effects of other linguistic predictors. Highly frequent verbs (e.g., *saber*, *creer*, *ser*) tend to develop stronger pronominal biases (Erker & Guy 2012; Dionne 2023). However, these effects are not uniform across all verbs, suggesting that frequency effects interact with other linguistic constraints rather

than functioning as a strict determinant of variation. Other important linguistic constraints include *tense, mood, and aspect (TMA)*, with imperfective forms generally favoring overt pronoun use more than other forms (Silva Corvalán 1982; Otheguy & Zentella 2012; Erker & Guy 2012; Orozco 2015; Shin & Montes-Alcalá 2014). The *semantic category of the verb* also plays a role: cognitive and mental activity verbs (e.g., *creer, pensar, saber*) exhibit the highest overt pronoun rates, followed by stative verbs (e.g., *ser, estar*), while external activity verbs (e.g., *estudiar, jugar*) disfavor pronoun expression (Otheguy & Zentella 2012; Erker & Guy 2012).

2.4.2.2 Subject Placement

In Spanish, subject placement relative to the verb exhibits systematic variation across both pronominal and non-pronominal subjects. Speakers commonly use both Subject-Verb (SV) and Verb-Subject (VS) word orders, as shown in example (2). While SV order (2a) is generally preferred (Vidal-Covas 2023), VS order (2b) remains a notable option in spontaneous speech.

- | | | | | | | |
|--------|------------------------|------|----|--------------------|------------|------|
| (2) a. | Ella/Mimi bailó. | (sv) | b. | Bailó | ella/Mimi. | (vs) |
| | She/Mimi dance-3SG.PST | | | dance-3SG.PST | she/Mimi | |
| | ‘She/Mimi danced.’ | | | ‘She/Mimi danced.’ | | |

Research on Spanish varieties suggests that both syntactic and information-structural factors shape subject placement. While SV order is generally preferred, VS order remains systematically influenced by verb type and pragmatic considerations (Vidal-Covas 2023). Similarly, Erker et al. (2017) reported that post-verbal subjects are more likely to occur with lexical noun phrases and clausal subjects than with pronominal subjects, reinforcing the notion that pronoun use plays a key role in syntactic variation. Additionally, linguistic

factors such as argument structure have been shown to influence subject placement. Studies grounded in the Unaccusative Hypothesis, developed by Perlmutter & Postal (1974), suggest that VS order is more common with unaccusative verbs than with unergative verbs, as unaccusative subjects behave more like direct objects syntactically. Supporting this claim, Vidal-Covas (2023) found that VS order was significantly more acceptable with unaccusative verbs, particularly in wide-focus contexts. Moreover, the study demonstrated that pragmatic conditions can override structural preferences, as VS order was more acceptable when the subject introduced new information. In contrast, unergative verbs overwhelmingly favored SV order, highlighting the interaction between verb class and word order preferences.

Speakers of Caribbean Spanish show a stronger preference for SV order than do speakers from other regions in the Hispanophone world, suggesting that word order may be more constrained in certain dialects (Ortiz López 2009). This tendency is particularly evident in the position of pronouns, which are more likely to appear in the pre-verbal position compared to non-pronominal subjects (Morales de Walters 1997; Morales 1999; Raña-Risso 2013; Zagona 2002). Additionally, increased use of SV order has been linked to longer residence in the United States, possibly due to convergence with English and bilingual optimization (Toribio 2004; Zapata et al. 2005; Raña-Risso & Barrera-Tobón 2018). Erker et al. (2017) examined Cuban Spanish speakers in New York and found generational shifts in subject placement, where heritage speakers exhibited different preferences from newcomers. Furthermore, (Raña-Risso & Barrera-Tobón 2018) identified a connection between the frequency and placement of pronouns, noting that an increase in *present* pronouns correlates with a rise in *pre-verbal* placement.

2.4.2.3 Filled Pauses

Filled Pauses (FPs), such as *[ə(m)]* or *[a(m)]* in English and *[e(m)]* or *[a(m)]* in Spanish, are common in spontaneous speech, playing a key role in both production and perception (Swerts 1998; Bell et al. 2003; Watanabe et al. 2008). For speakers, FPs help with planning upcoming utterances and holding the conversational floor (MacKay & Osgood 1959; Clark & Fox Tree 2002). For listeners, FPs act as cues that enhance attention, structure discourse, and aid syntactic processing (Watanabe et al. 2008; Bailey & Ferreira 2003).

Despite their prevalence, FPs have been marginalized in linguistic inquiry (though they have been extensively studied outside of linguistics for quite some time), often dismissed as mere performance errors (Chomsky 1965: 4). However, recent work suggests that FPs are not epiphenomenal to language but rather systematic and socially meaningful (Fruehwald 2016; Tottie 2014). They exhibit cross-linguistic variation, with speakers of different languages favoring distinct phonological realizations (Watanabe et al. 2004; Couper-Kuhlen & Ono 2007; Erker & Vidal-Covas 2022).

Studies show that FP placement is systematic, with a preference for utterance boundaries over internal positions (MacKay & Osgood 1959; Lipski 2005a; Vidal-Covas 2020; Graham 2013). FP usage also correlates with *utterance complexity*, with longer and more syntactically complex utterances favoring the use of FPs, particularly those with nasal codas (Watanabe et al. 2008; Vidal-Covas 2020). The presence or absence of *following silence* has been found to be a conditioning factor, with certain FPs more likely to be followed by pauses than others (Clark & Fox Tree 2002; Erker & Vidal-Covas 2022). In Spanish, phonological FPs with nasal codas (e.g., [em]) are more likely to be followed by silence

than those without (Vidal-Covas 2020).

Social and contact-related factors also influence FP variation. *Age, sex, and regional origin* all correlate with differences within and between speakers (Tottie 2014; Fruehwald 2016; Erker & Bruso 2017). Dialectal differences in FP preference have been observed across Spanish varieties (Erker & Bruso 2017; Galué 2002). Notably, younger Spanish speakers tend to favor lexical FPs over phonological ones (Erker & Bruso 2017; Graham 2013). Additionally, *contact with English* reportedly influences FP usage among U.S. Spanish speakers (Erker & Vidal-Covas 2022; Vidal-Covas 2020). Those with greater exposure to English favor centralized vowel realizations (i.e., [a]) over the traditional [e] of monolingual varieties (Erker & Vidal-Covas 2022). This pattern aligns with Silva-Corvalán's (1994)'s hypothesis that bilinguals adopt linguistic strategies to reduce cognitive load while avoiding radical structural changes.

2.4.2.4 Coda /s/

Variable production of syllable-final or *coda* /s/, as in example 3 (showing the most frequently attested segmental variants), has become one of the most widely studied sociolinguistic variables of Spanish in variationist research, as noted by Gradoville et al. (2022: 2).

- (3) Tú baila/s/ [s, h, Ø]
You dance-2SG.PRS
'You dance.'

The phoneme /s/ is a voiceless alveolar fricative that, depending on dialectal variation and phonological context, exhibits multiple allophonic realizations. The primary segmental variants of syllable-final /s/ include full retention [s], aspiration [h], and complete

deletion [Ø]. Aspiration entails a weakening of the alveolar fricative into a glottal fricative, which occurs frequently in Caribbean and coastal Spanish varieties, while deletion represents the most advanced stage of /s/ weakening.

The weakening of coda /s/ has been argued to be governed by well-documented phonetic processes, particularly *aspiration* ([s] → [h]), *deletion* ([s] → [Ø]), and *sonorization* ([s] → [z] before voiced consonants) (Hualde 2014: 155). Several intralinguistic factors systematically constrain coda /s/ production, including (i) *speech rate*, (ii) *word frequency*, (iii) *syllable stress*, and (iv) *preceding and following phonological environment* (Erker 2012). In addition, extralinguistic factors such as *sex*, *education level*, *regional origin*, and *immigration status* also shape its variation.

Dialectal variation in the production of coda /s/ is significant, particularly across Latin America. Retention of coda /s/ is characteristic of highland varieties, including non-coastal Mexico, Costa Rica, Guatemala, the Andean regions of Colombia, Ecuador, Peru, and Bolivia (Penny 2000; Hualde 2014). However, in Caribbean Spanish, including Cuba, the Dominican Republic, Puerto Rico, coastal Colombia, Mexico, Venezuela and Panama, /s/ routinely undergoes aspiration and deletion at a high rate (Lipski 2012: 4). Although most research on /s/ weakening has focused on coda position, it is important to note that aspiration can also occur in onset position in certain dialects, including parts of Andalucía, Colombia, Honduras, El Salvador, Mexico, and the Spanish of New Mexico (see Lipski 1999; Brown 2004; Flórez 1951; Brown & Torres Cacoullos 2003; Hualde 2014).

The geographic patterns of coda /s/ weakening in Spanish have been shaped by a range of sociolinguistic and historical factors, including African and Indigenous language contact, colonial settlement, and regional demographics. High rates of weakening in the

Caribbean, coastal Colombia, and parts of Central America have been linked to African substrate influence stemming from the legacy of slavery (Lipski 1996b; Lipski 2008), while the conservative retention of /s/ in the Andean region has been attributed to contact with Quechua. A similar trend is observed in Guatemala and parts of southern Mexico, where Mayan language contact appears to reinforce /s/ retention (Escobar 2012: 80). Beyond phonological and contact-based explanations, language ideologies also play a crucial role in shaping both the use and perception of coda /s/. In Caribbean varieties, /s/ weakening—through aspiration or deletion—is widely associated with informality, rurality, and regional speech, while its retention is linked to education, prestige, and formality (Hualde 2014: 293). These associations can lead to diverging linguistic behaviors: in some cases, speakers engage in hypercorrection—such as inserting an /s/ where it does not etymologically belong, a phenomenon known as *hablar fisno* in the Dominican Republic—as an attempt to align with perceived prestige norms (Lipski 1996b; Lipski 2008; Hualde 2014). In other cases, especially among Caribbean-origin speakers in the U.S., the weakening of /s/ functions as a marker of authenticity and dialectal pride, serving as a site of resistance to external linguistic pressures and reinforcing in-group identity (Erker 2017).

2.5 Salience of the Variables in the Study

As discussed in earlier sections, salience operates along a continuum and is shaped by both social and cognitive factors. Building on foundational insights from Labov, Trudgill, and Rácz, this section shifts from theory to application by examining how salience can be empirically observed through speaker behavior. To that end, I rely on explicit metalinguistic awareness, as evidenced by speakers' direct mentions of linguistic features in the *Spanish*

in *Boston Corpus*. By this measure, *coda /s/* and *liquids* emerge as the two most salient variables, as they are the only ones explicitly referenced in spontaneous conversations when speakers discuss dialectal differences. This places them at the highest end of the salience spectrum, aligning with Labovian *stereotypes* and Trudgill's criteria for heightened awareness. In contrast, *subject personal pronouns (SPPs)*, *subject placement*, and *filled pauses (FPs)* exhibit varying degrees of lower salience, with their placement on the continuum determined by the extent of their social and stylistic variation, as well as the level of speaker awareness they elicit. To visualize this gradience, Figure 2.2 integrates these five linguistic variables into the salience continuum previously introduced, positioning them according to the degree of social awareness and linguistic structuring they exhibit.

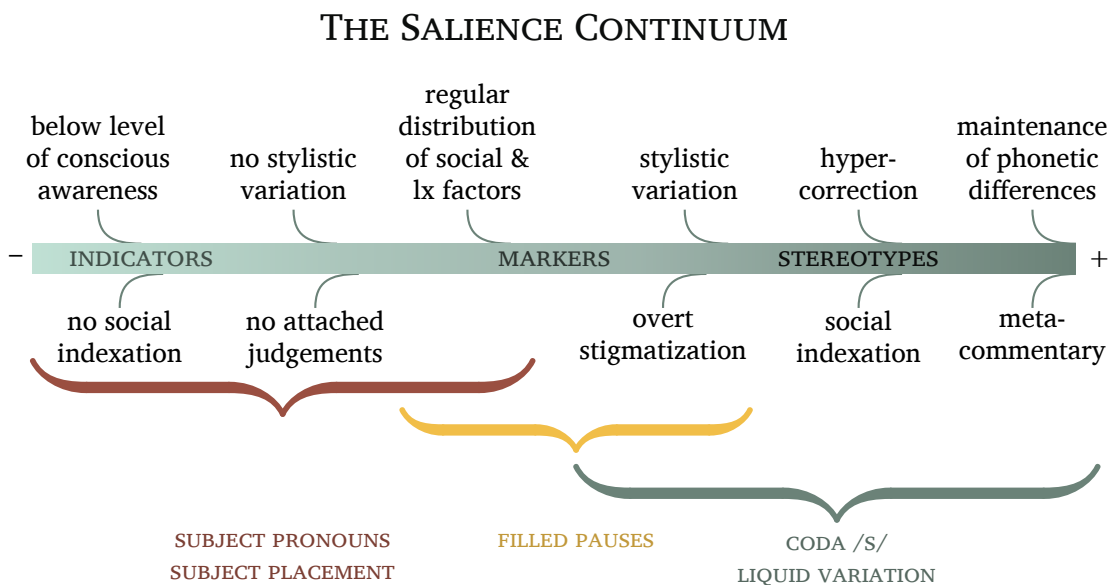


Figure 2.2: Placement of variables in the salience continuum

At the lower end of the salience spectrum, *subject personal pronouns (SPPs)* and *subject placement* exhibit different distributions across dialects of Spanish as well as between

Spanish and English but lack strong social indexation, stylistic variation, or explicit meta-commentary. *SPPs* are primarily conditioned by grammatical constraints such as switch reference, person and number, and verb semantics rather than social awareness. While regional dialects, particularly in the Caribbean, favor higher rates of overt pronoun use (Otheguy & Zentella 2012; Cameron 1993), speakers do not explicitly discuss or monitor their pronoun usage in spontaneous speech. Moreover, while cross-linguistic differences exist—Spanish being a pro-drop language, unlike English—these distinctions do not seem to carry overt social meaning among bilinguals. *Subject placement*, similarly, is primarily influenced by syntactic and discourse-pragmatic constraints, with Caribbean speakers favoring preverbal subjects more than other dialects (Ortiz López 2009; Erker et al. 2017). Despite exhibiting dialectal variation and crosslinguistic contrast, subject placement does not appear to trigger social awareness.

Slightly higher on the continuum, *filled pauses (FPs)* demonstrate some level of social variation and stylistic modulation, placing them closer to Labovian *markers*. Social variation has been observed, particularly among U.S. Spanish speakers, who favor centralized vowels [a] and [ə] over [e] (Erker & Vidal-Covas 2022), a shift linked to bilingual contact and convergence with English. Additionally, stylistic variation has been reported, with younger speakers and more formal settings favoring lexical FPs (e.g., “este...”) over phonological ones (e.g., [e(m)]) (Erker & Bruso 2017; Graham 2013). While overt awareness of *FPs* is limited, some evidence suggests that they can be perceived and evaluated by speakers. For instance, Puerto Rican rapper and medical doctor PJ Sin Suela notably released a song where the first 100 words began with [e], referencing how he was frequently criticized for using the filled pause [e] during medical school presentations. This

indicates that while *FPs* may not be as socially salient as other variables, their presence can be noticed in specific contexts.

At the highest end of the salience continuum, *coda /s/* and *liquid variation* exhibit all of the hallmarks of a highly salient sociolinguistic variable, including overt meta-commentary, hypercorrection, dialectal variation, phonological contrast maintenance, and social stratification. *Coda /s/* reduction is among the most perceptually salient features of Caribbean Spanish and frequently figures in folk ideologies of Spanish variation (Lipski 2008). Its high social salience is reflected in several ways. First, the process is subject to overt stigma and hypercorrection, as the weakening of */s/*—whether through aspiration or deletion—conflicts with orthographic representations. This sometimes results in hypercorrective insertions, a phenomenon known in the Dominican Republic as *hablar fisno* (Lipski 1996b; Lipski 2008; Hualde 2014). Second, *coda /s/* plays a crucial role in phonological contrast maintenance, as its deletion can neutralize grammatical distinctions such as plurality and verbal agreement, thereby affecting morphosyntactic transparency (Hualde 2014: 157). Third, its variation across dialects further contributes to its salience, as Caribbean and coastal Spanish varieties exhibit much higher rates of */s/* weakening than highland dialects, where retention is the norm (Lipski 2012; Penny 2000). Finally, and most crucially, its status as a highly salient linguistic feature is reinforced by metalinguistic awareness: speakers in the *Spanish in Boston Corpus* explicitly discuss *coda /s/* reduction in conversations, demonstrating that it is not only perceptually prominent but also socially meaningful.

Liquid variation, particularly in the Caribbean, is similarly recognized by speakers and is often subject to prescriptive correction, hypercorrection, and dialectal commentary. Like

coda /s/ reduction, it is frequently marked by overt stigmatization, as certain liquid variants are perceived as nonstandard. In Puerto Rico, for instance, explicit campaigns have discouraged deletion of liquids as well as the realization of /r/ as [r]. Additionally, its prominence stems from its role in phonological contrast maintenance. Liquid neutralization—such as the lateralization of /r/ to [l]—can eliminate phonemic contrasts in minimal pairs. As with *coda* /s/, *liquid variation* also exhibits clear dialectal distinctions, with Caribbean speakers lateralizing and neutralizing rhotics at significantly higher rates than speakers of other dialect groups (Lipski 2012; Hualde 2014). Importantly, speakers in the *Spanish in Boston Corpus* actively reference liquids when describing Puerto Rican and Dominican speech, reinforcing its status as a socially salient feature within this linguistic community.

These observations are explored further in Chapter 4, where I analyze how participants discuss (or overlook) specific features in spontaneous conversation. As I will show, *coda* /s/ and liquids stand out for their consistent appearance in speaker metacommentary, often surfacing without direct prompting. Their prominence in the discourse offers compelling evidence of their elevated salience in this contact setting. The next chapter will detail the methodology I developed for the analysis of liquids, further explaining the framework and approach adopted in this dissertation.

3 Data & Methods

To address the first research question—how linguistic and social factors shape variation in the production of liquids in Spanish—this dissertation analyzes a corpus of spontaneous speech, focusing on the phonetic realization of liquid consonants. The analysis includes a phonetic description of the segmental and acoustic characteristics of liquid variation, an examination of linguistic constraints that may influence production, and a consideration of contact-related factors and social characteristics of speakers.

Beyond analyzing *liquid variation* in isolation, this dissertation also investigates its covariation with additional linguistic variables—*filled pauses*, *subject personal pronouns*, *subject placement*, and *coda /s/*—to assess how salience influences their patterning in a contact setting. The data for these additional variables were previously collected as part of *The Spanish in Boston Project*, a collaborative research initiative involving a dedicated team of linguists.¹ Since these four variables have been extensively studied in prior research, their methodological treatment follows previously established procedures. Therefore, while their covariation is analyzed in this dissertation, their methodology is not discussed in detail here. The methodologies for analyzing these non-liquid variables can be found in Erker & Bruso (2017) for filled pauses, Otheguy & Zentella (2012), Erker (2017), and Vidal-

¹I wish to acknowledge the enthusiastic research team of the Spanish in Boston Project: Daniel G. Erker, Gretchen Anderson, Corey Angers, Yadirys Collado, E. J. Conlon, Gabriela Diaz-Quinones, Kendra Dickinson, Danielle Dionne, Daniela Estrella, Lily Fawcett, Lauren Gerrish, Morganne Goddard, Karina Hernández, Tommy Piernikowski, Madeline Reffel, Emma Temkin, Meda Sandu, Natalie Swiacki, Deb Waughtal, and Abbey Tolentino-Winter.

Covas (2013, *inter alia*) for subject pronouns, Erker et al. (2017) for subject placement, and Erker & Reffel (2021) for coda /s/. This chapter focuses exclusively on the methodology for *liquid variation*, outlining how the linguistic and social factors were coded and describing the statistical approach used in this study.

3.1 Data Collection

For the present dissertation, I selected a subsample of 22 participants from the original dataset and carried out entirely new data collection and analysis specific to the study of liquid variation. While the sociolinguistic interviews had already been conducted and transcribed as part of the original data collection, I systematically extracted and coded data on liquid variation from the interview recordings.

As part of the original data collection initiative (2014–2017) of *The Spanish in Boston Project*, each participant completed both (i) a sociolinguistic interview, conducted in Spanish, and (ii) a demographic and linguistic questionnaire. The questionnaire² was designed to gather detailed information about the participants' demographic characteristics, social and linguistic backgrounds, and attitudes toward language use. To ensure consistency and minimize variability in recording conditions, all interviews took place in the same quiet room using a *Zoom H4N* digital recorder and an *SM93* lavalier microphone. The audio recordings were captured at a sampling rate of 44.1 kHz and subsequently transcribed. This high-quality audio dataset serves as the basis for the novel analysis of liquid variation conducted in this dissertation.

²The questionnaire is included in Spanish in Appendix C and in English in Appendix D

3.1.1 Data Preparation and Coding

Lexical items containing liquid tokens were identified and segmented using Praat text grids. Each host word containing a liquid segment was manually segmented in a designated text grid tier. A minimum of 250 tokens were collected per speaker. While tokens for taps and laterals were relatively straightforward to identify, special attention was required for trill tokens due to their lower frequency in the dataset. After segmentation of the host words, an additional tier was created for annotating the liquid segments within each host word. The liquids were coded according to the segmentation protocol detailed in the following section. Subsequently, additional tiers were added for predictor variables, which were coded for all tokens. This approach ensures the dataset's integrity and facilitates acoustic and variationist analysis in subsequent stages of the study.

To ensure systematic and reproducible data analysis, a comprehensive coding manual was developed. This process began with the selection of a trial speaker to refine operational definitions and establish the coding protocol. The analysis relied on the audio recordings from the sociolinguistic interviews, which were transcribed directly in *Praat*. These orthographic transcriptions were used in conjunction with the audio to identify and annotate lexical items containing liquid tokens.

3.2 Study Design

A combined impressionistic and instrumental approach in the study of phonetic variables is advantageous, as it allows for greater accuracy and reliability in describing patterns of variation (Erker 2010; Labov 2001). Given the potential variability in perceptual analyses of liquids (Lawson et al. 2010), the use of waveform and spectrographic displays served

as a valuable tool for segmentation, helping to improve consistency in identifying boundaries and classifying tokens. Although factors such as periodicity, formant transitions, and intensity changes were considered to refine segmentation, the primary focus remains on perceptual coding and the classification of surface forms, as formant measurements were not analyzed in this dissertation.

3.2.1 Dependent Variable: Liquid Variation

The phenomenon of interest, *liquid variation*, was described, using the following variables:

PHONOLOGICAL FORM: This variable codes the presumed phonological identity of the token as one of three categories: /ɾ/, /r/, or /l/. The determination of the *Phonological Form* was based on a combination of factors, including the token's orthographic representation, its presumed phonological form within the host word, and, for the rhotics, the phonological environments outlined in Hualde's (2014: 183) guide to Spanish rhotic distribution. The use of orthographic spelling and Hualde's distribution guide, as summarized in Table 3.1, provided a structured framework for differentiating between the *vibrante simple* (tap) /ɾ/ and the *vibrante multiple* (trill) /r/. Specifically:

- **Contrastive Contexts:** These occur in specific word-dependent positions where the tap and trill are phonemically distinct. For example, in intervocalic and word-internal positions, words like *caro* (/ka.ro/, “expensive”) and *carro* (/ka.ro/, “car”) illustrate this contrast.
- **/r/ Environments:** These are contexts where only the trill /r/ occurs. Examples include word-initial or root-initial positions (*roca*, “rock”) and after a heterosyllabic consonant (*alrededor*, “around”).

- **/r/ Environments:** These are contexts where the tap /r/ occurs. Examples include positions after a tautosyllabic consonant (e.g., onset clusters like *broma*, “joke”) and word-finally when followed by a vowel (*ser amigos*, “to be friends”).
- **Variable Contexts:** These are environments where the realization of the rhotic may vary but is usually realized as the tap /r/. Examples include after a vowel and before a consonant (*parte*, “part”), word-finally followed by a consonant (*ser poeta*, “to be a poet”), and word-finally followed by a pause (*ser*, “to be”).

As shown in Table 3.1, specific notations such as # for word boundaries, ## for pauses or phonological phrase boundaries, and . for syllable boundaries were used to delineate these environments. In cases where the phonological environment alone did not provide sufficient clarity, the presumed phonological representation of the host word (based on lexical and phonotactic considerations) was also used to inform the coding. This framework defines the *envelope of variation* for Spanish rhotics, providing a structured way to analyze their phonological identity in context.

Table 3.1: Hualde’s (2014) Spanish Rhotic Distribution

ENVIRONMENT	CONTEXT	EXAMPLE	TRANSLATION
Contrastive (<i>word-dependent</i>)			
V__V	Intervocalic& Word-internal	/ka.ro/ vs. /ka.ro/	expensive vs. car
/r/			
#__	Word-initial (or root-initial)	/ro.ka/	rock
C.__	After heterosyllabic consonant	/al.re.de.dor/	around
/r/			
C__V	After tautosyllabic consonant (onset consonant cluster)	/bro.ma/	joke
V__#V	Word final followed by vowel	/ser#a.mi.gos/	to be friends
Variable (<i>usually /r/</i>)			
V__C	After vowel & Before consonant	/par.te/	part
V__#C	Word final followed by consonant	/ser#po.e.ta/	to be a poet
V__##	Word final followed by pause	/ser/	to be

¹ # represents word boundary

² ## indicates a pause or phonological phrase boundary

³ . represents a syllable boundary

SURFACE FORM: Each liquid was then labeled for the variants of each of the two types of liquids observed by the researcher when listening and inspecting the token acoustically:

LATERALS: All phonological lateral liquids were impressionistically coded as either [l], [ɫ], [ɾ], [h], [ʔ], or [Ø], as well as *ct* for *can't tell*. This last category was reserved for sounds that could not be confidently coded perceptually by the researcher.

RHOTICS: All phonological rhotics were impressionistically coded as either [ɾ], [r], [ɹ], [ɻ], [ɽ], [l], [ɫ], [x], [h], [ʔ], or [Ø], as well as *ct* for *can't tell*.

MATCH/MISMATCH: This variable describes whether the SURFACE FORM matches the PHONOLOGICAL FORM. As shown in Table 3.2, the coding is as follows:

MATCH: If the SURFACE FORM corresponded directly to the expected PHONOLOGICAL FORM, it was recorded as *match*.

MISMATCH: If the SURFACE FORM deviated from the PHONOLOGICAL FORM in its realization, it was recorded as *mismatch*.

Table 3.2: Examples of Match/Mismatch Dependent Variable for Liquids

HOST WORD	TRANSLATION	PHONOLOGICAL FORM	SURFACE FORM	MATCH/MISMATCH
<i>arriba</i>	up / above	/r/	[r]	MATCH
<i>correcto</i>	correct	/r/	[r]	MISMATCH
<i>humilde</i>	humble	/l/	[l]	MATCH
<i>mezcla</i>	mix	/l/	[l]	MATCH
<i>normal</i>	normal	/l/	[l]	MATCH
<i>perdón</i>	sorry / pardon	/ɾ/	[l]	MISMATCH
<i>pronunciación</i>	pronunciation	/ɾ/	[ɾ]	MATCH
<i>puertorriqueña</i>	Puerto Rican (fem.)	/r/	[r]	MISMATCH
<i>urbanización</i>	urbanization	/ɾ/	[Ø]	MISMATCH
<i>verduras</i>	vegetables	/ɾ/	[ɾ]	MATCH

For this analysis, a *mismatch*³ was defined strictly as any case where a liquid surfaced as

³While the match/mismatch approach may flatten some of the more subtle phonological distinctions between variants (such as different degrees of weakening or specific types of lateralization), it also allows for a

something other than its expected realization given its presumed phonological form. For example, the *vibrante simple* (tap, /ɾ/) or the *vibrante multiple* (trill, /r/) would have to surface as themselves in their respective contexts to be considered a case of phonological-phonetic match. Any deviation from these expected forms, such as substitution with a lateral, fricative, or glottal stop, would constitute a mismatch. It is important to note that this concept of *mismatch* is not intended to reflect notions of “non-standardness” or prescriptive norms. Instead, it refers to a divergence between the presumed phonological identity of a segment and its observed surface realization in this specific linguistic context. This distinction helped ensure consistency in the interpretation of the data.

3.2.2 Segmentation Procedure for Liquid Tokens

Following the data preparation process, the segmentation of host words and individual liquid tokens was carried out in Praat. Each liquid token was carefully segmented by closely examining both the waveform and the spectrogram to distinguish it from the surrounding sounds. In the segmentation process, two key objectives guided the protocol: (1) distinguishing the liquid from the surrounding sounds, and (2) classifying its identity based on its acoustic properties. These steps were essential for accurately analyzing the tokens, particularly in cases where the realization of liquids diverges from canonical expectations. The *Surface Form* variable, central to this study, captures such variation by providing a structured framework for classifying these sounds, even when they exhibit

more comprehensive view of the variation across all types of liquid realizations, incorporating the full range of patterns that occur in real-world speech. Many previous studies focus on isolated phonological phenomena (e.g., only deletion, only lateralization), which may miss broader patterns of variation. This approach is intended to capture all forms of liquid variation, though future research could develop more nuanced match/mismatch categories—such as distinguishing between rhoticity, weakening, and other types of realizational differences. In fact, I have already coded for some of these distinctions in my dataset, but limited the analysis here in order to maintain focus and scope. A more granular approach to mismatch categorization offers a promising direction for future research.

atypical acoustic characteristics.

The waveform, which represents frequency and amplitude over time, was analyzed to assess periodicity. Periodic speech sounds exhibit complex repeating waves, indicating regular vocal fold vibration, while aperiodic sounds display random, non-repetitive patterns. It should be noted that periodic and aperiodic components can co-occur within the same sound, as is often the case in speech. Spectrograms provided more detailed visual cues for identifying boundaries between sounds by displaying formant structure. For all liquids, changes in formant frequency—such as shifts in the first (F1), second (F2), or third (F3) formants—provided critical information for distinguishing between the preceding and following sounds. These changes often reflect the articulatory transitions between the liquid and adjacent sounds, such as the movement of the tongue or the opening and closing of the vocal tract, which can cause abrupt or gradual shifts in the formant structure. Additionally, intensity changes, such as a dip in amplitude when transitioning from a vowel to a liquid, were key cues for segmentation. The next section shows some examples to illustrate the segmentation and coding procedure for both rhotics and laterals.

Rhotics Consider the two rhotics in Figure 3.1. Figure 3.1a includes a flap in the word [para], while Figure 3.1b includes a trill in the word [para].

- Some possible cues to segment rhotics are:
 - There is aperiodicity in the waveform.
 - A steep fall of the third formant is a defining acoustic signature of rhotic sounds, particularly in retroflex or bunched articulations.
 - Duration helps in distinguishing rhotics from adjacent vowels. Rhotics, espe-

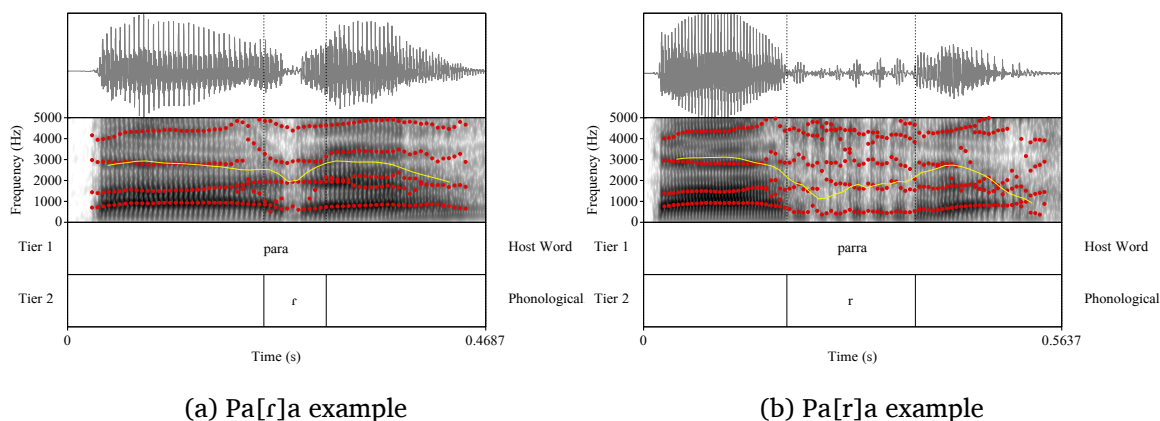


Figure 3-1: Waveform, spectrogram, and annotation of pa[r]a and pa[r]a

cially flaps, tend to have shorter durations.

- To differentiate from adjacent vowels, a dip in intensity can help (yellow line in Praat picture).
- It should be noted that a trill is much easier to differentiate from adjacent vowels than a flap because trills typically consist of multiple rapid closures of the tongue against the alveolar ridge, creating distinct bursts of energy visible in both the waveform and spectrogram. In contrast, flaps are characterized by a single, brief contact of the tongue, resulting in a shorter and less acoustically distinct interruption, which makes them more challenging to segment from adjacent vowels.
- A radical drop in amplitude can be a cue to where rhotics begin, as vowels have a higher amplitude than rhotics.
- F4-F5 range: Subtle acoustic differences in F4 and F5 provide additional information when coding the *Surface Form* of rhotics, particularly for distinguishing bunched and retroflex variants (Zhou et al. 2008: 4467).

Lateral Laterals are approximants, which means they are among the most sonorous consonants. As their classification hints, they are somewhere between a vowel and a consonant, a fact that can make it hard to differentiate between the sounds. Figure 3.2 includes the lateral /l/ in the word [pala].

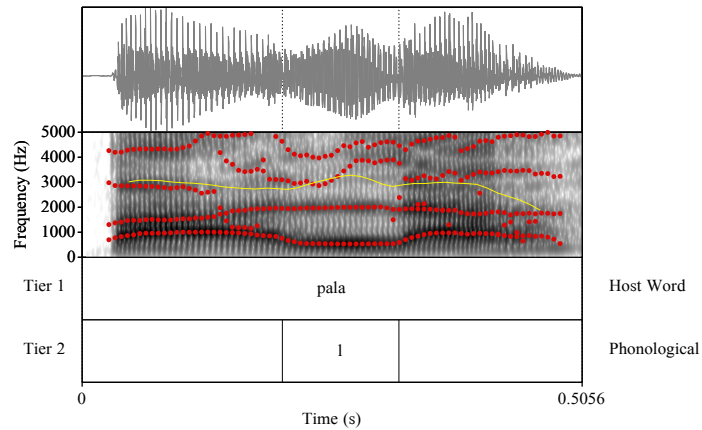


Figure 3.2: Waveform, spectrogram, and annotation of the word [pala]

- The following cues can be used to find and segment the lateral in order to differentiate between the previous and following sounds:
 - The lateral’s waveform is clearly periodic, a characteristic that can help differentiate it from flanking consonants. However, periodicity alone does not distinguish laterals from vowels, as both exhibit periodic waveforms. Additional acoustic cues, such as differences in intensity, formant transitions, and amplitude changes, are critical for segmentation.
 - For a lateral, there is no major dip in amplitude or intensity (*yellow line in Praat picture*) in contrast to rhotics, which tend to have lower intensity relative to vowels. This distinction is particularly relevant in contexts where liquids may exhibit overlapping acoustic properties, such as when a purportedly phonolog-

ical rhotic sound is realized with lateral-like qualities.

- When differentiating between a lateral and a vowel, the lateral typically shows slightly lower intensity due to the inherently higher sonority of vowels. This difference often appears as a subtle dip in intensity at the boundary between the two sounds. Additionally, while both vowels and laterals exhibit formant transitions, those of laterals tend to be more gradual and less pronounced.
- In post-vocalic positions, laterals often exhibit abrupt shifts in formant structure, providing a key cue for segmentation in these contexts.

It should be noted that the goal of the study was to examine variation in spontaneous speech. As such, “idealized” tokens, like the ones shown in Figures 3.1 and 3.2, were rarely encountered. The ultimate goal was to make the best segmentation possible and segmental identification, within reason.

3.2.3 Linguistic Factors

The study investigates liquid variation using variationist sociolinguistic methods, i.e. modeling variation that occurs in naturalistic speech by inspecting the relationship between the distribution of variants and a range of linguistic and social factors. Below, I list and briefly describe the linguistic factors used in the analysis of *liquid variation*. For a more detailed breakdown of how the data was coded, please refer to the Coding Manual, which can be found in Appendix B.

HOST-WORD - This variable refers to the carrier word for each liquid token.

SYLLABLE STRESS - Whether the syllable to which the token belonged bore primary stress was coded as either *stressed* or *unstressed*. Examples are provided in Table 3.3.

Table 3.3: Syllable Stress Example

HOST WORD	SYLLABIFICATION	STRESS
diferente	di.fe. ren .te	stressed
escuela	es. 'cue .la	unstressed
gustaría	gus.ta. 'rí .a	stressed
otras	'o .tras	unstressed
problema	pro. 'ble .ma	stressed
quieres	'quie .res	unstressed

- Primary stress is marked with a stress diacritic (') placed before the stressed syllable.
- Syllables containing the liquid consonant are **bolded**.
- Monosyllabic words were coded as *stressed*.

NUMBER OF SYLLABLES - The syllable count of each host word was coded to calculate speech rate (See SPEECH RATE below). Table 3.4 provides examples.

Table 3.4: Number of Syllables in Word

HOST WORD	TRANSLATION	# OF SYLLABLES
las	the (fem. plural)	1
puer.to	port	2
la.ti.no.a.mé.ri.ca	Latin America	7
o.to.rrí.no.la.rín.gó.lo.go	otolaryngologist	9

SYLLABLE TYPE - The type of syllable to which the token belonged was coded as either *open* or *closed*. Examples are provided in Table 3.5.

Table 3.5: Syllable Type Examples

HOST WORD	TRANSLATION	FIRST SYLLABLE	SYLLABLE TYPE	SECOND SYLLABLE	SYLLABLE TYPE
clase	class	cla	open	se	open
parte	part	par	closed	te	open

POSITION IN SYLLABLE - Each liquid was coded for its position within the syllable with the values *onset* or *coda*. Examples are listed in Table 3.6.

Table 3.6: Position in Syllable Examples

HOST WORD	TRANSLATION	FIRST SYLLABLE	POSITION IN SYLLABLE	SECOND SYLLABLE	POSITION IN SYLLABLE
real	royal / real	re	onset	al	coda
partir	to leave / to split	par	coda	tir	coda

POSITION IN WORD - Each liquid was coded for its position in the host word with the values *initial*, *internal*, or *final*. Refer to Table 3.7 for examples.

Table 3.7: Position in Word Examples

HOST WORD	TRANSLATION	FIRST SYLLABLE	POSITION IN WORD	SECOND SYLLABLE	POSITION IN WORD
real	royal / real	re	initial	al	final
partir	to leave / to split	par	internal	tir	final

WORD CLASS - The grammatical category of the host word to which each token belonged was coded as: *Noun*, *Verb*, *Determiner*, *Adjective*, *Adverb*, *Pronoun*, *Preposition*, *Conjunction*, or *Other*. The last category was reserved for anything that did not fall into any of the previous categories. See Table 3.8 for examples.

Table 3.8: Word Class Examples

CLASS	EXAMPLES	TRANSLATION
Noun	carro	car
Verb	celebramos, celebrar	we celebrate, to celebrate
Determiner	el, la(s), los un(a/o/s)	the, a/an
Adjective	grande, largo, corto	big, long, short
Adverb	rápidamente	quickly
Pronoun	yo, tú, él, ella	I, you, he, she
Preposition	bajo, con, hacia	under, with, toward
Conjunction	y, o, pero	and, or, but

PREVIOUS SOUND - The identity of the phonetic segment immediately preceding the liquid token.

- [a], [b], [β], [d], [ð], [e], [f], [g], [ɣ], [h], [i], [j], [k], [l], [m], [n], [ŋ], [ɲ], [o], [p], [k], [r], [ɾ], [ʀ], [ɽ], [ɻ], [s], [t], [u], [v], [x], [z]
- “CT” for when the sounds were impossible to determine confidently
- “Pause” referred to a period of silence lasting at least 50 milliseconds. This was identified as a segment of the waveform where no significant acoustic energy was present.
- These were re-coded to create new variables in R:
 - if *vowel*: Vowel Height, Vowel Frontness/Backness

– if *consonant*, Place & Manner of Articulation

FOLLOWING SOUND - The identity of the phonetic segment immediately following the liquid token was coded. (This variable had the same possible values as the variable PREVIOUS SOUND, above.)

RAW LEXICAL FREQUENCY - Each host word was coded for its frequency of occurrence within the dataset.

LOG LEXICAL FREQUENCY - The raw lexical frequency of each host word was log-transformed.

SPEECH RATE - The speech rate of each word was calculated using the following formula:

$$\text{SPEECH RATE (IN MS PER SYLLABLE)} = \frac{\text{DUR(MS) OF HOST-WORD}}{\# \text{ OF SYLLABLES}}$$

3.2.4 Social Factors and Language Contact Measures

The liquids were also coded for each of the following speaker characteristics:

AGE - Age at the time of the sociolinguistic interview.

SEX - Sex is a significant variable in classic sociolinguistic studies where females have traditionally been found to be more conservative speakers than males when it comes to stable variables, yet “lead in both the acquisition of new prestige patterns and the elimination of stigmatized forms” (Labov 2001: 266, 274). This variable was measured in terms of biological sex with the variants: *female* or *male*.

COUNTRY OF ORIGIN - This refers to the speaker’s place of birth or, for U.S.-born participants, the country of birth of their parents or grandparents. The two groups exam-

ined in this study are *Puerto Rico* and *Dominican Republic*. While Puerto Rico is a U.S. territory rather than an independent country, it is included here under Country of Origin due to its distinct sociolinguistic and cultural identity within the Hispanic Caribbean.

The speakers also vary in terms of their length of residence in the U.S. and their use of both English and Spanish. To assess variation in these dimensions, the following contact measures were examined. These measures were selected because they provide insight into how speakers' linguistic behaviors and experiences are shaped by their degree of exposure to English and Spanish, as well as the contexts in which these languages are used. Such factors are critical for understanding how contact with English as well as with speakers of varieties of Spanish other than their own may influence their use of linguistic variables.

PERCENTAGE OF LIFE IN THE UNITED STATES (PLUS) - This refers to the proportion of life that a speaker has spent living in the United States. This was measured by taking the years a person had spent in the U.S. at the time of their interview and dividing that number by their age. This measure serves as a quasi-continuous proxy for generational categories traditionally used to describe immigrant communities (e.g., first-generation, second-generation, etc.). Rather than assigning speakers to discrete generational groups, *PLUS* provides a more fine-grained measure of how much of a speaker's life has been spent in an English-dominant environment, allowing for a more nuanced assessment of contact-related effects on linguistic behavior.

SELF-REPORTED LANGUAGE USED WITH INTERLOCUTORS - This contact measure comes from speakers' responses to the following question (administered in Spanish):

‘Which language(s) do [or did] you speak with your: father, mother, siblings, younger children, older children, friends, boss, coworkers, classmates, spouse, or boy/girlfriend?’. Speakers were asked to respond: *Spanish*, *English*, or *both*. Given that some speakers did not provide an answer for every interlocutor, responses were converted to percentages to make comparisons between speakers possible. This study examines two variables derived from these responses: (i) the percentage of interlocutors with whom the speaker reported speaking *English only*, and (ii) the percentage of interlocutors with whom the speaker reported speaking *Spanish only*. These measures provide insight into the relative dominance and domain-specific use of each language.

These two variables were selected because they directly address the study’s focus on linguistic variation in contact settings. The *Percentage of Life in the United States* offers a quantifiable measure of prolonged exposure to English-dominant environments, while the *language used with interlocutors* captures the contexts in which English and Spanish are used, reflecting speakers’ habitual language use. Furthermore, these variables, alongside other questionnaire measures, provide a foundation for understanding how contact factors relate to both language use and ideologies, which are explored in subsequent chapters. The full list of questions used during the conversations is provided in Spanish in Appendix C and in English in Appendix D.

3.3 Statistical Model Selection Process

In order to quantitatively investigate variation in liquids (see Chapter 5), I employed a structured and iterative approach to statistical modeling. This section outlines the process

undertaken to construct and refine the models used to evaluate the relationship between the study's dependent and independent variables. The goal was to balance model complexity with interpretability, ensuring robust statistical conclusions while addressing the dissertation's research questions.

I began by conducting an initial assessment of multicollinearity among predictors to avoid confounding effects. This was done using the `vif()` function from Fox & Weisberg's (2019) `car` package, which calculated Variance Inflation Factors (VIFs) for all predictors. Predictors demonstrating significant multicollinearity were either removed or analyzed separately in distinct models to ensure convergence and stability. Following this, I fitted a series of fully saturated mixed-effects models, each incorporating all retained predictors and accounting for random effects associated with individual speaker variation. These models were constructed using the `glmer()` function from Bates et al.'s (2015) `lme4` package. Because each model was designed to examine variation within a specific phoneme (/r/, /ɾ/, /l/), the modeling process was not limited to a single overarching model but instead entailed multiple iterations, each focusing on a different subset of the data.

The model selection process proceeded in two stages. First, I conducted a comprehensive exploration of potential models using the `dredge()` function from Barton's (2023) `MuMIn` package, which systematically generated and compared all possible combinations of predictor variables based on Akaike Information Criterion (AIC) values. This step allowed for the identification of the best-fitting maximal model for each liquid subtype. Once the initial maximal models were determined, I refined them using the `drop1()` function from base R (2024). This step involved iteratively testing the removal of individual predictors to arrive at a final set of models that balanced explanatory power with parsimony. Thus,

the models ultimately selected were not fully saturated models but rather simplified versions that retained only the most informative predictors, ensuring both interpretability and robustness in explaining liquid variation.

4 Speaker Profiles: Language Use and Attitudes

This chapter examines the speakers who were selected for the current study, focusing on their demographic profiles, linguistic backgrounds, and sociocultural experiences. The first section introduces the speakers, detailing key aspects such as their geographic origin, educational attainment, and linguistic backgrounds, offering a foundation for understanding the sociolinguistic context of their speech. The subsequent sections explore patterns of language use and attitudes toward linguistic and dialectal variation. This exploration will be important when discussing the results later, as it provides insights into how participants' language choices and perceptions shape and reflect broader sociolinguistic patterns. By examining how participants utilize and reflect on their linguistic repertoires and express their perceptions of language contact, this chapter highlights the interplay between individual experiences, community dynamics, and broader social ideologies, setting the stage for subsequent analysis and discussion.

4.1 Speaker Selection Criteria and Description

The speakers for this project were selected based on their geographic origins. For this study, all Puerto Rican and Dominican speakers whose interviews had been transcribed and whose coding for additional variables had been largely completed were included. A total of 22 speakers are included in the current study. They represent two places: Puerto Rico and the Dominican Republic. Since *liquid variation* has been well documented for

Puerto Ricans, this group can serve as a comparison baseline to (i) see if they have similar behaviors as other Puerto Rican groups and (ii) see how Dominican speakers compare.

Table 4.1: Speakers included in Study

PSEUDONYM	SEX	AGE	PLUS
Dominican Republic			
Daniel	M	33	3
Daniela	F	38	16
David	M	47	23
Damaris	F	45	24
Diana	F	43	30
Danilo	M	46	57
Diego	M	73	71
Donaldo	M	21	81
Ernesto	M	21	100
Dafne	F	21	100
Delia	F	23	100
Puerto Rico			
Pascal	M	37	4
Paloma	F	20	15
Pilar	F	45	22
Pablo	M	21	38
Patricio	M	36	42
Paola	F	23	87
Pacifica	F	33	88
Priscila	F	24	100
Ponce	M	25	100
Pamela	F	18	100
Prospero	M	42	100

¹ PLUS = Percent Life in US

² Ernesto's parents are from the Dominican Republic and El Salvador

ficient in both English and Spanish, with several speakers demonstrating a preference for one language over the other depending on context or social interactions. The majority learned Spanish first, with English being acquired later in life, often through formal education or family contact.

In terms of language and dialect contact, participants' experiences vary considerably. Some have lived in the U.S. for decades, engaging with speakers of different Spanish varieties, while others are more recent arrivals with differing levels of exposure to U.S.-based Spanish-speaking communities. These differing experiences shape not only their use of

As summarized in Table 4.1, nine of them are US-born, while the rest were either born in one of the two places above or their parents emigrated from them. The sample includes an even distribution of male and female speakers (11 each), and participants range in age from 18 to 73 years old. Educational backgrounds also vary: seventeen speakers have completed university, three have obtained postgraduate degrees, and two have completed only primary and secondary school. Linguistically, most participants reported being pro-

Spanish and English but also their awareness and evaluation of dialectal distinctions within the broader Spanish-speaking community. As indicated in Table 4.1, this variation is reflected in participants' *Percentage of Life in the U.S. (PLUS)* scores. Seven speakers have a *PLUS* of 25 or less, meaning they have spent a relatively small proportion of their lives in the U.S. In contrast, seven participants have a *PLUS* of 100, indicating they have lived their entire lives in the U.S. mainland. The remaining eight speakers fall between these two extremes, representing a range of linguistic and cultural experiences, as well as differing levels of exposure to both English-dominant settings and diverse Spanish-speaking communities.

For a more detailed overview of each speaker, including their self-reported linguistic background, patterns of language use, and attitudes toward dialectal variation, see Appendix A.

4.2 Language Use & Attitudes Toward Dialectal Variation

This section lays the groundwork for addressing the second research question: whether salient and non-salient features pattern differently in a context of language and dialect contact. By examining participants' habitual language use and their attitudes toward dialectal variation, this section provides context for understanding the social and linguistic factors at play. These insights will be important when discussing the results in the covariation chapter, where the relationship between liquid variation and other linguistic variables is explored in detail.

In terms of language use, the data reveal a clear correlation between greater life experience in the mainland United States and increased use of English. Participants' responses

about their usual language of interaction with various interlocutors (Spanish, English, or both) were converted into percentages for three categories: *Percent Interlocutors Spanish Only*, *Percent Interlocutors Both*, and *Percent Interlocutors English Only*. Figure 4.1 illustrates

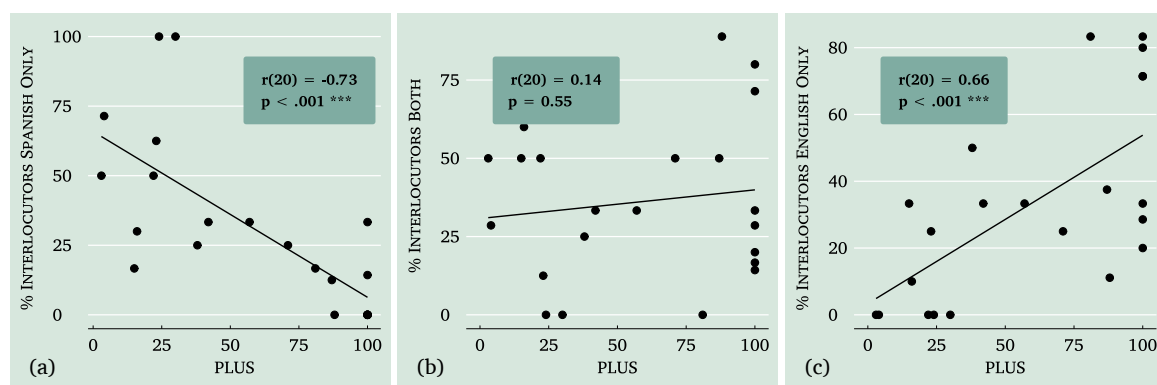


Figure 4.1: Habitual language use with interlocutors among 22 Bostonians (a) More life experience in the U.S. is associated with a decrease of Spanish only with Interlocutors. (b) A higher percentage of time spent in the use is associated with more interlocutors of both languages. (c) Greater life experience in the U.S. is associated with an increased use of English with Interlocutors.

the relationships between these variables and the contact measure *Percent Life in the United States (PLUS)*. Two of these relationships are statistically significant. As PLUS increases, the exclusive use of Spanish decreases significantly ($r(20) = -0.73, p < .001$), while the exclusive use of English increases significantly ($r(20) = 0.66, p < .001$). Although not statistically significant, it is noteworthy that the use of both English and Spanish shows a slight positive correlation with PLUS ($r(20) = 0.14, p = 0.55$).

Regarding dialectal variation, Figure 4.2 illustrates a unanimous belief among the 22 participants in this dissertation that Spanish differs between national groups. This finding aligns not only with the responses from the larger participant group in the Spanish in Boston Corpus, but also with research demonstrating that national group differences play a central role in the categorization of Spanish dialectal variation (Erker & Vidal-Covas 2024).

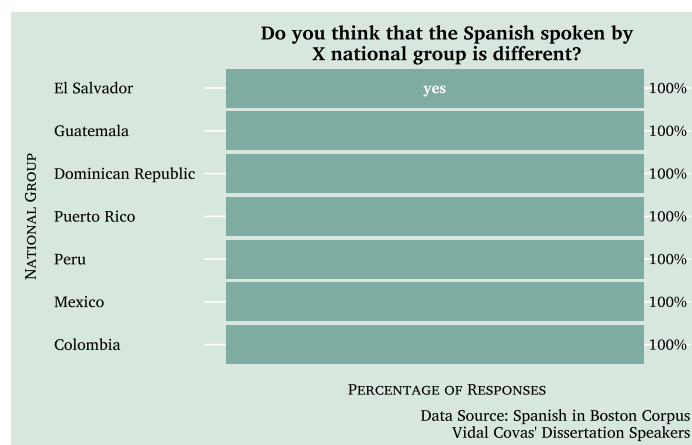


Figure 4.2: Perceptions of whether Spanish differs across individual national groups.

When it comes to potential national misidentification based on their speech, Figure 4.3 provides a snapshot of the attitudes of the Dominicans and Puerto Ricans in the study. In response to the question *'If someone told you that you sounded like X [X being a person of a different Hispanophone nation-state than your own], would you like it?'*, a majority of the speakers expressed indifference towards misidentification. One notable pattern is that Puerto Rican participants in the study are more likely to dislike being identified as Puerto Rican or Dominican than as Cuban, whereas Dominican participants are more likely to dislike being identified as Cuban than as Puerto Rican or Dominican.

Finally, consider Figure 4.4 and the participants' perspectives on preserving, adapting, and incorporating national group differences into Spanish language instruction in Boston schools. When asked *Should they learn how to speak Castilian Spanish?*, 90% of participants responded no. Similarly, 76% disagreed with the question *Should they avoid words that others might not know?*, 86% supported the idea that speakers should maintain their national group's Spanish, and 58% agreed with the question *Should the kind of Spanish you speak be taught in schools?*. Notably, all participants affirmed that US Spanish speakers

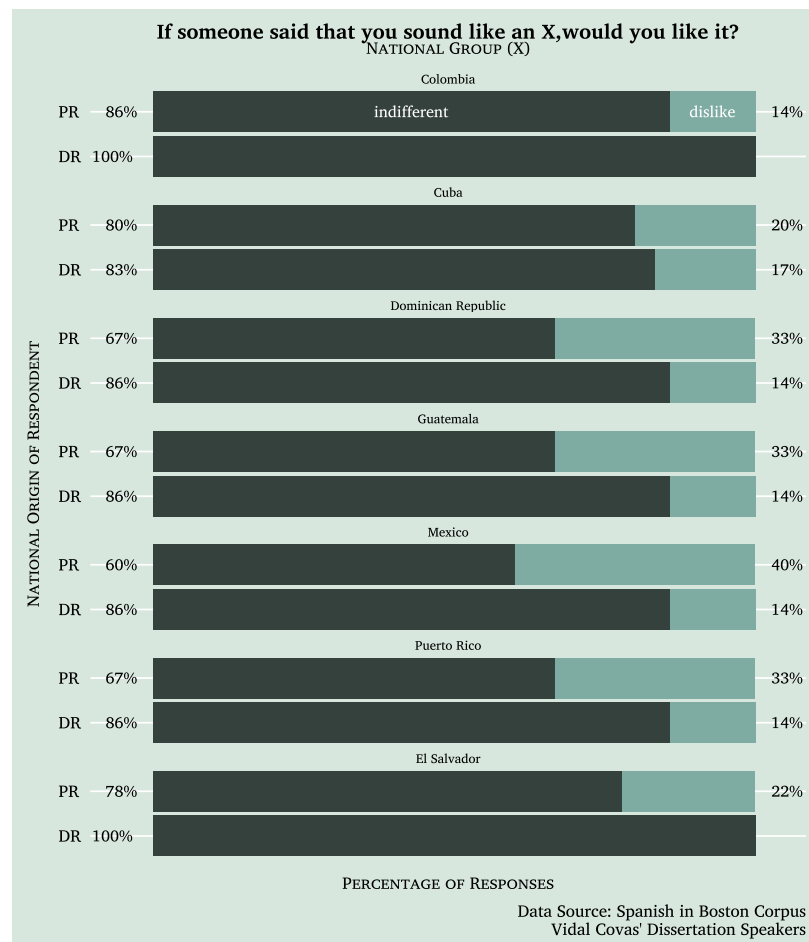


Figure 4.3: Attitudes towards speech-based misidentification of national origin.

should preserve their Spanish. These responses strongly suggest a pronounced sense of national group pride and a commitment to maintaining distinct ways of speaking within the Spanish-speaking community.

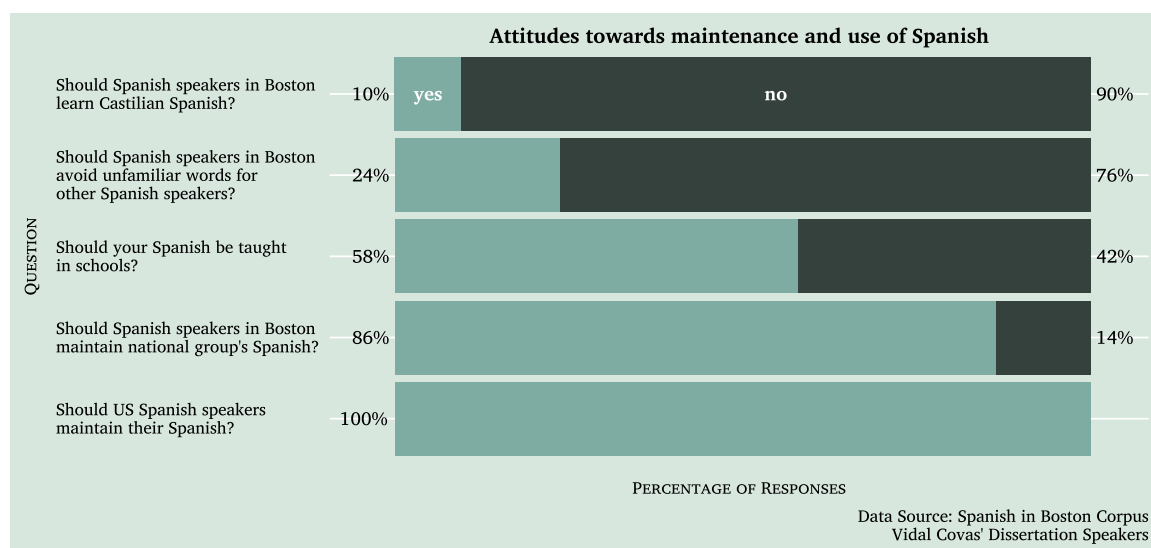


Figure 4.4: Attitudes toward the preservation, adaptation, and education of Spanish varieties.

These quantitative results provide a broad understanding of how speakers in Boston perceive and engage with dialectal variation. The overwhelming preference for maintaining national varieties of Spanish suggests that dialectal distinctiveness carries social significance within the community. However, these broad trends do not reveal which specific linguistic features speakers associate with dialect differences—or how they articulate these perceptions in their own words.

To explore this further, we turn to the speakers' own descriptions of Spanish dialects. The following qualitative data provide key insights into which aspects of variation speakers notice, how they distinguish between varieties, and—just as importantly—what features they do not mention at all.

4.3 Saying and Not Saying: The Role of Salience in Speaker Commentary

A striking pattern in the interviews is the consistency with which speakers focus on two key features when discussing dialectal differences: *coda /s/ weakening* and *liquid variation*. Despite the wide range of linguistic variables that differentiate Spanish varieties, no speaker explicitly mentions features such as *subject pronoun usage*, *subject placement*, or *filled pauses*—all of which are widely studied in variationist sociolinguistics but appear to hold little metalinguistic salience for these speakers.

This selective attention to */s/* and *liquids* is significant because it highlights the asymmetries in linguistic awareness. Not only do speakers consistently identify these variables as meaningful, but they also provide phonetic detail about their realization across dialects. Many do so without being directly prompted—a strong indication that these features carry a high degree of social meaning. Moreover, the specific variants they highlight—such as Puerto Rican */r/-velarization* to [x], Dominican */r/-vocalization* to [i], or widespread */s/-weakening*—reveal a structured understanding of how phonetic traits pattern across national groups.

At the same time, not all speakers describe these features with equal specificity. Some, like Dafne, Danilo, and David, offer detailed linguistic observations, while others recognize variation but struggle to articulate concrete differences. This variation in metalinguistic awareness reflects broader differences in exposure, engagement, and the social meanings attached to particular forms.

The following section examines these patterns in depth, first exploring the perspectives of speakers who provide explicit descriptions of variation, and then considering those who

acknowledge dialectal differences but struggle to pinpoint specific features. Finally, it turns to the features that remain unmentioned, shedding light on what these absences reveal about linguistic salience.

4.3.1 Explicit Descriptions of Variation

Some of the most compelling moments in the interviews occur when speakers articulate specific features as markers of regional speech. These speakers not only recognize salient traits but also associate them with specific social or geographic distinctions. Consider Dafne (Table 4.2), a US-born Dominican (21F) with strong bilingual proficiency who has spent her entire life in the U.S. Despite her U.S. upbringing, she exhibits an acute awareness of how Caribbean Spanish differs from more overtly “prestigious” varieties like Colombian or Castilian Spanish. She describes Dominican and Puerto Rican speech as characterized by the deletion of “letters,” specifically mentioning lateralization and velarization of rhotics.

Table 4.2: Dafne (21F, US-born Dominican, 100% life in the U.S.)

SPEAKERS: DAFNE (D) & INTERVIEWER (I)	
SPANISH	ENGLISH
I ¿Cree que el español que hablan uh, los diferentes grupos colombianos, dominicanos, guatemaltecos, todos ellos, es diferente?	Do you think the Spanish spoken by different groups—Colombians, Dominicans, Guatemalans, all of them—is different?
Sí, yo diría que los dominicanos especialmente son más, yo no sé, más locos con el-con español. Nosotros no-no decimos algunas letras y como “Que voy pa’l campo” que nosotros juntamos mucho las palabras, dejamos a un “ele” por acá un una “erre” de una palabra... Umm... que es diferente a los puertorriqueños que dicen oh voy al caxo o algo así... So entonces, yo creo que los dominicanos y los puertorriqueños definitivamente son los más locos en hablar español. Pero si vas como a Colombia es más estricto y como se habla el español en España algo así.	Yes, I’d say that Dominicans especially are more, I don’t know, wilder with Spanish. We don’t say some letters, like, “Que voy pa’l campo,” (I’m going to the farm/field/rural area) where we merge words a lot, dropping an “l” here or an “r” at the end of a word... Umm... which is different from Puerto Ricans who say, “Oh voy al caxo,” (I’m going to the car) or something like that... So I think Dominicans and Puerto Ricans are definitely the wildest in speaking Spanish. But if you go to Colombia, it’s stricter, like how Spanish is spoken in Spain or something.

Similarly, Danilo, a 46-year-old Dominican who immigrated to the U.S. at age 20, provides a highly detailed account of variation, noting that rhotic vocalization occurs in the northern regions of the Dominican Republic. He also identifies the uvular [x] variant as a feature of Puerto Rican Spanish, which he associates with speakers from rural areas (see Table 4.3).

Table 4.3: Danilo (46M, established Dominican, 57% life in the U.S.)

SPEAKERS: DANILO (D) & INTERVIEWER (I)	
SPANISH	ENGLISH
I ¿Cómo es diferente el español de República Dominicana en comparación con la de Puerto Rico?	How is Dominican Republic Spanish different from Puerto Rican Spanish?
D El español de Puerto Rico? El acento del puertorriqueño tiene, tienen un cantadito cómo se le va y viene... la erre ellos, el acento en la erre pronuncia- muy pronunciado la erre como carro y ellos dicen cajo cajo especialmente la gente del campo dice cajo. La arrastran la erre “caxo” o y la erre también realiza la ele “puelto” en vez de puerto “puelto” le pone la ele.	Puerto Rican Spanish? The Puerto Rican accent has a kind of sing-song quality to it... the “r,” their accent on the “r” is pronounced—very pronounced, like “carro” (car), and they say “cajo, cajo,” especially people from rural areas. They drag the “r” like “caxo,” or they also turn the “r” into an “l,” like “puelto” (port) instead of “puerto.”
I ¿No hacen en RD?	They don’t do that in DR?
D En Santo Domingo hay sitios si que en el campo que también hacen eso también le pone la i pone la i donde no va la i. Como... “Comer” ellos dicen “comei”. Eso es en el noroeste de la isla que hablan así “comei” con la i no sé por qué ellos dicen “coméi” “trabajái”. Ah... estudiai y esas son la gente del campo la gente de la montaña lo que está trabajando que hablan con la i no sé explicarlo so mucho tiempo tienen ese su acento yeah yeah pronunciado ahí.	In Santo Domingo, there are places, yes, in the countryside, where they also do that—they add an “i” where it doesn’t belong. Like... “comer” (to eat) becomes “comei.” That’s in the northwest of the island where they speak like that—“comei” with the “i.” I don’t know why they say “coméi,” “trabajái” (work). Ah... “estudiai,” and those are the people from the countryside, the people from the mountains, the workers who speak with the “i.” I don’t know how to explain it; it’s been like that for a long time—that’s their accent, yeah, yeah, pronounced there.

Other speakers confirm these observations but add further nuance. Pilar, a 45-year-old Puerto Rican who immigrated at 24, describes similar rhotic variations in Puerto Rican Spanish while also recognizing vocalization and the rhotization of /l/ in Dominican Spanish. Importantly, she provides geographical specifics, distinguishing where certain variants

are more common within the Dominican Republic (see Table 4.4).

Table 4.4: Pilar (45F, established Puerto Rican, 22% life in the U.S.)

SPEAKERS: PILAR (P) & INTERVIEWER (I)		
	SPANISH	ENGLISH
I	¿Cómo es diferente, um, el español de Puerto Rico con Dominica República?	How is Puerto Rican Spanish different from Dominican Republic Spanish?
P	Sí, obviamente el acento. Yo creo que en todos países hay palabras que se hablan mal. En Puerto Rico se arrastra la erre.	Yes, obviously the accent. I think in every country there are words that are mispronounced. In Puerto Rico, the “r” is dragged.
I	Las “r”.	The “r.”
P	Por ejemplo, en vez de decir “arroz”, dicen “axo”.	For example, instead of saying “arroz,”(rice) they say “axo.” Um, “voy a buscar el carro” (I’m going to get the car), becomes “voy a buscar el caxo.”
P	Ehm, “voy a buscar el carro”. “Voy a buscar el caxo”.	Um, “voy a buscar el carro” (I’m going to get the car), becomes “voy a buscar el caxo.”
I	¿En Puerto Rico?	In Puerto Rico?
P	En Puerto Rico se habla así- especialmente la gente que vive más en la isla o que le llaman más fuera del área metropolitana, más en los campos. También hay sus regionalismos. Por ejemplo, en República Dominicana, la gente de la capital menciona la “ele” muy fuerte... Sin embargo la gente del sur menciona la “erre” muy fuerte. Por ejemplo, eh, la palabra correcta es “sol”. “Hace un sol muy caliente”. Si tú te vas al sur, te pueden decir, “Pero qué sorr más caliente”.	In Puerto Rico, people speak like that—especially those who live further inland or outside the metropolitan area, more in rural areas. There are also regionalisms. For example, in the Dominican Republic, people from the capital pronounce the “l” very strongly... However, people from the south pronounce the “r” very strongly. For example, the correct word is “sol” (sun). “Hace un sol muy caliente” (It’s a very hot sun). If you go to the south, they might say, “Pero qué sorr más caliente.”
I	¿En dónde? ¿En cuál país?	Where? In which country?
P	República Dominicana. Todo depende del lugar.	The Dominican Republic. It depends on the region. And if you go to Cibao, they’ll say, “Oh, pero este soi tan caliente.” They end with an “i.”
I	¿Crees que la gente de habla hispana en Boston debe evitar palabras o dichos que los otros grupos no conocen?	Do you think Spanish speakers in Boston should avoid words or expressions that other groups don’t know?
P	No. No lo creo. Creo que debe explicar qué significa cuando lo utilice porque eso es bueno para tener más información de- por ejemplo, eh- como el ejemplo que te mencioné. Lo de México. “Oye, carnal, mira qué linda está esa chava”.	No. I don’t think so. I think they should explain what it means when they use it because it’s good to have more information. For example, like the example I mentioned to you. The one from Mexico. “Oye, carnal, mira qué linda está esa chava.”
I	¿Qué?	What?
P	Exacto. Y, claro, yo le entiendo ahora porque estoy familiarizada un poquito con mexicanos. Pero si, por ejemplo, yo no entiendo, yo pregunto, “¿Qué significa esto?” Y entonces ellos me explican y para otra vez que ellos lo utilicen, ya yo entiendo lo que significa.	Exactly. And, of course, I understand it now because I am a little familiar with Mexicans. But if, for example, I don’t understand, I ask, “What does this mean?” And then they explain it to me, and the next time they use it, I already understand what it means.
I	Sí, claro.	Yes, of course.
P	Siento que es bueno que tú expreses lo que tú eres, lo que tú sientes, pero que también expliques porque es así, para que otros también puedan entender.	I feel that it’s good for you to express who you are, what you feel, but also explain why it is like that, so others can understand as well.

(CONTINUED) SPEAKERS: PILAR (P) & INTERVIEWER (I)

	SPANISH	ENGLISH
I	Sí, claro. ¿Es difícil muchas veces para aprender las diferencias entre diferentes grupos o muchas veces no hay un problema?	Yes, of course. Is it difficult sometimes to learn the differences between different groups, or is there usually no problem?
P	Bueno, si son- si no es cosas profundas, yo creo que no hay ningún problema. A veces son conversaciones ligeras, que no tienen sentido profundo para que pienso que, bueno, ¿por qué no?	Well, if they are not deep topics, I think there is no problem. Sometimes they are light conversations that don't have deep meaning, so I think, well, why not?
I	Okay. Um, ¿crees que la gente de habla hispana en Boston deben aprender a hablar como los españoles de España?	Okay. Um, do you think Spanish speakers in Boston should learn to speak like Spaniards from Spain?
P	¿Por qué?	Why?
I	No- porque a veces, um, enseñan el tipo de español de España en escuelas en los Estados Unidos. Y por eso alguna gente cree que es tipo de de español- ¿Qué se debería hablar? Mira, yo siento que no importa donde tú vengas. Si tu español es correcto, puedes hablarlo... Ves O sea, tiene que ver mucho la manera cómo tú puedes... cómo tú hablas el español. Si lo hablas de manera correcta. Siempre van a ver excepciones en que dijiste una palabra mal, eh, como nosotros le llamamos, te comiste una ese o pusiste una ese	No—because sometimes, um, they teach the Spanish from Spain in schools in the United States. And because of that, some people think that's the type of Spanish- That should be spoken? Look, I feel that it doesn't matter where you come from. If your Spanish is correct, you can speak it... You see, it really depends on how you speak Spanish. If you speak it correctly. There will always be exceptions where you say a word incorrectly, like we say, you skip an "s" or put an "s" where it doesn't belong. That can happen. But I feel that each person has the right to speak their accent, their Spanish. Look, like I just said now: "e[h]pañol" (Spanish). But the correct word is "e[s]pañol." If I say it correctly. But there are other circumstances not worth mentioning—uh, but I feel no. Each person has the right to speak their Spanish, as long as it is correct.
P	donde no va. Eso sí puede pasar. Pero siento que cada persona tiene derecho a hablar su acento, su español. Mira como yo te dije ahora: "e[h]pañol". Pero la palabra correcta es "e[s]pañol". Si yo lo digo correctamente. Pero hay otras circunstancias que no valen la pena mencionar- eh, pero siento que no. Que cada persona tiene derecho a hablar su español, siempre y cuando esté correcto.	

David, a 47-year-old Dominican who immigrated at 36, further differentiates between how *liquid variation* and *coda /s/ deletion* manifest in speech. He suggests that */s/ deletion* is tied to speech rate, while rhotic variation follows distinct regional patterns. Notably, his observations, like Pilar's and Danilo's, extend beyond his own dialect, as he also describes Puerto Rican variants in detail (see Table 4.5).

Table 4.5: David (47M, established Dominican, 23% life in the U.S.)

SPEAKERS: DAVID (D) & INTERVIEWER (I)		
	SPANISH	ENGLISH
I	¿Cómo se sabe que una persona es dominicana al hablar?	How can you tell that a person is Dominican by the way they speak?
D	Por la, eh, casi siempre por la ese	By the, uh, almost always by the “s.”
I	¿ese?	“s”?
D	sí, por ejemplo.. Y por las abreviaciones eh, tú sabes...	Yes, for example... and by abbreviations, you know...
I	abreviaciones como... por ejemplo...	Abbreviations like... for example...
D	de una abreviación... eh, decir “como tú tá”... y es una abreviación de... normalmente sería “¿cómo tú estás?”. Entonces eh, decir “como tú tá” es una forma bien...	An abbreviation... uh, saying “como tú tá”... which is an abbreviation of... normally it would be “¿cómo tú estás?”(how are you?). So, saying “como tú tá” is a very...
I	¿dominicana?	Dominican?
D	Sí... y ¿cuál sería otra? no sé, pero hay muchas abreviaciones	Yes... and what would be another one? I don’t know, but there are many abbreviations.
I	Y ¿cómo se distingue a un puertorriqueño por su forma de hablar?	And how do you distinguish a Puerto Rican by the way they speak?
D	la erre, ellos dicen por ejemplo um... entre otras cosas tienen mucho mucho... eh, la forma de hablar es bien... también rítmica, eh, muy fácil de distinguir en en el español. Pero sobre todo la erre ellos por ejemplo van decir un carro... dirían un ca[x]o un ca[x]o. Es un si una forma un poco... ehm Tú sabes de.. de mala pronunciación del español, pero ellos dicen... en vez de decir un ca[r]o ellos dicen... un ca[x]o	The “r,” they say, for example, um... among other things, they have a very... uh, their way of speaking is also very rhythmic, uh, very easy to distinguish in Spanish. But especially the “r,” for example, if they are going to say “un carro”(a car), they would say “un ca[x]o,” “un ca[x]o.” It’s kind of a... uh, you know, a mispronunciation of Spanish, but instead of saying “un ca[r]o,” they say “un ca[x]o.”
I	¿ca[l]o?	“ca[l]o”?
D	ca[x]o, con la jota eh se escucha como si fuera una jota... ca[h]o	“ca[x]o,” with the “j,” uh, it sounds as if it were a “j”... “ca[h]o.”
I	okay	Okay
D	eh, y tú escuchas una persona de Puerto Rico que... tú le dice[Ø]... ¿qué ca[r]o ustedes tienen y dicen “bueno, yo tengo un ca[h]o” eh... “[ho.ho]” (rojo)... [ho.ho] porque la erre ellos ellos... La la letra erre para ellos es bien... No sé... no sé, porque la pronuncian tan mal. La pronuncian como si fuera una jota	Uh, and you hear a person from Puerto Rico and you ask them, “what ca[r]o do you have?” and they say, “bueno, yo tengo un ca[h]o” uh... “[ho.ho]”(red)... [ho.ho] because the “r,” they... they... the letter “r” for them is very... I don’t know... I don’t know why they pronounce it so badly. They pronounce it as if it were a “j.”

While speakers generally agree that *coda* /s/ *variation* is a defining feature of Dominican and Puerto Rican Spanish, they describe it in less precise terms compared to *liquid variation*. This is likely due to the fact that /s/ has fewer possible realizations than liquids. Whereas *liquid variation* involves multiple socially and geographically conditioned variants (e.g., lateralization, vocalization, uvularization, etc.), *coda* /s/ in

Caribbean Spanish typically follows a more predictable pattern of aspiration, deletion, or other forms of reduction. As a result, speakers tend to discuss /s/ *variation* in broader terms. For example, Delia, a 23-year-old US-born Dominican, comments that Dominicans shorten sounds, particularly *coda* /s/. Yet she also states that Puerto Ricans and Dominicans sound similar to her, implying that while she recognizes the phenomenon, she does not see it as a key distinguishing factor between the groups (see Table 4.6).

Table 4.6: Delia (23F, US-born Dominican, 100% life in the U.S.)

SPEAKERS: DELIA (D) & INTERVIEWER (I)		
	SPANISH	ENGLISH
I	Es decir. ¿Cómo se sabe que una persona es dominicana al hablar?	I mean, how can you tell someone is Dominican by the way they speak?
D	Okay yo creo que mm no sé de tu experiencia pero muchos de los dominicanos no dicen algunas de las letras... Le falta la “s” al fin de la palabra... En vez de decir... ¿Cómo te digo? algunas cosas están como shortened... no voy a decir que una forma bien o mal como hablarlo pero...	Okay, I think, um, I don’t know about your experience, but many Dominicans don’t pronounce some of the letters... They drop the “s” at the end of a word... Instead of saying... how do I explain it? Some things are just shortened... I’m not going to say it’s a right or wrong way to speak but...
I	¿Y entonces cómo tú crees que se distingue un puertorriqueño?	And how do you think a Puerto Rican is distinguished?
D	Los puertorriqueños que yo conozco... me suena igual como los dominicanos.	The Puerto Ricans I know... they sound the same to me as Dominicans.
I	¿O crees que el dominicano debe de tratar de adaptarse a otro tipo de español?	Or do you think Dominicans should try to adapt to another type of Spanish?
D	Eh no. Porque yo creo uno se entiende. Esas cosas que uno dice o la forma que habla es parte de la cultura como te digo.	Um, no. Because I think we understand each other. These things we say or the way we speak are part of the culture, as I was saying.

Likewise, Ernesto, a 21-year-old US-born Dominican-Salvadoran, acknowledges that Dominicans and Puerto Ricans “don’t pronounce everything” and admits difficulty understanding his Dominican family members. However, he struggles to articulate exactly what makes these varieties distinct from each other(see Table 4.7).

Table 4.7: Ernesto (21M, US-born Dominican-Salvadoran, 100% life in the U.S.)

SPEAKERS: ERNESTO (E) & INTERVIEWER (I)		
	SPANISH	ENGLISH
I	¿Cómo se distingue a un dominicano por su forma de hablar?	How can you tell a Dominican by the way they speak?
E	Bueno mi padre es dominicano. Sí, entonces tengo familia dominicanos y no se entienden. para nada, no no lo entiendo. Cuando me hablan, sólo me quedo sentado diciendo “sí, sí, sí sí”	Well, my father is Dominican. Yes, so I have Dominican family, and they can’t be understood. Not at all, I don’t understand them. When they talk to me, I just sit there saying “yes, yes, yes, yes.”
I	¿y los puertorriqueños?	And Puerto Ricans?
E	Son como los dominicanos casi pero no sé no me gusta su forma de hablar de los puertorriqueños. No.	They are almost like Dominicans, but I don’t know, I don’t like the way Puerto Ricans speak. No.
I	¿Me puedes dar un ejemplo de la forma de hablar de puertorriqueños o mexicanos?	Can you give me an example of how Puerto Ricans or Mexicans speak?
E	los mexicanos, no puedo decir una forma específica pero hablan diferente. y los puertorriqueños son como los dominicanos que no pronuncian todo pero también su forma de hablar... parece feo	Mexicans, I can’t describe a specific way, but they speak differently. And Puerto Ricans are like Dominicans in that they don’t pronounce everything, but also the way they speak... it sounds ugly.

Similarly, Pamela, an 18-year-old US-born Puerto Rican, states that different Spanish-speaking groups sound distinct to her, but she cannot imitate their speech unless she hears them first. She does note, however, that Salvadorans speak much slower than Puerto Ricans and Dominicans, reinforcing that speech rate may be another perceptually salient site of dialectal differentiation (see Table 4.8).

Table 4.8: Pamela (18F, US-born Puerto Rican, 100% life in the U.S.)

SPEAKERS: PAMELA (P) & INTERVIEWER (I)		
	SPANISH	ENGLISH
I	¿Cómo se distingue a un colombiano por su forma de hablar?	How do you distinguish a Colombian by the way they speak?
P	ah, Es muy diferente. Cuando yo escucho a los ah... colombianos a veces es muy difícil para entender un poco porque los uh los formas de hablar hay muchas palabras que son diferentes y también tienen como acentos muy diferentes también um. Es muy diferente. Cuando yo lo escucho, yo lo puedo como entender. Cuando yo escucho a alguien que está hablando español, yo puedo decidir si ellos son salvadoreño, colombiano, cubano...	Ah, it’s very different. When I hear Colombians, sometimes it’s very difficult to understand a little because, uh, the way they speak, there are many words that are different, and they also have very different accents too. It’s very different. When I hear it, I can kind of understand. When I hear someone speaking Spanish, I can tell if they are Salvadoran, Colombian, Cuban...

(CONTINUED) SPEAKERS: PAMELA (P) & INTERVIEWER (I)

	SPANISH	ENGLISH
I	¿Sí Puedes? ¿O no puedes?	You can? Or you can't?
P	Sí, yo puedo.	Yes, I can.
I	¿Puede darme un ejemplo de palabras o la forma de que hablan?	Can you give me an example of words or the way they speak?
P	Cuando pienso, no. Y yo no puedo como, yo no puedo- <i>I can't imitate, you know.</i> Yo no puedo, como es tan diferente yo no puedo- <i>I like...</i> cuando lo, cuando, sí cuando lo oigo yo puedo pero ahora no.	When I think about it, no. And I can't, like, I can't- <i>I can't imitate, you know.</i> I can't, it's so different that I can't- <i>I like...</i> when I, when, yes when I hear it I can, but right now I can't.
I	¿Cómo se distingue a un dominicano por su forma de hablar?	How do you distinguish a Dominican by the way they speak?
P	Yo creo que los dominicanos son lo más similar que los puertorriqueños, um ya también cuando yo, cuando yo estaba- cuando yo crecí, yo me crecí con muchos dominicanos, todas mis amigas son dominicanas. Y es muy similar, um yo no creo que son muy diferente, como, yo no me he dado... yo no sé. No hay mucho, como... como con los colombianos son muy diferentes cuando... pero cuando yo oigo a los dominicanos yo no, no pienso en mucho.	I think Dominicans are the most similar to Puerto Ricans, um, and also when I, when I was- when I grew up, I grew up with a lot of Dominicans, all my friends are Dominican. And it's very similar, um, I don't think they are very different, like, I haven't noticed... I don't know. There's not much, like... like with Colombians, they are very different when... but when I hear Dominicans, I don't think much about it.
I	¿Y puertorriqueños? ¿Cuál es su forma de hablar?	And Puerto Ricans? What is their way of speaking?
P	Um, eso es muy difícil porque cuando yo hablo con mis amigos ellos siempre me dicen "Tú hablas diferente de nosotros." Es muy diferente, y pero yo no puedo oír, como yo, yo hablo español, soy puertorriqueña, yo no sé la diferencia. Pero yo sé con nosotros usamos spanglish mucho mucho mucho. Y yo aprendí eso cuando yo empecé mis clases de español, y cuando yo estaba lo estaba oyendo, mira yo no uso esa palabra, y mi maestro me dijo "sí, eso es spanglish." Y yo le digo: "Oh, pues yo no sabía." Sí, porque es una mezcla increíble en los Estados Unidos. La misma pregunta con salvadoreños, ¿cómo se distingue?	Um, that's very difficult because when I talk to my friends, they always tell me, "You speak differently from us." It's very different, but I can't hear it, like, I speak Spanish, I'm Puerto Rican, I don't know the difference. But I do know that we use Spanglish a lot, a lot, a lot, a lot. And I learned that when I started my Spanish classes, and when I was hearing it, I was like, I don't use that word, and my teacher told me, "Yes, that's Spanglish." And I said, "Oh, well, I didn't know."
I	Unidos. La misma pregunta con salvadoreños, ¿cómo se distingue?	Yes, because it's an incredible mix in the United States. Same question with Salvadorans, how do you distinguish them?
P	Um, solamente tengo una amiga que es salvadoreña, y cuando ella habla yo oigo muchas diferencias, es como... um... Más lento como elloa hablo, porque en otros países siempre usa el español es muy muy rápido, pero en los salvadoreños encuentro que siempre hablan con lentísimo y le puedes oír cada palabra, y um... sí, eso es...	Um, I only have one friend who is Salvadoran, and when she speaks, I hear many differences, it's like... um... they speak more slowly, because in other countries, Spanish is always spoken very, very fast, but with Salvadorans, I find that they always speak very slowly, and you can hear every word, and um... yeah, that's it...

As the preceding excerpts clearly show, speakers consistently orient to a narrow set of highly salient features—particularly liquid variation and coda /s/—when discussing

dialectal differences. An equally important finding, however, is what speakers do not mention. No speaker identifies any of the other variables analyzed in this study, such as subject pronoun usage, despite the fact that overt pronoun expression constitutes a well-documented site of regional differentiation in Spanish. Caribbean varieties, in particular, have been shown to exhibit high rates of overt pronoun use, with this variation shaped primarily by discourse and syntactic factors (Otheguy & Zentella 2012; Erker 2012; Vidal-Covas 2013; Torres Cacoullos & Travis 2018).

Instead, when speakers lack specific knowledge about another dialect, they often contrast it with their own. Diego, a 73-year-old Dominican, provides a clear example of this. When asked about how Colombian Spanish differs from other varieties, he notes their strong /s/ pronunciation—not necessarily because this is their most defining feature, but because it contrasts with his own variety, in which /s/ is frequently neutralized (see Table 4.9).

Table 4.9: Diego (73M, established Dominican, 71% life in the U.S.)

SPEAKERS: DIEGO (D) & INTERVIEWER (I)		
	SPANISH	ENGLISH
I	¿Cómo se sabe que una persona es colombiana al hablar?	How can you tell that a person is Colombian by the way they speak?
D	Ssss Porque, ellos <i>pronounce...</i> <i>th-their S's are more pronounced</i> . Las eses de ellos son más pronunciadas. Entonces hay cie[Ø]tas palabras que ellos usan que son exclusivamente de ellos.	Ssss Because, they <i>pronounce...</i> <i>th-their S's are more pronounced</i> . Their “s’s” are more pronounced. So, there are certain words they use that are exclusive to them.
I	¿Cómo se sabe que una persona es dominicana al hablar?	How can you tell that a person is Dominican by the way they speak?
D	Bueno, <i>his tempo</i> , <i>his tempo is er-</i> , <i>is, is, is different</i> . Ah, you know, hablan más alto <i>and...</i> you know, just like the Cuban. The Cuban is different.	Well, <i>his tempo</i> , <i>his tempo is er-</i> , <i>is, is, is different</i> . Ah, you know, they speak louder <i>and...</i> you know, just like the Cuban. The Cuban is different.
I	¿Y cómo se distingue a un puertorriqueño por su forma de hablar?	And how do you distinguish a Puerto Rican by the way they speak?
D	Bueno, habla rápido. Habla. yeah. Eh, no, cubanos habla-hablan más rápido. <i>The fastest-talking is the Cuban, and the longest. Because, yeah longest, a Cuban could talk for days. For daaays. For days.</i>	Well, they speak fast. They speak. Yeah. Uh, no, Cubans speak even faster. <i>The fastest-talking is the Cuban, and the longest. Because, yeah longest, a Cuban could talk for days. For daaays. For days.</i>

(CONTINUED) SPEAKERS: DIEGO (D) & INTERVIEWER (I)

	SPANISH	ENGLISH
I	¿Y cómo se distingue a un salvadoreño por su forma de hablar?	And how do you distinguish a Salvadoran by the way they speak?
	<i>Some things you got to tell by the features. ah... ah... ah... you know, just to... s-say that they're like, from Central America, cause I don't ask nobody their</i>	<i>Some things you got to tell by the features. Ah... ah... ah... you know, just to... s-say that they're like, from Central America, cause I don't ask nobody</i>
D	<i>nationality actually. Cause I don't like people even asking mine, right? But I can tell their nationalities... You know, by looking at them. Even the clothes that they wear will tell me, would tell me, right?</i>	<i>their nationality actually. Cause I don't like people even asking mine, right? But I can tell their nationalities... You know, by looking at them. Even the clothes that they wear will tell me, would tell me, right?</i>
I	¿con todas personas?	With everyone?
D	No, but some people, they jump at you, you know... de a donde ser... you know.	No, but some people, they jump at you, you know... where are you from?... you know.
I	Yeah. It's like "¿S-So eres?" Muchas personas. Y-Yeah, yeah, they- Like it's a good thing-, you know but, then when you tell them where you're	Yeah. It's like "¿So are you...?" A lot of people. Y-Yeah, yeah, they- Like it's a good thing-, you know but, then when you tell them where you're
D	<i>from, they don't want to hear it. you know. Because that's not what they were expecting. You know what I'm saying? That's not what they were expecting.</i>	<i>from, they don't want to hear it. You know. Because that's not what they were expecting. You know what I'm saying? That's not what they were expecting.</i>

Paola, a 23-year-old Puerto Rican, takes a similar approach when asked about dialectal distinctions. While she confidently claims she can distinguish between Puerto Rican, Dominican, and South American Spanish, she struggles to articulate precisely how. When pressed for specifics, she describes Puerto Rican rhotics as having variants like [x] and [l], but she does not provide comparable details for Colombian Spanish (see Table 4.10). Her response highlights an important pattern: while speakers may have an intuitive sense of linguistic differences, their ability to verbalize these distinctions varies, often depending on their familiarity with a given variety.

Table 4.10: Paola (23F, US-born Puerto Rican, 87% life in the U.S.)

SPEAKERS: PAOLA (P) & INTERVIEWER (I)		
	SPANISH	ENGLISH
I	¿Piensa que cómo se distingue a una persona de su grupo por su forma de hablar como colombiano, puertorriqueño, y eso?	Do you think that a person can be distinguished by their way of speaking, like Colombian, Puerto Rican, and so on?

(CONTINUED) SPEAKERS: PAOLA (P) & INTERVIEWER (I)

	SPANISH	ENGLISH
P	Sí, de verdad que yo sé cuando hay un puertorriqueño, y yo sé cuando hay un dominicano, y yo sé cuando es de América del Sur, y yo- yo no sé por qué. Yo creo que son algunas palabras y cómo lo dicen, um, más que otra cosa.	Yes, honestly, I can tell when someone is Puerto Rican, and I can tell when someone is Dominican, and I know when they're from South America, and I—I don't know why. I think it's certain words and how they say them, um, more than anything else.
I	Entonces, ¿puede dar un ejemplo de cómo español un colombiano?	So, can you give an example of Colombian Spanish?
P	Sí- oh. Oh, hace tiempo que no hablo con colombiano, pero yo sé en en Puerto Rico, eh, los colombianos que yo conocía me se reían porque yo digo Puerto “[x]ico”... y no entendían porque a veces la “ere” es la arrastra como “[x]”... O a veces es una “ele” como “puelto”.	Yes—oh. Oh, it's been a while since I've spoken with a Colombian, but I know in Puerto Rico, um, the Colombians I knew would laugh at me because I say Puerto “[x]ico”... and they didn't understand because sometimes the “r” is dragged like “[x]”... Or sometimes it's an “l”, like “puelto” (port).

4.4 Evaluating Dialects: Attitudes and Social Meaning

In addition to describing variation, several speakers reveal attitudes toward language variation, sometimes framing some dialects as more “correct” or prestigious than others. Pacífica, a 33-year-old US-born Puerto Rican, provides a particularly striking example. She describes Puerto Rican Spanish as “improper” and attributes this to what she sees as linguistic laziness—Puerto Ricans “drag their words” and fail to pronounce final segments properly. By contrast, she views Colombian Spanish as a prestigious variety, stating that Colombians “put the /s/ where it goes” and pronounce words fully, thereby speaking a “proper” form of Spanish (see Table 4.11).

Table 4.11: Pacífica (33F, US-born Puerto Rican, 88% life in the U.S.)

	SPEAKERS: PACÍFICA (P) & INTERVIEWER (I)	
	SPANISH	ENGLISH
I	¿Cree que el español que hablan, los puertorriqueños es diferente?	Do you think the Spanish spoken by Puerto Ricans is different?
P	Sí... um <i>it's improper, I don't know how to say that in Spanish</i>	Yes... um <i>it's improper, I don't know how to say that in Spanish</i>
I	Está bien	It's okay

(CONTINUED) SPEAKERS: PACÍFICA (P) & INTERVIEWER (I)

	SPANISH	ENGLISH
	Mucho slang, yeah, mucho slang, mucho... yeah.	A lot of slang, yeah, a lot of slang, a lot... yeah.
P	No pronuncian las palabras como son, <i>you know</i> , son como, como vagos a hablar, arrastran las palabras	They don't pronounce the words as they are, <i>you know</i> , they are kind of lazy when they speak, they drag words.
I	¿Me puedes decir cómo se distinguen diferentes grupos al hablar español? ¿So, cómo se distingue un colombiano?	Can you tell me how different groups can be distinguished when they speak Spanish? So, how do you distinguish a Colombian?
P	Por las palabras que usan, el, además del acento, las palabras que usan, como los pronuncian, <i>so, I guess</i> , es el acento, pero los son diferentes que nosotros.	By the words they use, besides the accent, the words they use, how they pronounce them, <i>so, I guess</i> , it's the accent, but they are different from us.
I	Okay um y, ¿cuáles cosas diferentes usan los colombianos que tú puedes decir que si tú lo oyes hablando, puedes decir ese sí es colombiano?	Okay, um, and what are some different things Colombians use that make you say, if you hear them speaking, "that person is definitely Colombian"?
P	<i>Umm... can I say it in English?</i>	<i>Umm... can I say it in English?</i>
I	Yeah, if you want	Yeah, if you want
P	<i>It's they use like a more proper Spanish, like instead of tú, usan vos, or, um, they pronounce the word the way it's supposed to be pronounced, versus we, Puerto Ricans, tend to like cut off half of it, or like add ah... oa or ao or eo at the end. They don't tend to do that. They put the "s" where it goes, and, and they, they're more fluent, like their fluency is better. Yeah, like nosotros, you know, like, estoy cansa[Ø]o, ellos dicen estoy cansado, you know, so, no le co[l]tan.</i>	<i>They use like a more proper Spanish, like instead of tú, they use vos, or, um, they pronounce the word the way it's supposed to be pronounced, whereas we, Puerto Ricans, tend to like cut off half of it, or like add ah... oa or ao or eo at the end. They don't tend to do that. They put the "s" where it goes, and, and they, they're more fluent, like their fluency is better. Yeah, like we, you know, like, estoy cansa[Ø]o, they say estoy cansado(I'm tired), you know, so, they don't cut it off.</i>
I	Y los dominicanos, ¿cómo se distinguen, cómo hablan?	And Dominicans, how do they distinguish themselves, how do they speak?
P	<i>um they're just the accent</i>	<i>um they're just the accent</i>
I	Okay, y ¿me puedes dar un ejemplo del acento?	Okay, and can you give me an example of the accent?
P	<i>I'm trying to think, um I should know this... I can't think of anything... oh my god, okay, so like em, it kinda goes back to the Colombians like they, a lot of you guys pronounce the whole word, like we, cuando nosotros decimos "arroz"(rice), muchos puertorriqueños dicen "[axo]", you know, so you guys say "[aros]" or actually you guys call it "moro", or whatever it is, but</i>	<i>I'm trying to think, um I should know this... I can't think of anything... oh my god, okay, so like em, it kinda goes back to the Colombians like they, a lot of you guys pronounce the whole word, like we, when we say "arroz", many Puerto Ricans say "[axo]", you know, so you guys say "[aros]" or actually you guys call it "moro", or whatever it is, but</i>
I	¿Piensa que la gente que habla español en Boston debe de hablar el español de su país?	Do you think people who speak Spanish in Boston should speak the Spanish of their home country?
P	Yo creo que cada cual usa el español a su costumbre...	I think that everyone speaks Spanish according to their custom...
I	¿Y piensa que deben evitar las frases o palabras o los dichos que los otros grupos no conocen?	And do you think they should avoid phrases or words or expressions that other groups don't know?

(CONTINUED) SPEAKERS: PACÍFICA (P) & INTERVIEWER (I)

SPANISH	ENGLISH
No creo que deberían de omitirlo por completo porque eso es parte de quienes son, pero sí deberían de sonar, <i>you know, like, to sound</i>, un poquito más, como más profesional, pero todo depende donde, en el lugar donde estén, <i>you know?</i>	I don't think they should omit it completely because it's part of who they are, but they should sound, <i>you know, like, to sound</i>, a little bit more, like more professional, but it all depends on where they are, <i>you know?</i>

Pilar, despite asserting that people should speak their own variety, still acknowledges a “right” way to speak, suggesting an internalized hierarchy of linguistic prestige (see Table 4.4 above). As discussed above in Table 4.2, Dafne also makes an evaluative distinction, albeit a more neutral one. She characterizes Dominican and Puerto Rican speech as “crazy” compared to Colombian Spanish, reinforcing the perception that Caribbean varieties are marked by rapid or reduced articulation.

At the same time, not all speakers frame dialect differences in hierarchical terms. Pascal (see Table 4.12), a 37-year-old Puerto Rican who recently arrived in the U.S., explicitly resists the idea that any variety is superior to another. Before even addressing dialectal distinctions, Pascal makes a point of stating that “Spanish is Spanish,” signaling a belief in the fundamental unity of the language despite regional variation. When he does discuss differences, he notes that Puerto Ricans “eat their r’s” and lateralize them but does so without attaching overt value judgments.

Table 4.12: Pascal (37M, recently arrived Puerto Rican, 4% life in the U.S.)

SPEAKERS: PASCAL (P) & INTERVIEWER (I)		
	SPANISH	ENGLISH
I	¿Crees que el español que hablan los colombianos es diferente? De los puertorriqueños.	Do you think the Spanish spoken by Colombians is different? From Puerto Ricans?
P	Bueno, español es español. Pero es diferente el acento. Es lo que entiendo yo.	Well, Spanish is Spanish. But the accent is different. That's how I understand it.
I	¿Y cómo te suena esa diferencia?	And how does that difference sound to you?

(CONTINUED) SPEAKERS: PASCAL (P) & INTERVIEWER (I)

SPANISH	ENGLISH
El puertorriqueño, nosotros, nos comemos como le dicen mucho la “r”. Nosotros, si vamos a decir eh... como te diría yo eh, una palabra que fuera con “r” como cual, “comer”, por ejemplo. Nosotros decimos “comer, comer”, nosotros decimos “comer (comel)... Nosotros la palabra que lleva, eh, una “r” a lo último nosotros decimos como “l”.	Puerto Ricans, we kind of, as they say, drop the “r” a lot. For example, if we are going to say, um... how should I put this, a word that ends with “r,” like “comer,” (to eat) for example. We say “comer, comer,” we say “comer (comel)... For words that end in “r,” we say it more like an “l.”

Donaldo (Table 4.14) and Priscila (Table 4.13) both emphasize the importance of maintaining one’s dialect, but their perspectives differ in intensity. Priscila is particularly vocal about linguistic loyalty, insisting that Puerto Ricans “should speak like Puerto Ricans” and explicitly advocating for these differences to be preserved and even taught in schools. In contrast, Donaldo expresses support for linguistic maintenance but with a more flexible stance, viewing language change as a natural process rather than something speakers should actively control.

Table 4.13: Priscila (24F, US-born Puerto Rican, 100% life in the U.S.)

SPEAKERS: PRISCILA (P) & INTERVIEWER (I)	
SPANISH	ENGLISH
I ¿Cree usted que la gente de habla hispana en Boston debe seguir hablando el español de sus compatriotas? P Si tú eres de Puerto Rico, debes hablar como los puertorriqueños. I ¿Crees que se deba enseñar el español de los diferentes grupos en la escuela? P Como te dije, que los colombianos y los cubanos no hablan como los puertorriqueños so yo creo que si alguien va a enseñarle a los estudiantes español, deben decir: “esto es como los cubanos hablan”, “esto es como los puertorriqueños hablan”, y “esto es como los mexicanos hablan”	Do you think Spanish-speaking people in Boston should continue speaking Spanish like their countrymen? If you’re from Puerto Rico, you should speak like Puerto Ricans. Do you think the Spanish of different groups should be taught in schools? Like I said, Colombians and Cubans don’t speak like Puerto Ricans, so I think, yes, if someone is going to teach students Spanish, they should say: “this is how Cubans speak,” “this is how Puerto Ricans speak,” and “this is how Mexicans speak.”

Table 4.14: Donaldo (21M, established Dominican, 81% life in the U.S.)

SPEAKERS: DONALDO (D) & INTERVIEWER (I)		
	SPANISH	ENGLISH
I	¿Cree que el español que hablan los dominicanos es diferente?	Do you think the Spanish spoken by Dominicans is different?
D	Um, si. las palabras son diferentes y muchas de las letras se pierden en la palabra cuando hablan. ¿Cree usted que la gente que habla español en Boston debe seguir hablando el español de sus compatriotas?	Um, yes. The words are different, and many of the letters are lost in the word when they speak. Do you think people who speak Spanish in Boston should continue speaking the Spanish of their countrymen?
	Deben. Como cambia cambia. Si cambia, cambia... ellos no deben tener ningún... Yo no creo que ellos deben tener ningún intento. Cómo sale, sale y ya.	They should. If it changes, it changes. If it changes, it changes... they shouldn't have any... I don't think they should make any effort. However it comes out as it comes out.

Finally, Ponce (Table 4.15), despite being U.S.-born and having spent 100% of his life in the mainland U.S., firmly aligns himself with Puerto Rican identity through his speech. When discussing rhotic variation, he explicitly claims ownership of a particular feature, stating that the trill is pronounced as [x] in word-initial position because “that’s something we Puerto Ricans do”. His assertion highlights how features can serve as an in-group marker, reinforcing linguistic identity even in contexts of diaspora.

Table 4.15: Ponce (25M, US-born Puerto Rican, 100% life in the U.S.)

SPEAKERS: PONCE (P) & INTERVIEWER (I)		
	SPANISH	ENGLISH
I	Umm, ¿y los puertorriqueños, cómo se sabe que una persona es de Puerto Rico?	Umm, and Puerto Ricans, how can you tell that a person is from Puerto Rico?
	Ah, como pronunciamos la erre a veces muchos hispanos dicen erre pero nosotros hacemos “exe”	Ah, the way we pronounce the “r”—sometimes many Hispanics say “r,” but we make it “exe”
P	[risa]. No en vez de decir puertorriqueño nosotros decimos puertoxiqueño [risa]. En vez de xojo rojo nosotros decimos xoxo [risa].	[laughs]. Instead of saying “puertorriqueño,” we say “puertoxiqueño” [laughs]. Instead of “rojo,”(red) we say “xoxo” [laughs].
I	“[xoxo]”.	“[xoxo]”
P	Como decimos la erre, es bien como que nos comemos la erre [risa].	The way we say the “r,” it’s really like we swallow the “r” [laughs].

4.5 Dialect Perceptions and Identity: Key Takeaways

The preceding excerpts highlight several important patterns. First, *liquid variation* and *coda /s/ variation* emerge as salient features in commentary on dialectal variation in the Hispanophone world, but speakers describe the two phenomena with differing levels of specificity. Some—like Dafne, Danilo, and David—offer detailed linguistic observations, while others struggle to articulate concrete differences even when they recognize them. This variation suggests that dialect awareness is not uniform; rather, it is shaped by factors such as personal experience, regional exposure, and linguistic metaconsciousness.

Second, language attitudes play a significant role in shaping these perceptions. While some participants, such as Pacifica, describe Puerto Rican Spanish in explicitly negative terms—associating it with “improper” or “lazy” speech—others, like Pilar, demonstrate a more nuanced perspective, recognizing variation while still referencing a “correct” way to speak. Priscila, for instance, expresses a strong commitment to maintaining Puerto Rican Spanish, both in principle and in practice. When asked whether Spanish speakers in Boston should continue speaking Spanish like their countrymen, she responds unequivocally: “*Si tú eres Puertorriqueño, debes hablar como los puertorriqueños. Yo voy a hablar español como los puertorriqueños.*” (*If you’re from Puerto Rico, you should speak like Puerto Ricans. I’m gonna speak like the Puerto Ricans*)

Donaldo, on the other hand, acknowledges the importance of maintaining one’s way of speaking but adopts a more flexible stance. Rather than emphasizing linguistic loyalty, he suggests that change should not be resisted: “*Si cambia, cambia... ellos no deben tener ningún... Yo no creo que ellos deben tener ningún intento. Cómo sale, sale y ya.*” (*If it changes, it*

changes... I don't think they should make any effort. However it comes out, it comes out.) While he does not dismiss dialectal distinctiveness, his perspective reflects a broader acceptance of language variation and change, in contrast to Priscila's firm stance on preserving Puerto Rican speech as a marker of identity.

Finally, perhaps just as revealing as what speakers do describe is what they do not. No participant highlights any of the other variables in the study. Instead, when struggling to identify distinguishing features of another dialect, many default to comparing it to their own. For example, Diego characterizes Colombian Spanish by its strong /s/ pronunciation, not necessarily because this is its most defining feature, but because it contrasts with his own Dominican variety.

Taken together, these excerpts underscore two key insights. First, speakers are acutely aware of certain linguistic features, particularly when it involves highly salient features like *liquids* and *coda* /s/. Their ability to provide examples suggests an implicit, and at times explicit, understanding of the social and regional dimensions of these features. Second, the accuracy and detail of these descriptions are often shaped by speakers' proximity to and interaction with specific varieties. While Pilar articulates clear distinctions between Puerto Rican and Dominican speech, Delia acknowledges similarities, reflecting her limited exposure to Dominican Spanish outside the US context.

5 Results: An Examination of Liquids

This chapter delves into the variable production of Spanish liquids—the rhotics (/r/) & /r/) and the lateral /l/—within a *comparative variationist framework* (Poplack & Levey 2010). This approach, widely applied in studies of Spanish in the U.S. (Otheguy & Zentella 2012; Erker 2017; Torres Cacoullos & Travis 2010; Erker & Otheguy 2016; Torres Cacoullos & Travis 2018), enables a systematic examination of variation in contact settings by comparing patterns of variability across measures of contact, such as immigrant generation or duration of life spent in a given locale. Building on the conceptual foundation laid in previous chapters, this analysis addresses key questions: What linguistic and social factors drive variation in the production of liquids? How do liquids align with the continuum of salience, and what does this reveal about their patterns of stability or change?

By “change,” I refer to differential usage across speakers as a function of linguistic and social predictors, including Percentage of Life in the U.S. (PLUS), language use with interlocutors, and internal phonological constraints. If liquid variation aligns with patterns that have been observed among low-salience features, speakers with greater exposure to English—whether measured by PLUS, interactional patterns, or other contact-related measures—may exhibit variants more closely resembling English norms, reflecting the output of purported processes of bilingual optimization (Muysken 2013). Conversely, if liquid variation patterns with coda /s/, its use may function as a symbol of dialectal identity, resisting English-aligned variants despite contact. The potential for liquids to serve as

an *identity shibboleth*—a marker of national pride or in-group membership—offers an important test of the relationship between social salience and linguistic behavior in contact settings.

To investigate these questions, I examine the distribution of liquid forms, assess their alignment with phonological expectations, and evaluate predictors of variation through statistical modeling. While the focus here is on identifying the linguistic and social factors that condition variation in liquids, the following chapter explores their covariation with other sociolinguistic variables, offering a broader perspective on coherence in linguistic systems.

5.1 General Overview of the Variation in the Dependent Variable

A total of 7,354 tokens of *liquids* were analyzed, with /ɾ/ (tap) having the highest number of tokens ($n = 3,871$), followed by /l/ (lateral, $n = 2,626$) and /r/ (trill, $n = 857$). As outlined in Chapter 3 (see Table 3.1), the phonemic distinctions and expected allophonic distributions for these liquids are based on standard accounts of Spanish phonology, particularly following Hualde (2014). Figure 5.1 below displays the proportional distribution of surface forms for each of the three liquids under study. In this context, *match* denotes surface forms that are identical to the phoneme, while *mismatch* refers to those that differ from the phonological form.

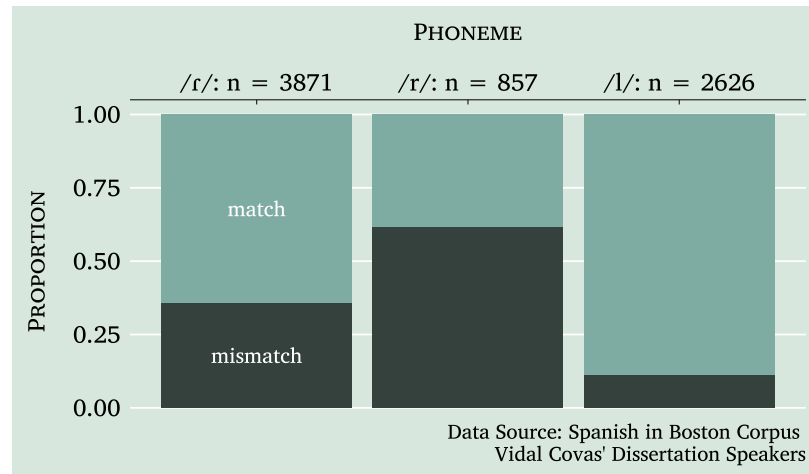


Figure 5.1: Rates of phonological and phonetic (mis)match for all liquids

Figure 5.2 expands upon Figure 5.1 by providing a detailed breakdown of distribution of the surface forms associated with each of the three phonemes: /r/ (tap), /r/ (trill), and /l/ (lateral). The visualization illustrates the distribution of each surface form and categorizes them based on whether they align with the phonological form (*match*) or deviate from it (*mismatch*). The light shading indicates instances of *match*, highlighting surface forms that conform to the phonological expectations, while the dark shading represents *mismatch*, emphasizing phonetic realizations that are different from the phoneme.

Each panel corresponds to one of the three phonemes and reflects their relative token frequencies. Within each panel, surface forms are ordered by their frequency, revealing notable patterns in phonetic variation. For instance, the lateral /l/ demonstrates a high proportion of conformity to the canonical [l], with relatively few tokens exhibiting *mismatch*. In contrast, the tap /r/ shows greater phonetic diversity, with [r] as the dominant form but substantial representation of mismatched realizations such as [ɾ], [ʔ], and [∅]. Similarly, the trill /r/ displays a high frequency of the mismatch form [r̄], which is even

more common than the trill [r] itself, with the latter being the next most frequent surface form.

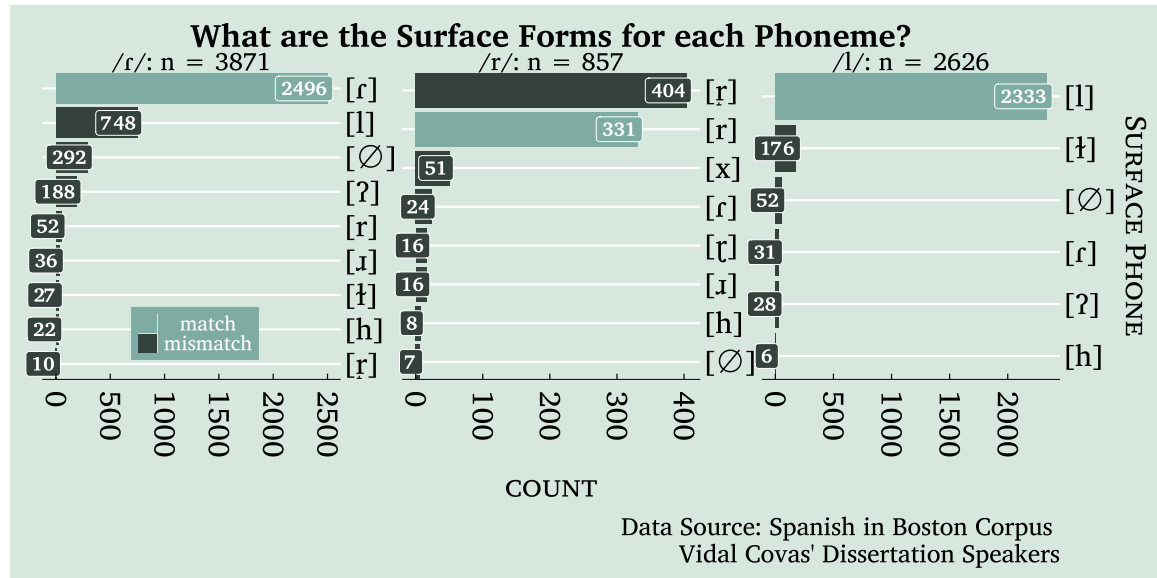


Figure 5-2: Rates of liquids

Having established the overall proportional distribution of surface forms, we now turn our attention to a more detailed examination of the three phonemes under study: /r/, /r/, and /l/. In this analysis, I use the match/mismatch variable to categorize surface forms based on their correspondence with the purported phonological form. As previously defined, a *match* occurs when the surface form aligns with the phonological form, while a *mismatch* reflects a discrepancy between the two. This categorical distinction allows for the use of logistic regression, facilitating a structured statistical evaluation of the relationship between surface and phonological forms across the data set. Having outlined the model selection process in Section 3.3, we now turn to the detailed results of statistical modeling of variation for each phoneme.

5.2 Analysis of Tap, Trill, and Lateral

For each liquid, I present the best-fitting mixed-effects logistic regression model, identified using the `dredge()` and `drop1()` functions, and discuss key findings related to mismatch. The goal of these regressions is to estimate the probability of a mismatch as a function of several social and linguistic predictors, while accounting for both *fixed effects* (the linguistic and social predictors) and *random effects* (speaker). Including *speaker* as a random effect allows us to account for individual variation between speakers that might influence the outcome but are not directly related to the fixed effects. In other words, by incorporating random effects, we can partition the variability into that attributable to the speakers versus the predictors. For this analysis, I opted not to include *word* as an additional random effect because the primary focus is on inter-speaker variability rather than variability associated with specific lexical items. This decision balances model complexity with interpretability as well as model convergence likelihood, but future work could explore the potential influence of word-level variability. Relatedly, random slopes, which would allow the effect of specific predictors to vary by speaker, were not included in this analysis to avoid overparameterizing the models given the size of the dataset.

The r^2 values provide an indication of the model's explanatory power. The *marginal r^2* reflects the proportion of variance explained by the fixed effects, while the *conditional r^2* represents the variance explained by both the fixed and random effects (e.g., by-speaker variability). The *Intraclass Correlation Coefficient (ICC)* measures the proportion of total variance in the outcome attributable to differences between speakers, as opposed to variability within speakers (i.e., between observations from the same speaker).

In each model summary, the β (*estimate*) represents the change in the odds of a mismatch associated with a one-unit increase in a continuous predictor variable or, in the case of a categorical variable, the change relative to the reference level, holding all other variables constant. It is important to note that β is not a direct measure of effect size; rather, its magnitude reflects the relationship between the predictor and the odds of the outcome. A positive estimate indicates that the predictor increases the likelihood of a mismatch, while a negative estimate indicates a decrease. The *standard error (SE)* reflects the variability of the estimate due to sampling, while the *confidence interval (CI)* provides a range within which the true value of the parameter is likely to fall, given a specified level of confidence (e.g., 95%). The *z-statistic* reflects the standardized distance of the estimate from zero, and is used to calculate the *p-value*, which indicates whether the predictor's effect is statistically significant.

5.2.1 Statistical Analysis of Variation in the Tap — /ɾ/

The best-fitting model for tap variation, as shown in Table 5.1, includes the predictors *country of origin*, *following sound category*, *log lexical frequency*, *milliseconds per syllable*, *PLUS (percent life in the US)*, *position in syllable*, *sex*, and *stress*. It is important to note that inclusion in the best-fitting model does not necessarily imply that a variable is a significant predictor; for instance, *PLUS* exhibits a weak effect and is not statistically significant in this model (see Table 5.1). However, as demonstrated in Figure 5.4, even non-significant predictors can still contribute to the overall fit of the model. This figure visually illustrates the impact of each predictor on model AIC, highlighting that removing *PLUS* results in a small but measurable change in model fit, which justifies its retention in the analysis.

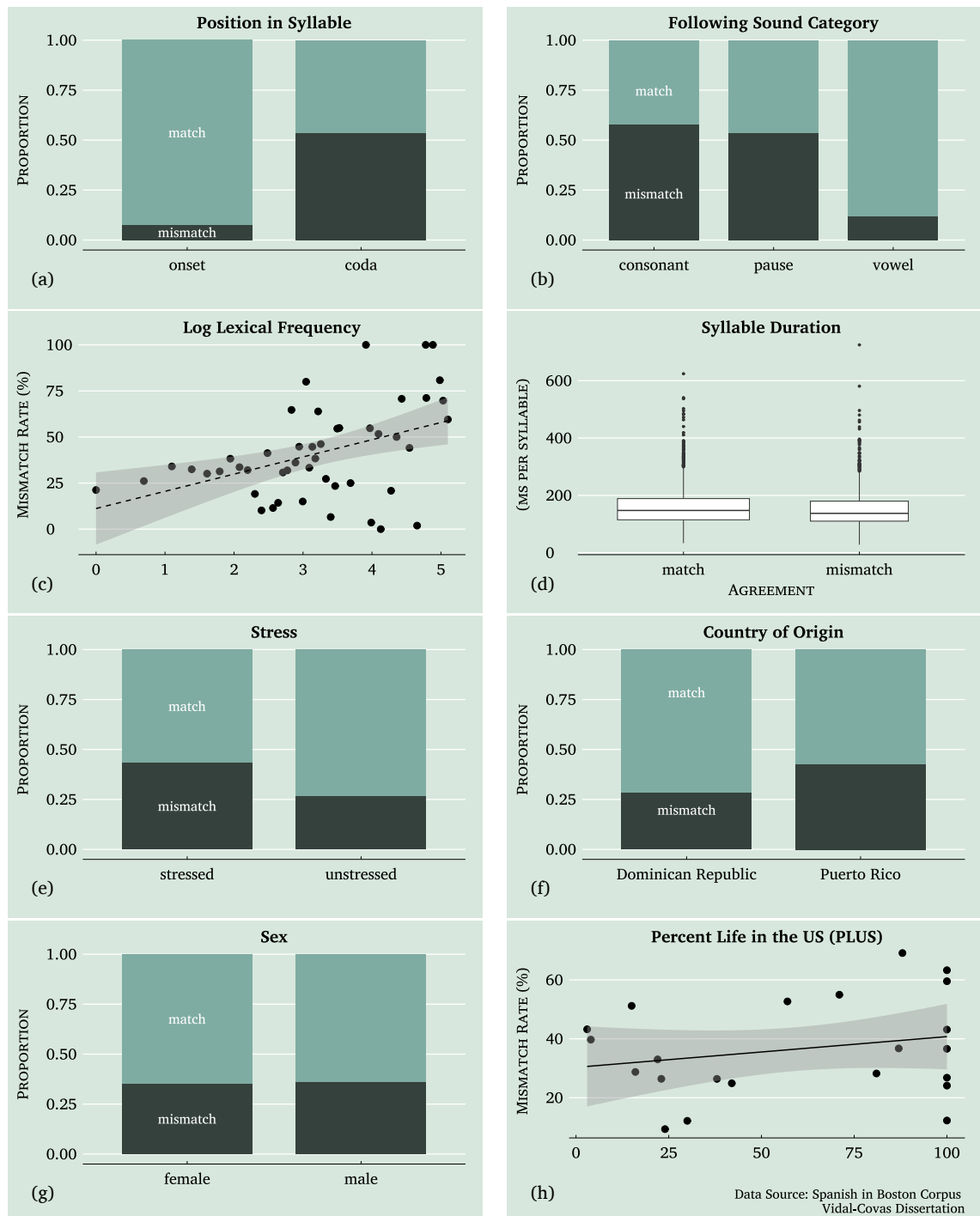


Figure 5-3: Visual representation of the effects of the predictors included in the best-fitting mixed effects model for tap mismatch

Table 5.1: Mixed Effects Model: Tap Mis-(match) Prediction

PREDICTORS	DEP. VARIABLE: MISMATCH				
	β	SE	CI 95 %	Z-STAT	P-VALUE
Intercept	-3.25	.51	[-4.24--2.26]	-6.42	.000***
Country of Origin (<i>ref:DR</i>)					
Puerto Rico	1.03	.39	[.27--1.79]	2.65	.008***
Following Sound Category (<i>ref:cons</i>)					
Pause	-.35	.17	[-.68--.03]	-2.13	.033**
Vowel	-1.51	.14	[-1.79--1.24]	-10.67	.000***
Log Lexical Frequency	.16	.03	[.11--.22]	5.70	.000***
Milliseconds per Syllable	-.22	.05	[-.32--.12]	-4.29	.000***
PLUS	.01	.01	[-.00--.02]	1.56	.119
Position in Syllable (<i>ref:onset</i>)					
Coda	2.05	.18	[1.70--2.41]	11.21	.000***
Sex (<i>ref:female</i>)					
Male	.85	.39	[.09--1.61]	2.20	.028**
Stress (<i>ref:stressed</i>)					
Unstressed	-.43	.10	[-.62--.23]	-4.29	.000***
RANDOM EFFECTS					
σ^2	3.29				
τ_{00} SPEAKERS	.77				
ICC	.19				
N _{SPEAKERS}	22				
SD _{SPEAKERS}	.88				
Observations	3871				
Marginal R ² / Conditional R ²	.46 / .56				
AIC	3364.36				

* $p < .1$, ** $p < .05$, *** $p < .01$

The fixed effects in the model reveal several significant results. *Position in syllable: coda* ($\beta = 2.05$, $p < .001$) significantly increases the likelihood of a mismatch, indicating that codas are associated with a higher probability of mismatch. Likewise, *following sound category: vowel* ($\beta = -1.51$, $p < .001$) and *log lexical frequency* ($\beta = .16$, $p < .001$) demonstrate significant effects, with vowels following taps reducing mismatch likelihood, while increased lexical frequency raises it. These relationships are visually represented in Figure 5-3, which illustrates the distribution of mismatch patterns across the different predictors. Specifically, Subfigure 5-3a highlights the increased likelihood of mismatches in coda positions, and Subfigure 5-3b shows the distribution of phonetic (mis)match across following sound categories.

To assess the impact of each predictor on model fit, I examined the increase in *Akaike information criterion (AIC)* if each predictor were removed from the model, summarized in Figure 5.4. The largest increases in AIC were observed for *position in syllable* (129.8), *following sound category* (118.3), and *log lexical frequency* (30.9), indicating that these predictors contribute substantially to the model’s explanatory power. Conversely, *PLUS* showed a negligible AIC increase (0.3) and a non-significant p-value, suggesting its limited effect on model fit.

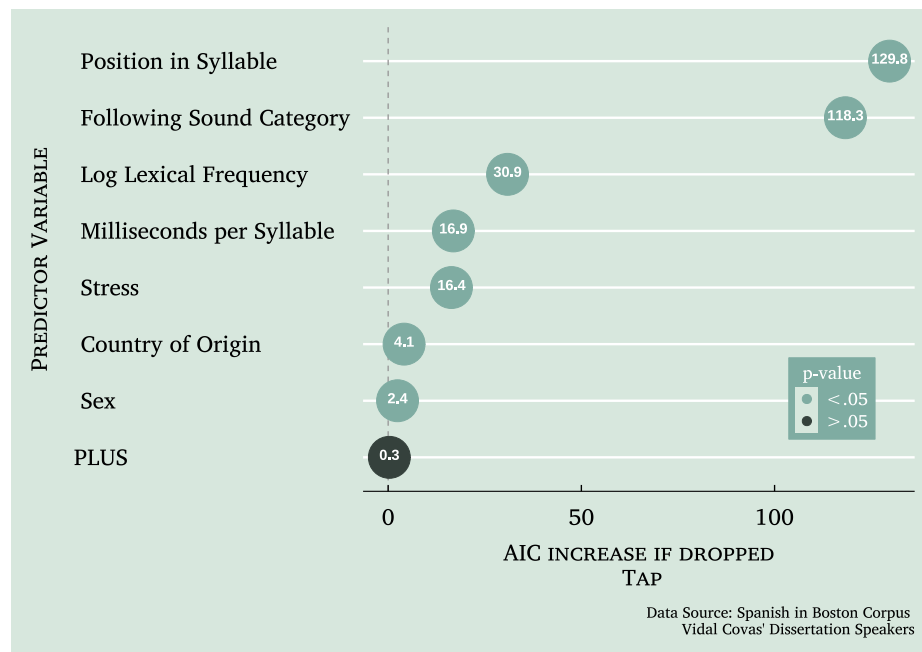


Figure 5.4: Impact of predictor variable removal on AIC in mixed effects model — /r/

Conversely, *milliseconds per syllable* ($\beta = -.22$, $p < .001$) and *stress: unstressed* ($\beta = -.43$, $p < .001$) have negative regression coefficients, indicating that longer syllable durations and unstressed syllables are associated with a decreased likelihood of a mismatch. Although *PLUS* ($\beta = .01$, $p = .119$) is not statistically significant, it was retained in the model because removing it would increase the AIC, indicating a reduction in overall model

fit. Subfigure 5.3d shows that longer syllable durations correspond to fewer mismatches, and Subfigure 5.3e shows that unstressed syllables are more prone to mismatches compared to stressed syllables.

The random effects for *speaker* show a variance of $\tau_{00} = 0.77$, with the ICC of .19, indicating that 19% of the variance in mismatch is due to differences between speakers. The model's *marginal* r^2 is .46, representing the variance explained by fixed effects alone, while the *conditional* r^2 is .56, showing that the full model (including random effects) explains 56% of the variance.

5.2.1.1 Summary of Findings for Tap Variation

The results from the statistical modeling of tap variation reveal key patterns in how mismatches occur and the factors that influence them.

First, *position in syllable* plays a crucial role: taps occurring in *coda* position are much more likely to surface as a mismatch than those in onset position. This is not surprising and aligns with previous findings on segmental weakening in Spanish, where coda consonants are more susceptible to phonetic reduction or modification (Poplack 1979; Alba 1990; Quilis 1993; Ramos-Pellicia 2004; Beaton 2016; Gilbert & Rohena-Madrado 2017, *inter alia*). Similarly, *following sound category* has a strong effect. Taps followed by *vowels* are much more likely to match their phonological target, while those followed by *consonants* or *pauses* have a higher probability of mismatch. This suggests that coarticulatory effects and articulatory constraints influence realization patterns, with vowel-adjacent taps benefiting from greater stability. Lexical factors also shape variation: *higher-frequency words* show an increased likelihood of mismatch.

Temporal factors are also relevant. The results show that *longer syllable durations* reduce mismatch likelihood. This suggests that taps produced in *faster speech* are more prone to non-canonical realizations, reinforcing the idea that mismatch rates may be linked to speech rate and articulatory effort. This pattern aligns with claims by Katz (Katz 2021), who argues that many lenition effects may be explained in terms of duration. The effects of *stress* further support this trend. *Unstressed syllables* are significantly more likely to show matches than stressed ones. This pattern, however, reflects a tendency *within the Spanish tap data presented here*, rather than a general phonological trend. In fact, it runs counter to common cross-linguistic patterns, where reduction is typically favored in prosodically weaker positions. The fact that, in this dataset, Spanish taps show increased reduction in prosodically stronger positions (i.e., stressed syllables) highlights a point of typological interest and suggests that tap realization in this variety of Spanish may diverge from more general lenition patterns — a finding that warrants further investigation.

Sociolinguistic factors also emerge as significant predictors. Speakers from *Puerto Rico* exhibit higher mismatch rates compared to those from the *Dominican Republic*, suggesting dialectal differences in the realization of the tap. Additionally, *male speakers* show higher rates of mismatch than female speakers, aligning with prior sociophonetic research on gender-based variation in liquid reduction and articulation (Beaton 2005). Finally, the effect of *PLUS* was weak and not statistically significant. However, it was retained in the model because removing it increased AIC, indicating that—even if not individually significant—it still contributed to model fit in some capacity.

5.2.2 Tap Subset: Laterals Mismatch

To better understand the factors contributing to tap variation, I conducted a subset analysis focusing on lateral mismatches—the most frequent type of mismatch observed in the dataset. The full dataset comprised 3,871 tap tokens, of which 3,244 were included in this subset. Within the subset, 2,496 tokens were realized as [ɾ], making up 76.94% of the data. The most common mismatch, [l], was produced in 748 cases, occurring in 23.06% of the subset. Given its relatively high occurrence and its classic association with so-called *liquid neutralization*, this subset analysis aimed to identify the linguistic and social factors that most strongly influenced this particular realization. Table 5.2 presents the optimal model for predicting lateral mismatches for tap.

Table 5.2: Mixed Effects Model: Tap Match vs. Lateral Mismatch Prediction

PREDICTORS	DEP. VARIABLE: MISMATCH				
	β	SE	CI 95 %	Z-STAT	P-VALUE
Intercept	-6.05	.88	[-7.77--4.33]	-6.88	.000***
Country of Origin (<i>ref:DR</i>)					
Puerto Rico	1.83	.51	[.84--2.82]	3.62	.000***
Following Sound Category (<i>ref:cons</i>)					
Pause	-.31	.20	[-.71--.09]	-1.50	.132
Vowel	-2.11	.20	[-2.50--1.72]	-10.60	.000***
Log Lexical Frequency	.13	.04	[.06--.20]	3.43	.001***
Milliseconds per Syllable	-.19	.07	[-.32--.05]	-2.74	.006***
Position in Syllable (<i>ref:onset</i>)					
Coda	2.55	.34	[1.88--3.22]	7.45	.000***
Vowel	1.74	.76	[.24--3.23]	2.28	.023**
Sex (<i>ref:female</i>)					
Male	1.21	.51	[.22--2.20]	2.39	.017**
Stress (<i>ref:stressed</i>)					
Unstressed	-.82	.13	[-1.08--.55]	-6.06	.000***
RANDOM EFFECTS					
σ^2	3.29				
τ_{00} SPEAKERS	1.29				
ICC	.28				
N _{SPEAKERS}	22				
SD _{SPEAKERS}	1.13				
Observations	3244				
Marginal R ² / Conditional R ²	.66 / .75				
AIC	1955.22				

* $p < .1$; ** $p < .05$; *** $p < .01$

The predictors included in this model are *country of origin*, *following sound category*, *log lexical frequency of word*, *milliseconds per syllable*, *position in syllable*, *previous sound category*, *sex*, and *stress*. Model selection was conducted using the `dredge()` function, which identified the best-fitting model based on AIC. Additionally, the `drop1()` function confirmed that all retained predictors contributed meaningfully to model fit, as removing any one of them resulted in an increase in AIC.

The random effects structure indicates substantial between-speaker variability, with the *ICC* at .28, suggesting that nearly a third of the variation stems from differences across speakers. The *marginal r^2* of 0.66 highlights the significant explanatory power of fixed effects alone, while the *conditional r^2* of .75 underscores the role of both fixed and random effects.

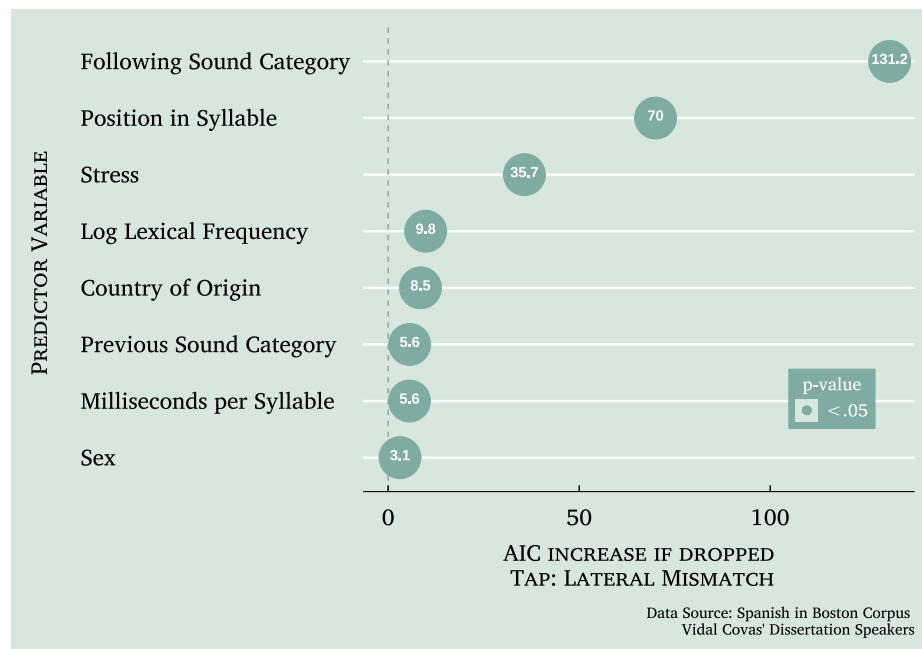


Figure 5-5: Impact of predictor variable removal on AIC in mixed effects model —
/r/: [r] & [l]

The relative impact of each predictor on the model's AIC is further visualized in Fig-

ure 5.5, which provides a clear representation of how dropping each variable affects the overall model fit. In this model, *following sound category* proved to be the most influential predictor, as shown by its large AIC increase (131.2) if dropped. This suggests that the phonetic context following a tap plays a critical role, particularly in determining the likelihood of surfacing as a lateral form. Position in Syllable, also significant in the broader tap mismatch model, exerts a strong influence here, with coda positions strongly favoring lateral mismatch, though its effect is secondary to Following Sound Category.

The effect of *stress* is also significant, with unstressed syllables less prone to lateral mismatches—a pattern consistent with findings in the broader tap mismatch analysis. Although predictors like *log lexical frequency* and *milliseconds per syllable* are less impactful, they still contribute meaningfully, highlighting the relevance of lexical and temporal dynamics in mismatch occurrences.

In both the overall tap mismatch model and the lateral subset, two social factors—*country of origin* and *sex*—stand out as significant contributors. In the subset analysis, Puerto Rican speakers and males are more prone to lateral mismatches, with these social factors ranking higher in their influence on AIC than some linguistic factors, such as *previous sound category* and *milliseconds per syllable*. This reinforces the importance of social variables in explaining variation particularly for this subset of the data, adding depth to the analysis beyond the purely linguistic factors.

Interestingly, *previous sound category*, included in this subset model, provides additional insights by emphasizing the impact of the preceding phonetic environment. Vowel sounds before the tap significantly predict lateral mismatch, highlighting a nuanced aspect of phonetic context for this specific surface form. Finally, unlike in the overall tap mismatch

model, the *PLUS* predictor was excluded here, as it did not significantly affect the subset model's fit.

5.2.2.1 Summary of the Results for Tap Subset: Laterals Mismatch

In terms of the model examining lateral mismatches for taps, the strongest predictor in the model was *following sound category*, indicating that the segment following the tap significantly affects whether it surfaces as a lateral. Specifically, mismatches were most likely when the following sound was a consonant, while vowels and pauses significantly reduced mismatch probability.

Additionally, *position in syllable* was a key factor, with coda positions favoring lateral mismatches. Similarly, *stress* influenced mismatch likelihood, with unstressed syllables less prone to lateralization. While less influential than in the larger analyses of tap, *log lexical frequency* and *milliseconds per syllable* also contributed to the model analyzing lateral realizations of tap, suggesting that more frequent words and faster speech rates may slightly influence lateralization patterns. Interestingly, *previous sound category* was included as a predictor, revealing that lateral mismatches were more likely when the preceding sound was a vowel.

Social factors also played an important role in lateral mismatches. *Country of origin* was a significant predictor, showing that Puerto Rican speakers exhibited more lateral mismatches than Dominican speakers. Likewise, *sex* had a measurable effect, with male speakers more likely to produce lateral mismatches. These findings highlight the importance of incorporating social factors, which contribute to variability in tap realization beyond purely linguistic conditioning.

5.2.3 Statistical Analysis of Variation in the Trill — /r/

The best-fitting model for the trill, shown in Table 5.3, includes the predictors *country of origin*, *log lexical frequency*, *milliseconds per syllable*, *PLUS (percent life in the US)*, and *position in word*.

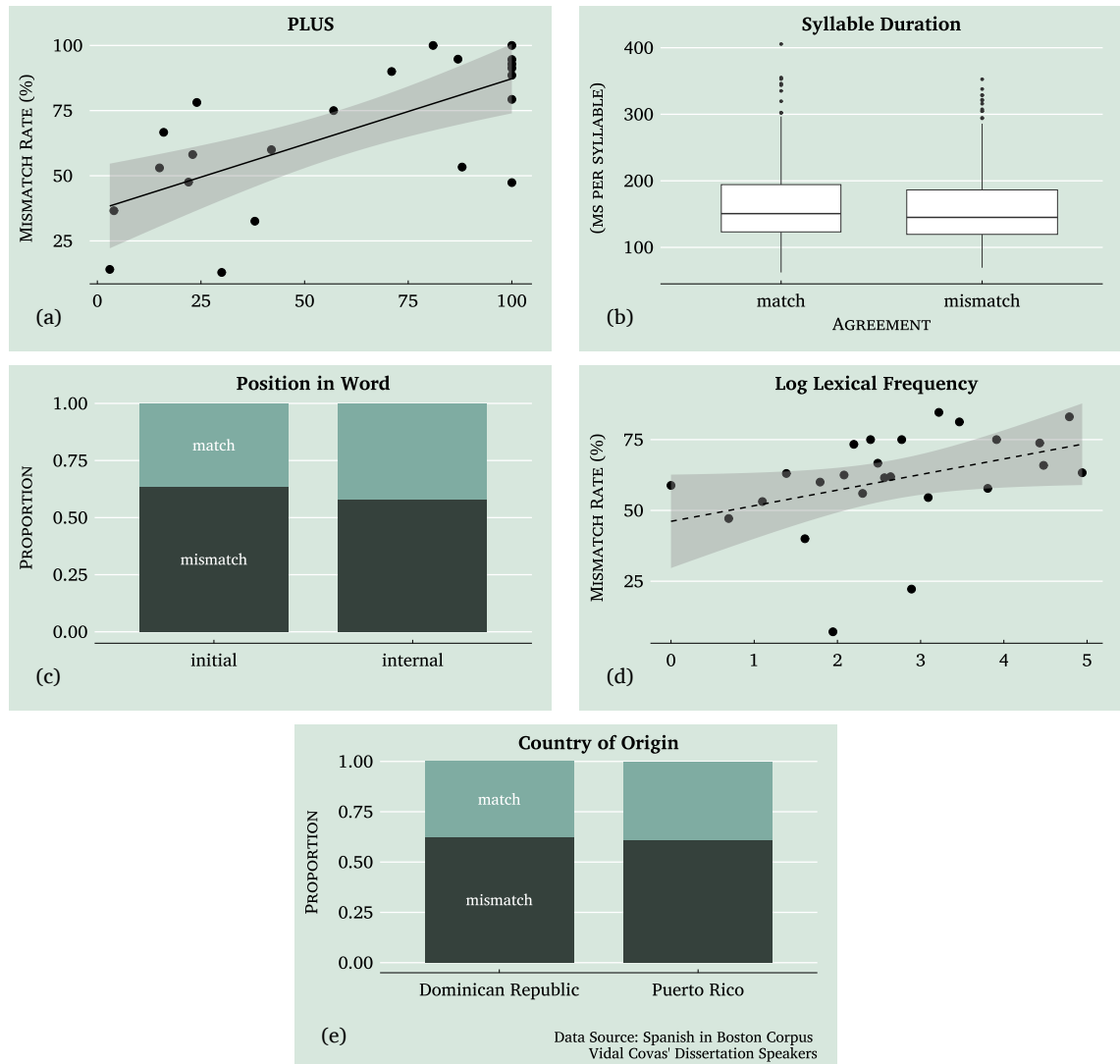


Figure 5-6: Visual representation of the effects of the predictors included in the best-fitting mixed effects model for trill mismatch

The fixed effects in the model reveal several significant effects. Higher *log lexical fre-*

quency ($\beta = .15$, $p = .006$) significantly increases the likelihood of a mismatch, suggesting that as lexical frequency rises, so does the probability of mismatch. Similarly, *PLUS* ($\beta = .03$, $p < .001$) shows a positive relationship with mismatch probability, suggesting that a higher percentage of life spent in the US is associated with more frequent mismatches. This relationship is visually represented in Subfigure 5-6a.

Table 5.3: Mixed Effects Model: Trill Mis-(match) Prediction

PREDICTORS	DEP. VARIABLE: MISMATCH				
	β	SE	CI 95 %	Z-STAT	P-VALUE
Intercept	-.74	.51	[-1.73–.26]	-1.44	.149
Country of Origin (<i>ref:DR</i>)					
Puerto Rico	-.81	.51	[-1.80–.19]	-1.59	.112
Log Lexical Frequency	.15	.05	[.04–.25]	2.76	.006***
Milliseconds per Syllable	-.30	.09	[-.48–.13]	-3.39	.001***
PLUS	.03	.01	[.02–.05]	4.81	.000***
Position in Word (<i>ref:initial</i>)					
Internal	-.57	.18	[-.93–.22]	-3.15	.002***
RANDOM EFFECTS					
σ^2	3.29				
τ_{00} SPEAKERS	1.07				
ICC	.25				
N SPEAKERS	22				
SD SPEAKERS	1.03				
Observations	857				
Marginal R ² / Conditional R ²	.28 / .45				
AIC	907.35				

* $p < .1$; ** $p < .05$; *** $p < .01$

Conversely, *milliseconds per syllable* ($\beta = -.30$, $p = .001$) and *position in word: internal* ($\beta = -.57$, $p = .002$) have negative regression coefficients, indicating that slower speech and internal word positions reduce the likelihood of a mismatch. Subfigures 5-6b and 5-6c depict these effects, illustrating how mismatches are less likely in internal word positions and with longer syllable durations. Additionally, *country of origin: PR* has a negative but non-significant effect ($\beta = -0.81$, $p = .112$), implying fewer mismatches among Puerto Rican speakers. Although this effect is not statistically significant, it is included to maximize model fit, as demonstrated in Subfigure 5-6e, which shows patterns of phonetic/phono-

logical (mis)match by country.

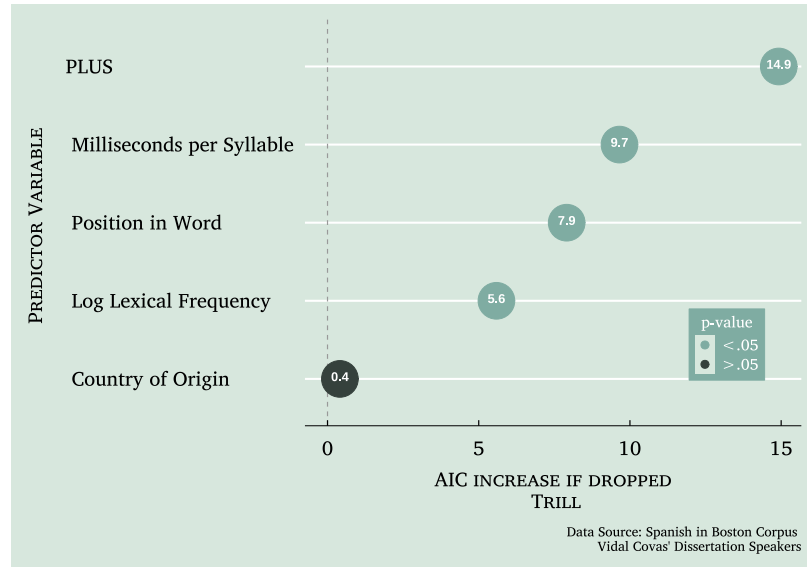


Figure 5-7: Impact of predictor variable removal on AIC in mixed effects model — /r/

The random effects for *speaker* show a variance of $\tau_{00} = 1.07$, and the *ICC* is .25, indicating that 25% of the total variance in mismatch is attributable to differences between speakers. The model's explanatory power is reflected in the *marginal* r^2 of .28, representing the variance explained by the fixed effects, and a *conditional* r^2 of .45, indicating that the full model (including random effects) explains 45% of the variance.

As shown in Figure 5-7, subsequent analysis using the `drop1()` function confirmed that all significant predictors in the final model contributed meaningfully to explaining the variation in mismatch outcomes. Although *country of origin* was not a statistically significant predictor ($p = .112$), its inclusion is justified because removing it from the model results in a marginal increase in AIC (.4 points), suggesting that its presence, while not contributing significantly in isolation, helps maintain the overall model fit.

5.2.3.1 Summary of the Results for Trill

The findings from this model highlight key patterns in trill mismatch variation, emphasizing the role of both linguistic and social factors in shaping this phonetic outcome. The significance of *log lexical frequency* suggests that trills in more frequently used words are more prone to mismatch. The relationship between *PLUS* and mismatch probability indicates that speakers who have spent a greater proportion of their lives in the U.S. are more likely to exhibit trill mismatches.

Phonetic factors such as *milliseconds per syllable* and *position in word* appear to reduce mismatch likelihood. The association between *milliseconds per syllable* and mismatch probability indicates that slower speech, characterized by longer syllable durations, corresponds with more consistent trill productions. Similarly, trills in *internal word positions* are less likely to undergo mismatch.

Although *country of origin* was not statistically significant, its inclusion in the model suggests some degree of differentiation between Puerto Rican and Dominican speakers in their realization of the trill.

5.2.4 Trill Subset: [r̥] Mismatch

To better understand the factors influencing trill variation, I conducted a subset analysis focusing on the most frequent mismatch type, [r̥]. The full dataset, as discussed above, contained 857 trill tokens, of which 735 were included in this subset. Within this subset, the match category—meaning tokens realized as [r]—accounted for 331 tokens or 45.03%, while the most common mismatch, [r̥], occurred in 404 tokens, representing 54.97% of the subset. Given that this mismatch type was the predominant realization, this analysis

aimed to identify the linguistic and social factors that contributed to its occurrence. Key predictors in this model are *country of origin*, *log lexical frequency*, *milliseconds per syllable*, *PLUS* (*percent life in the US*), *position in word*, and *sex*.

Table 5.4: Mixed Effects Model: /r/ Match vs. [r] Mismatch Prediction

PREDICTORS	DEP. VARIABLE: MISMATCH				
	β	SE	CI 95 %	Z-STAT	P-VALUE
Intercept	-.07	.53	[-1.10–.97]	-.12	.901
Country of Origin (<i>ref:DR</i>)					
Puerto Rico	-.90	.46	[-1.81–.01]	-1.95	.052*
Log Lexical Frequency	.14	.05	[.04–.25]	2.63	.009***
Milliseconds per Syllable	-.35	.09	[-.53–-.16]	-3.70	.000***
PLUS	.02	.01	[.01–.04]	3.85	.000***
Position in Word (<i>ref:initial</i>)					
Internal	-.72	.19	[-1.09–-.34]	-3.72	.000***
Sex (<i>ref:female</i>)					
Male	-.87	.46	[-1.77–.02]	-1.91	.056*
RANDOM EFFECTS					
σ^2	3.29				
τ_{00} SPEAKERS	.83				
ICC	.20				
N _{SPEAKERS}	22				
SD _{SPEAKERS}	.91				
Observations	735				
Marginal R ² / Conditional R ²	.24 / .39				
AIC	834.54				

* $p < .1$; ** $p < .05$; *** $p < .01$

Notably, *milliseconds per syllable* and *position in word* emerged as the strongest internal predictors, with significant negative effects ($\beta = -.35$, $p < .001$ and $\beta = -.72$, $p < .001$, respectively). This suggests that longer syllable durations and internal word positions substantially reduce the likelihood of a mismatch to [r]. Moreover, *PLUS* had a significant positive effect ($\beta = .02$, $p < .001$), indicating more time in the US leads to a higher probability of /r/ surfacing as [r].

Conversely, other predictors, such as *log lexical frequency*, *country of origin: PR*, and *sex:male* were moderately significant ($\beta = .14$, $p < .05$, $\beta = -.90$, $p < .1$, and $\beta = -.87$, $p < .1$ respectively). These results indicate that mismatches are more likely when the word

has a higher lexical frequency, in the speech of Dominican speakers compared to Puerto Rican speakers, and in the speech of females compared to males.

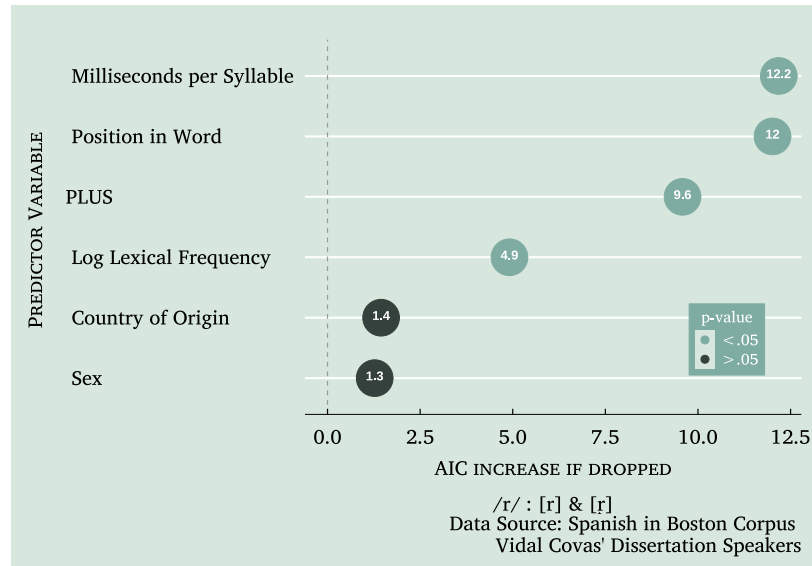


Figure 5-8: Impact of predictor variable removal on AIC in mixed effects model —
/r/ : [r] & [r]

Using the `drop1()` function, I evaluated the impact of each predictor on the model's fit, as visualized in Figure 5-8. The most influential predictors were *milliseconds per syllable*, *position in word* and *PLUS*, each resulting in a notable increase in AIC (12.3, 12.1, and 9.6, respectively) when excluded.

The model explains a moderate portion of the variance, with a *marginal* r^2 of .24 for the fixed effects and a *conditional* r^2 of .39, including random effects. This indicates that the fixed effects alone account for 24% of the variance in mismatch outcomes, and the full model captures 39% when speaker-level variability is considered.

5.2.4.1 Summary of the Results for Trill Subset: [r] Mismatch

The results of the trill subset analysis reveal several key patterns in the production of [r] as a realization of trill. Among the strongest predictors, *milliseconds per syllable*, *position in word*, and *PLUS* are highly significant, with longer syllable durations (i.e., slower speech rates), internal word positions, and a lower PLUS associated with a lower likelihood of a trill being realized as [r]. This suggests that faster speech, word-initial positions, and more time in the US favor the production of this particular mismatch.

Additionally, *log lexical frequency* was associated with a higher likelihood of the mismatch, suggesting that more frequently used words are more likely to surface as [r]. *Country of origin* and *sex* were moderately significant predictors, with Puerto Rican and male speakers producing fewer instances of the variant than Dominican and female speakers.

5.2.5 Statistical Analysis of Variation in the Lateral — /l/

For the lateral liquid subtype, the best-fitting model, displayed in Table 5.5, includes the predictors *following sound category*, *stress*, *syllable type*, *% English only with interlocutors*, and *previous sound category*. As shown in Figure 5.10, *following sound category* emerged as the most influential predictor, with an AIC increase of 27.9 if omitted. Other significant predictors, such as *stress* (9.8) and *syllable type* (9.6), also show considerable impacts on the model's fit. By contrast, *previous sound category* contributed minimally, with an AIC increase of only 1.5, suggesting a limited role in explaining mismatch variability.

The fixed effects analysis highlighted that *following sound category: pause* ($\beta = -0.83$, $p = .007$) and *following sound category: vowel* ($\beta = -1.44$, $p < .001$) both significantly decrease the likelihood of a mismatch compared to consonants, indicating that mismatches

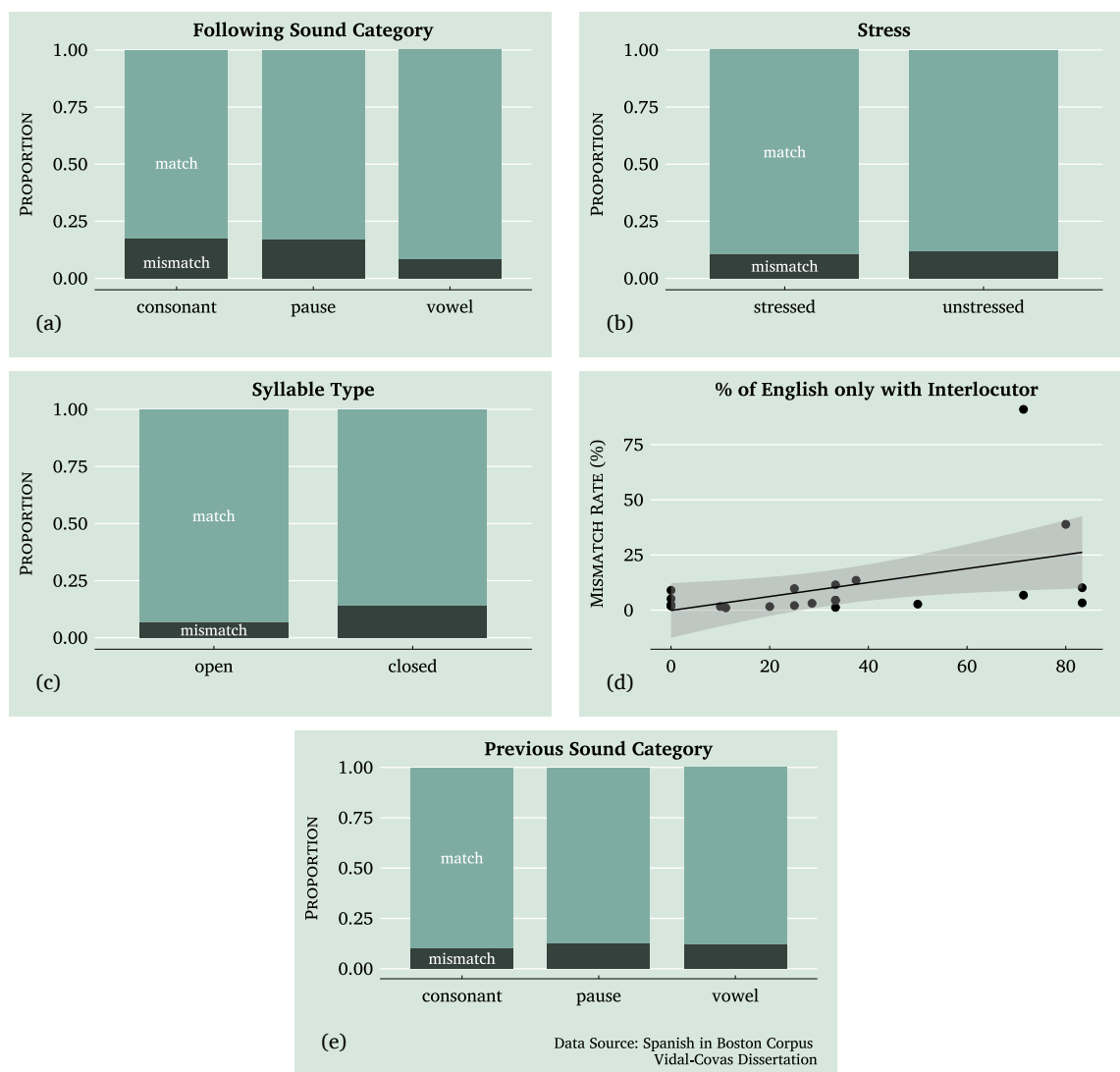


Figure 5-9: Visual representation of the effects of the predictors included in the best-fitting mixed effects model for lateral mismatch

are more probable when the following sound is a consonant. Additionally, *stress: unstressed* ($\beta = 0.65$, $p < .001$) and *syllable type: closed* ($\beta = 0.83$, $p = .001$) have positive regression coefficients, suggesting that mismatches are more frequent in unstressed and closed syllables. The % *English only with interlocutors* ($\beta = 0.03$, $p = .004$) also showed a positive effect, indicating that higher percentages of English-only interactions increase mismatch likelihood.

Table 5.5: Mixed Effects Model: Lateral Mis-(match) Prediction

PREDICTORS	DEP. VARIABLE: MISMATCH				
	β	SE	CI 95 %	Z-STAT	P-VALUE
Intercept	-3.73	.63	[-4.96--2.50]	-5.93	.000***
Following Sound Category (<i>ref:cons</i>)					
Pause	-.83	.31	[-1.43---.23]	-2.72	.007***
Vowel	-1.44	.27	[-1.97--.92]	-5.38	.000***
% English Only w Interlocutors	.03	.01	[.01--.05]	2.90	.004***
Previous Sound Category (<i>ref:cons</i>)					
Pause	-.86	.43	[-1.69--.03]	-2.02	.043**
Vowel	-.38	.24	[-.86--.09]	-1.58	.115
Stress (<i>ref:stressed</i>)					
Unstressed	.65	.19	[.28--1.01]	3.49	.000***
Syllable Type (<i>ref:open</i>)					
Closed	.83	.24	[.36--1.30]	3.44	.001***
RANDOM EFFECTS					
σ^2	3.29				
τ_{00} SPEAKERS	1.88				
ICC	.36				
N SPEAKERS	22				
SD SPEAKERS	1.37				
Observations	2626				
Marginal R ² / Conditional R ²	.23 / .51				
AIC	1104.61				

* $p < .1$; ** $p < .05$; *** $p < .01$

Visual representations of these effects, shown in Figure 5.9, further illustrate the relationships between each predictor and mismatch rates. For instance, they show that mismatch likelihood is notably reduced in contexts where pauses or vowels follow the lateral. Unstressed syllables, conversely, are more likely to result in mismatches compared to stressed ones, and closed syllables tend to exhibit more mismatches than open syllables.

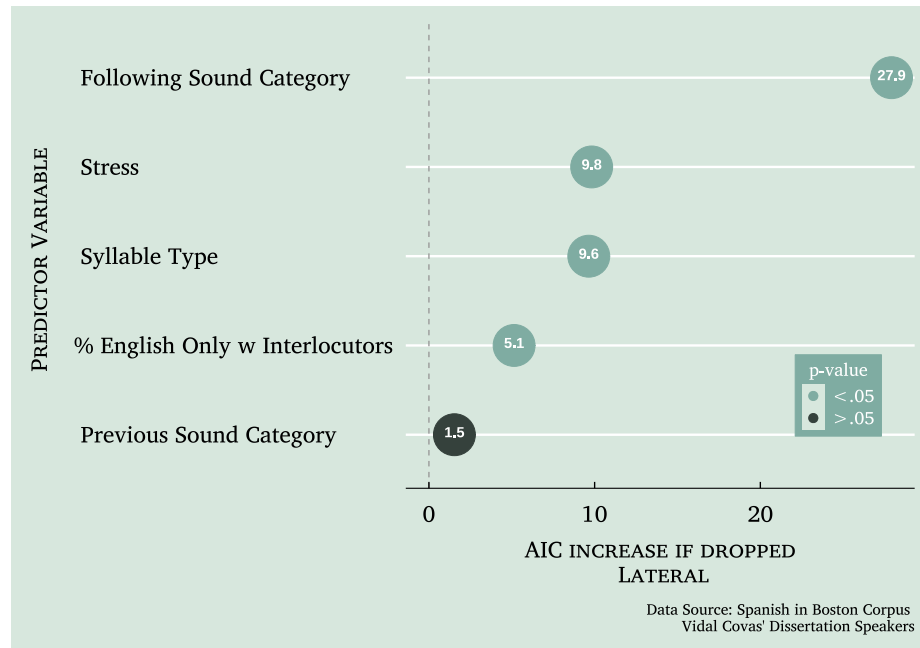


Figure 5-10: Impact of predictor variable removal on AIC in mixed effects model — /l/

The random effects variance for *speaker* is $\tau_{00} = 1.88$, with an *ICC* of .36, indicating that 36% of the variance in mismatches is due to differences between individual speakers. The *marginal* r^2 of .23 reflects the variance explained by fixed effects alone, while the *conditional* r^2 of .51 suggests that the full model accounts for 51% of the variance in mismatch outcomes.

5.2.5.1 Summary of the Results for Lateral

Among the most significant predictors, *following sound category* played a major role, with mismatches more likely when the following segment was a consonant, while vowels and pauses were associated with a reduced likelihood of mismatch.

Phonological factors such as *stress* and *syllable type* were also relevant. The positive regression coefficient for *stress: unstressed* indicates that laterals occurring in unstressed

syllables were more prone to mismatch. Similarly, the effect of *syllable type: closed* suggests that closed syllables foster more frequent lateral mismatches than open syllables, potentially due to increased articulatory constraints in coda positions.

Social factors further contributed to variation in lateral mismatch. The significant effect of *% English only with interlocutors* indicates that speakers who engage in more English-only interactions exhibit a higher likelihood of mismatch. Additionally, while *previous sound category* was included in the model, its overall effect was minor, with mismatches slightly more likely after a consonant than after vowels or pauses.

5.2.5.2 Lateral Subset: Dark-L Mismatch

To better understand the factors influencing lateral variation, I conducted a subset analysis focusing on the most frequent mismatch type, [ɫ]. The full dataset, as discussed above, contained 2,626 lateral tokens, of which 2,509 were included in this subset. Within this subset, the match category—meaning tokens realized as [l]—accounted for 2,333 tokens, making up 92.99% of the data. In contrast, the most common mismatch, [ɫ], was found in 176 tokens, representing 7.01% of the subset. Given that this mismatch type was the predominant realization among non-matching tokens, this analysis aimed to identify the linguistic and social factors that contributed to its occurrence.

The model for this subset, shown in Table 5.6, examines the variation between non-velarized lateral [l] matches and velarized lateral [ɫ] mismatches. It includes predictors that capture both phonetic and social influences, specifically *milliseconds per syllable*, *PLUS (percent life in the US)*, *stress*, and *syllable type*.

In this model, *PLUS* ($\beta = .05$, $p = .004$) and *syllable type: closed* ($\beta = .83$, $p = .005$)

are particularly influential, both showing positive effects on mismatch likelihood. This indicates that a higher percentage of life spent in the US and closed syllables are associated with a greater chance of dark-l mismatch. Though *stress: unstressed* and *milliseconds per syllable* approached significance, their impacts are more modest ($p = .089$ and $p = .092$, respectively).

Table 5.6: Mixed Effects Model: Non-Velarized Lateral [l] Match vs. Velar Lateral [ɫ] Mismatch Prediction

PREDICTORS	DEP. VARIABLE: MISMATCH				
	β	SE	CI 95 %	Z-STAT	P-VALUE
Intercept	-9.42	1.48	[-12.32—-6.53]	-6.38	.000***
Milliseconds per Syllable	.18	.10	[-.03—.38]	1.68	.092*
PLUS	.05	.02	[-.02—.09]	2.90	.004***
Stress (<i>ref:stressed</i>)					
Unstressed	.48	.29	[-.07—1.04]	1.70	.089*
Syllable Type (<i>ref:open</i>)					
Closed	.83	.29	[-.26—1.41]	2.83	.005***
RANDOM EFFECTS					
σ^2	3.29				
τ_{00} SPEAKERS	4.95				
ICC	.60				
N SPEAKERS	22				
SD SPEAKERS	2.22				
Observations	2509				
Marginal R ² / Conditional R ²	.33 / .73				
AIC	504.71				

* $p < .1$; ** $p < .05$; *** $p < .01$

Results shown in Figure 5.11 highlight *PLUS* and *syllable type* as having the largest impacts on model fit, with AIC increases of 6.4 for each if dropped. Meanwhile, *stress* and *milliseconds per syllable* had comparatively small effects, indicating their roles may be more peripheral in this specific context.

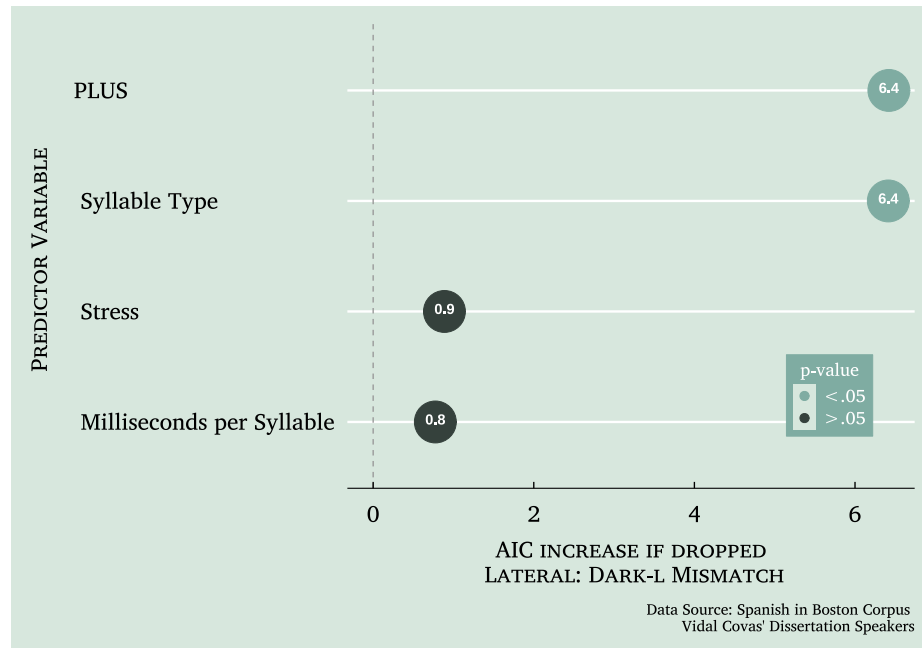


Figure 5-11: Impact of predictor variable removal on AIC — /l/: [l] & [ɫ]

The random effects show substantial variability between speakers, with an *ICC* of .60, suggesting that speaker-level differences account for 60% of the variance in dark-l mismatches. Additionally, the model's *marginal* r^2 is .33, while the *conditional* r^2 is .73, indicating that fixed effects alone explain 33% of the variance, with the full model explaining 73% when random effects are included.

5.2.5.3 Summary of the Results for Lateral Subset: Dark-L Mismatch

Among the factors influencing the likelihood of a lateral being realized as [ɫ] instead of [l], *PLUS* and *syllable type* emerge as the strongest predictors. Speakers who have spent a greater percentage of their lives in the U.S. are more likely to produce [ɫ] mismatches. Additionally, closed syllables favor the production of this mismatch, highlighting the role of syllable structure in the realization of laterals.

Although *stress* and *milliseconds per syllable* did not reach conventional significance thresholds, they still contribute to the model’s overall fit. The direction of their effects suggests that unstressed syllables and shorter syllable durations (i.e., faster speech rates) may be associated with a higher likelihood of dark-l realization, though their influence is weaker compared to *PLUS* and *syllable type*.

The random effects indicate substantial speaker-level variation, with an *ICC* of .60, meaning that a significant portion of the variation in dark-l mismatch patterns is attributable to individual differences among speakers. This suggests that while linguistic and social predictors play a role, speaker-specific factors—such as idiosyncratic variation in articulation or sociolinguistic identity—may also contribute to the realization of this variant.

5.2.6 Summary of Findings Across Liquid Models

In this section, I summarize the key findings across the six models analyzed, focusing on the trill, tap, and lateral liquids, along with their respective subsets. Each model reveals unique insights into the predictors influencing mismatch rates and highlights both shared and distinct factors across liquid types. Table 5.7 presents an overview of the main effects across the models. Here, a “+” indicates an increased likelihood of mismatch, while a “-” corresponds to a decreased likelihood of mismatch. Cells shaded in green reflect predictors that showed significant effects. A pattern observed in the rhotic models is that higher lexical frequency tends to increase the likelihood of mismatch, aligning with prior research on frequency-driven phonetic reduction. This reflects broader usage-based findings that frequent forms are more likely to undergo reduction—such as intervocalic /d/ deletion in

Spanish (Bybee 2002) or phonetic shortening in Dutch affixes (Pluymaekers et al. 2005). However, this effect did not appear in the lateral models, underscoring that frequency alone is not a universal predictor of variation. This mirrors inconsistencies found in morphosyntactic variation research, where frequency sometimes acts as an amplifier of other constraints rather than exerting a direct effect (Poplack 2001; Erker & Guy 2012; Dionne 2023).

Beyond these frequency-related effects observed in the rhotic models, a range of articulatory, contextual, and prosodic factors—such as *syllable structure*, *position in the word or syllable*, *stress*, and *speech rate*—also shaped liquid variation to varying degrees across models. These findings align with prior research on segmental weakening and phonetic reduction in Spanish (Poplack 1979; Alba 1990; Quilis 1993), particularly in highlighting the greater susceptibility of coda liquids to mismatch. While *PLUS* did not significantly predict tap variation, it consistently emerged as a significant factor in the models for trill mismatches and in the subset of laterals realized as [ɬ], suggesting that increased U.S. life experience influences variability in certain liquid types. However, this effect does not indicate wholesale convergence with English patterns; rather, it suggests selective adaptation, with different liquids exhibiting distinct trajectories of stability and change.

For taps, the strongest social predictors were *country of origin* and *sex* rather than contact measures, pointing to the maintenance of dialectal patterns rather than direct influence from English. The clearest example of this is lateralization of the tap, where *PLUS* had no effect: a Puerto Rican male is significantly more likely to lateralize than not, underscoring the persistence of region-specific articulations rather than contact-driven change.

Table 5.7: Summary of Results Across Liquids Models

PREDICTOR VARIABLE	TAP	/ɾ/ - [l]	TRILL	/r/ - [r] _ɿ	LATERAL	/l/ - [ɫ]
Log Lexical Frequency	+	+	+	+		
Milliseconds per Syllable	-	-	-	-		+
Position in Word (<i>ref:initial</i>)						
Internal			-	-		
Position in Syllable (<i>ref:onset</i>)						
Coda	+	+				
Previous Sound (<i>ref:cons</i>)						
Pause					-	
Vowel		+			-	
Following Sound (<i>ref:cons</i>)						
Pause	-	-			-	
Vowel	-	-			-	
Stress (<i>ref:stressed</i>)						
Unstressed	-	-			+	+
Syllable Type (<i>ref:open</i>)						
Closed					+	+
Country of Origin (<i>ref:DR</i>)						
Puerto Rico	+	+	-	-		
PLUS	+		+	+		+
% English Only					+	
Sex (<i>ref:female</i>)						
Male	+	+		-		

+ indicates an increased likelihood of mismatch

- indicates a decreased likelihood of mismatch

shaded cells reflect predictors that showed significant effects: $p < 0.1$; $p < 0.05$; $p < 0.01$

In contrast, the trill and lateral models tell two distinct stories regarding contact effects. For the trill, *PLUS* is the only significant social predictor when considering the entire dataset. However, when narrowing the focus to the subset of trill tokens realized as [r]_ɿ, a more nuanced picture emerges. Here, *country of origin* and *sex* become moderately significant, revealing that a Dominican female who has spent a greater portion of her life in the U.S. is more likely to produce [r]_ɿ—a variant that already exists in Spanish—than the canonical [r]. Rather than signaling English influence, this suggests that speakers are making a sociolinguistic choice to use an available Spanish variant, possibly as a means of indexing group belonging within a Caribbean Spanish identity.

The lateral, on the other hand, presents a clearer case of contact-induced variation, as

the only significant social predictors are related to English exposure. Contact appears to drive the likelihood of lateral mismatches, with the subset analysis of dark-*l* realization further reinforcing this. Not only is *PLUS* a significant predictor in this subset, but it also has the strongest effect on model fit, highlighting the role of U.S. exposure in shaping lateral variation. Crucially, [ɫ] is not a variant that exists in Spanish outside of contact with English, distinguishing it from the other liquid variations examined. However, this does not mean that speakers are unaware of /l/ as a site of variation. Rather, it may be that /l/ does not carry the same weight in signaling national or regional belonging, leading speakers to devote less effort to maintaining a distinctively “Puerto Rican” or “Dominican” realization of the lateral while focusing more on the rhotics.

Taken together, these results suggest that while liquids are often discussed collectively as a variable feature of Caribbean Spanish, speakers appear to differentiate them in their sociolinguistic behavior. All three liquids exhibit variation, but rhotics seem to carry greater social significance, making them stronger candidates for identity marking within Caribbean Spanish. By contrast, /l/, which lacks the same social weight, may be more susceptible to bilingual optimization strategies, as speakers prioritize variation in features that provide greater social leverage. In other words, speakers may direct their effort toward linguistic variables that yield the most symbolic capital within their social and linguistic networks, while allowing /l/ to shift more freely under the influence of English contact.

6 Covariation and Coherence: Relationships between Linguistic Variables

As discussed previously, the study of linguistic variation in contact settings offers a unique lens through which to understand the interplay between salience, agency, and social meaning. Previous studies have emphasized the importance of examining relationships between variables to uncover patterns of structural convergence, resistance, or independence in bilingual communities (Erker & Otheguy 2016; Guy et al. 2022). Building on these insights, this chapter investigates the co-variation of the linguistic features in this study among Spanish-speaking Bostonians of Puerto Rican and Dominican origin, focusing on how salience potentially shapes linguistic behavior in a contact setting.

To assess covariation, I explored patterns of variation in five variables: *liquid variation* (n = 7,354), *coda /s/* (n = 3,806), *filled pauses* (n = 1,718), *subject personal pronouns* (n = 7,240), and *subject placement* (n = 4,079). By framing these variables within a continuum of salience, this analysis explores how linguistic features potentially differ in their relationship to the pressures of contact. The following sections detail the relationships among these variables and how they interact within the broader linguistic landscape under investigation.

This co-variation analysis directly addresses the second research question of the study: whether salient and non-salient features pattern differently in contexts of language and dialect contact. Variables on the lower end of the salience continuum, such as filled

pauses and subject placement, show evidence of structural convergence with English, reflecting bilingual optimization strategies and the pressures of cognitive efficiency. In contrast, higher-salience variables like coda /s/ and liquids appear comparatively more stable across contact-related metrics. This relative stability is not due to any inherent resistance to change, but rather to the fact that salient features are more likely to be noticed, talked about, and strategically deployed by speakers. As such, they are more responsive to social rather than cognitive pressures—and their trajectories depend on the specific ideologies and identity goals of a given community. In the case of Spanish-speaking Boston, speakers exhibit strong own-group loyalty and high dialectal awareness, conditions that favor the maintenance of nationally recognizable speech patterns. Thus, the comparative patterning of convergence and stability observed here reflects a broader dynamic: while lower-salience features tend to shift in predictable, cognitively motivated ways, the treatment of higher-salience features is more variable, shaped by social meaning, ideological orientation, and local histories of contact.

As the final results chapter, this analysis, when considered alongside the findings from earlier chapters, provides an overview of how salience informs our understanding of linguistic variation and covariation in bilingual contact settings. By examining how features co-vary—or fail to co-vary—this chapter illuminates the mechanisms through which linguistic systems vary and change in contact settings. These insights are critical for understanding how speakers utilize their linguistic repertoires to construct identity, manage group affiliations, and respond to sociolinguistic pressures. The patterns observed in this chapter lay the groundwork for the subsequent discussion and conclusion chapter, where their broader implications for sociolinguistic theory, language contact, and the role of

salience in shaping linguistic variation and change will be explored. Before examining the relationships between these features, Section 6.1 establishes their individual distributions.

6.1 Overview of Empirical Patterns

While the previous chapter provided an analysis of liquid variation, this section shifts focus to four additional variables that have not yet been discussed in detail: *coda /s/*, *filled pauses*, *subject personal pronouns*, and *subject placement*. Building on prior work in these four variables, this section presents a general overview of the distribution of their variants in the speech of the 22 Spanish-speaking Bostonians analyzed in this dissertation.

Subject Pronoun Expression The analysis of subject personal pronouns followed the methodology of Otheguy & Zentella (2012), categorizing eligible finite verb tokens as either *present* or *absent* with respect to pronominal forms. A pronoun was classified as *present* if it explicitly appeared in speech, while it was considered *absent* if it did not, as shown in Table 6.1. The dataset included a total of 7,240 sites of pronominal variation, of which 2,698 (37.27%) were categorized as *present* and 4,542 (62.73%) as *absent*.

Table 6.1: Examples of Subject Pronoun Expression Coding

(1) Yo bailo salsa en la plaza los jueves. También (2) ø bailo merengue. Los miércoles (3) ø tomo clases de gimnasia. Ignacio nada. (4) Él también alza pesas pero (5) ø no hace gimnasia. (1) I dance salsa in the plaza on Thursdays. (2) I (ø) also dance merengue. On Wednesdays, (3) I (ø) take gym classes. Ignacio swims. (4) He also lifts weights but (5) He ødoesn't do gymnastics.			
TOKEN #	PRONOUN	PRESENT/ ABSENT	TRANSLATION
(1)	yo	present	I
(2)	ø	absent	(I)
(3)	ø	absent	(I)
(4)	él	present	he
(5)	ø	absent	(he)

Subject Placement Subject placement was analyzed across both pronominal and non-pronominal subjects, following the approach used in previous studies (Erker et al. 2017; Erker 2017). Subjects were categorized based on word order: *pre-verbal* subjects followed *SV* order, while *post-verbal* subjects followed *VS* order, as illustrated in Table 6.2. Of the 4,079 subject tokens analyzed, 3,613 (88.58%) appeared in the *pre-verbal* position, while 466 (11.42%) were *post-verbal*.

Table 6.2: Examples of Subject Placement Coding

(1) **Ignacio** es muy rápido en natación, pero sus mejores estilos son la mariposa y el estilo libre, donde siempre gana medallas (2) **él**. En el relevo de estilos, (3) **Ignacio** nada la parte de estilo libre, mientras hacen pecho, mariposa y espalda (4) **sus compañeros**.
 (1) *Ignacio is very fast at swimming, but his best strokes are butterfly and freestyle, where (2) he always wins medals. In the medley relay, (3) Ignacio swims the freestyle part, while (4) his teammates do breaststroke, butterfly, and backstroke.*

TOKEN #	SUBJECT	PRE/POST -VERBAL	TRANSLATION
(1)	Ignacio	pre-verbal	Ignacio
(2)	él	post-verbal	he
(3)	Ignacio	pre-verbal	Ignacio
(4)	sus compañeros	post-verbal	his teammates

Filled Pauses The methodology used for describing variation in filled pauses can be found in Erker & Bruso (2017) and Erker & Vidal-Covas (2022). As demonstrated in Table 6.3, tokens were categorized based on vowel quality, distinguishing between centralized and not centralized variants. Filled pauses containing [a] or [ə] were classified as *centralized*, while those containing [e] were classified as *not centralized*. Across the 22 interviews, a total of 1,718 filled pauses were identified, with 999 (58.15%) categorized as *centralized* and 719 (41.85%) as *not centralized*.

Table 6.3: Examples of Filled Pause Categorization

(1) [e]	Yo entreno todos los días porque quiero mejorar mi tiempo y (2) [e] ganar más medallas. (3) [ə̃m] A veces estoy muy cansado, (4) [ə̃], pero (5) [ə̃] como quiera (6) [am] voy a la piscina y (7) [a] entreno.
(1) Eh	I train every day because I want to improve my time and (2) eh win more medals. (3) Um sometimes I'm really tired, (4) uh, but (5) uh anyway (6) um I go to the pool and (7) uh train.

TOKEN #	VOWEL	CENTRALIZATION	TRANSLATION
(1)	[e]	not centralized	eh
(2)	[e]	not centralized	eh
(3)	[ə̃m]	centralized	um
(4)	[ə̃]	centralized	uh
(5)	[ə̃]	centralized	uh
(6)	[am]	centralized	um
(7)	[a]	centralized	uh

Coda /s/ The classification of coda /s/ in this dissertation built on the methodological foundation established in Erker (2017) and Erker & Reffel (2021), but introduced a novel coding approach. Specifically, a *match/mismatch* framework—like the one used in the analysis of liquid variation—was employed to evaluate whether the presumed PHONOLOGICAL FORM and observed SURFACE FORM matched. If the surface realization was anything other than [s], it was classified as a *mismatch*. This includes realizations such as [z], [h], [Ø] (deletion), or less common variants like [g]. In contrast, tokens that retained [s] as their surface form were classified as *match*. Table 6.4 provides examples of this classification. Across the dataset, a total of 3,806 tokens of coda /s/ were analyzed, with 2,071 (54.41%) classified as *match* and 1,735 (45.59%) as *mismatch*.

Table 6.4: Examples of Match/Mismatch Dependent Variable for Coda /s/

HOST WORD	TRANSLATION	PHONOLOGICAL FORM	SURFACE FORM	MATCH/MISMATCH
despertar	to wake up	/s/	[s]	MATCH
desproporcionado	disproportionate	/s/	[Ø]	MISMATCH
escuela	school	/s/	[g]	MISMATCH
estilo	style	/s/	[h]	MISMATCH
inestabilidad	instability	/s/	[s]	MATCH
responsabilidad	responsibility	/s/	[z]	MISMATCH
transcripción	transcription	/s/	[s]	MATCH

With these baseline distributions established, we now turn to the relationships between these variables.

6.2 Variation in Lower Salience Variables in a Contact-Setting

Keeping in mind the discussion of Spanish-English bilingualism and language use patterns presented in Section 4.2, let us explore how low-salience features are used in the contact setting of interest. Specifically, I reconstruct previously published analyses of three variants—pronoun presence, pre-verbal subjects, and centralized filled pauses—to provide context within the broader sample of 80 speakers from the Spanish in Boston Corpus. These variables have been shown in prior studies to exhibit patterns of change that, while modest, increasingly align their usage with the grammatical properties of analogous structures in English (Erker 2012). The broader patterns observed among these speakers provide a foundation for interpreting the behavior of the 22 speakers analyzed in detail throughout this dissertation.

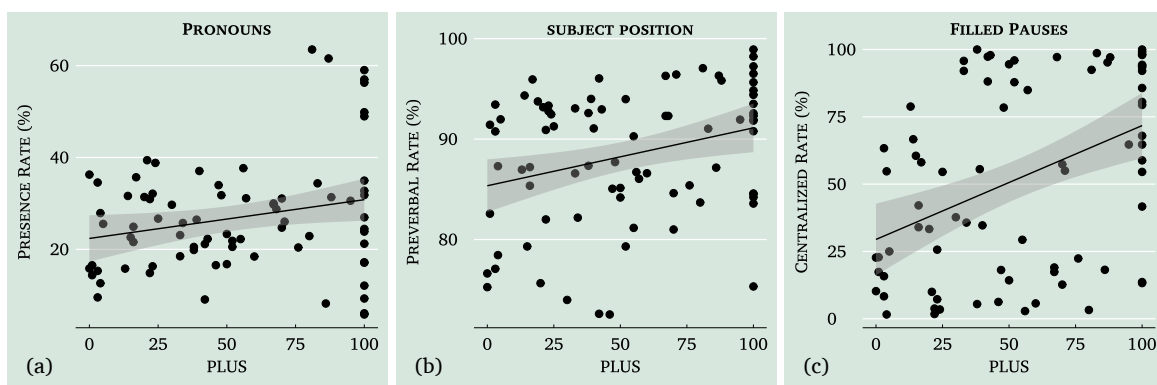


Figure 6-1: Relationship of pronoun presence, pre-verbal subjects, and centralized filled pauses to PLUS among 80 Bostonians. These three figures represent a larger sample from the Spanish in Boston Corpus, compared to the 22 speakers analyzed in most parts of the dissertation.

Figure 6-1 highlights the trends in the use of these features among the 80 speakers.

The specific patterns observed are as follows:

PRONOUN PRESENCE (6.1A) - Higher rates of pronoun presence are associated with an increase in *Percent Life in the United States (PLUS)*, reflecting structural alignment with English, where pronoun use is nearly obligatory.

PRE-VERBAL SUBJECTS (6.1B) - Speakers with greater mainland U.S. residency demonstrate a higher frequency of pre-verbal subjects, a characteristic feature of English syntax.

CENTRALIZED FILLED PAUSES (6.1C) - The use of centralized filled pauses increases alongside PLUS, further supporting the evidence of cross-linguistic influence.

6.2.1 Zooming In: Low-Salience Features Among 22 Speakers

To more closely examine the linguistic behavior of the 22 Spanish-speaking Bostonians analyzed in this dissertation, Figure 6.2 provides a zoomed-in view of three lower-salience features—pronoun presence, subject position, and centralized filled pauses—in relation to *Percent Life in the United States (PLUS)*. The patterns observed in this subset generally mirror the broader trends seen in the larger corpus: as speakers' time in the U.S. increases, their usage of these features shifts in ways that more closely align with English norms.

Specifically, speakers with greater U.S. residency tend to use pronouns more frequently (Figure 6.2a), favor pre-verbal subject constructions (Figure 6.2b), and show increased use of centralized filled pauses (Figure 6.2c)—features that parallel English usage. In the case of filled pauses, speakers increasingly favor the centralized variant [a], which is available in both English and Spanish, potentially as a way to reduce processing costs in bilingual

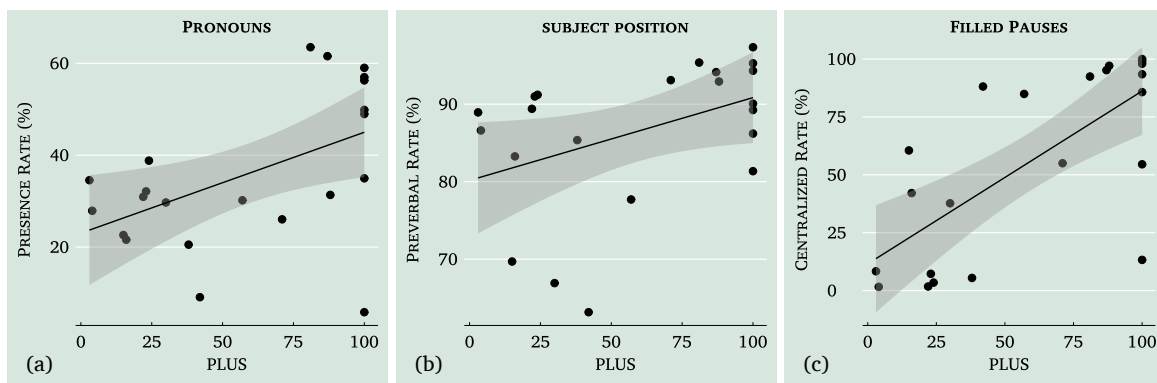


Figure 6-2: Relationship of pronoun presence, pre-verbal subjects, and centralized filled pauses to PLUS among 22 Spanish-speaking Bostonians. These three figures focus on the 22 speakers analyzed in detail throughout the dissertation.

speech. These observations are consistent with prior findings that socially weak variables may shift in ways that ease cross-linguistic processing, potentially as part of bilingual optimization strategies (Muysken 2013; Erker & Bruso 2017).

These patterns, while suggestive, should be interpreted cautiously. Given the absence of a full constraints-based analysis, we cannot determine whether these surface-level trends reflect changes in underlying grammars or shifts in usage conditioned by other social or linguistic factors. To probe this further, the next section presents regression analyses at the token level, offering a more fine-grained account of how social predictors like U.S. residency correlate with the use of low-salience variables.

Regression Model Results: Low-Salience Features Among 22 Speakers

The patterns observed are supported by regression model results, summarized in Table 6.5. These analyses are conducted at the token level, examining individual instances of pronoun presence/absence, subject positioning, and pause filling rather than aggregate speaker-level patterns. The table presents findings for *pronoun presence*, *pre-verbal subjects*, and *centralized filled pauses* among the 22 Spanish-speaking Bostonians analyzed.

Unlike the liquids chapter, these models focus exclusively on social predictors, rather than linguistic ones. This reflects the fact that linguistic constraints vary across variables, making direct comparisons difficult. More importantly, the aim of this analysis is to examine how social factors shape patterns of linguistic variation and covariation in a contact setting, rather than to model internal linguistic constraints. By centering social influences, this approach highlights the ways in which external factors structure speakers' linguistic behavior across multiple variables.

Table 6.5: Summary of Model Results Corresponding to Pronoun Presence, Pre-Verbal Subjects, and Centralized Filled Pauses among 22 Spanish-speaking Bostonians

PREDICTORS	STAT	DEPENDENT VARIABLE:		
		PRONOUNS PRESENT ¹	PREVERBAL SUBJECTS ²	CENTRALIZED FPS ³
Intercept	β	-.62	1.58**	-1.43
	SE	(.59)	(.67)	(1.77)
Country of Origin (<i>ref:DR</i>)				
Puerto Rico	β	-.12	.03	.06
	SE	(.29)	(.31)	(.89)
PLUS	β	.01**	.02***	.05***
	SE	(.00)	(.00)	(.01)
Age	β	-.01	.00	-.03
	SE	(.01)	(.01)	(.04)
Sex (<i>ref:female</i>)				
Male	β	-.49*	-.12	-.57
	SE	(.29)	(.30)	(.86)
RANDOM EFFECTS				
SD _{SPEAKERS}		.64	.59	1.87
ICC		.29	.26	.78
Observations		7240	3888	1718
Marginal R ² / Conditional R ²		.06/.17	.09/.18	.44/.73

* $p < .1$; ** $p < .05$; *** $p < .01$

MODEL SPECIFICATIONS:

¹ `glmer(PRONOUNS PRESENT ~ COUNTRY OF ORIGIN + PLUS + AGE + SEX + (1|SPEAKER))`

² `glmer(PREVERBAL SUBJECTS ~ COUNTRY OF ORIGIN + PLUS + AGE + SEX + (1|SPEAKER))`

³ `glmer(CENTRALIZED FPS ~ COUNTRY OF ORIGIN + PLUS + AGE + SEX + (1|SPEAKER))`

Key results include:

PLUS - A significant positive effect on all variables, indicating that increased U.S. residency aligns with greater structural convergence to English.

COUNTRY OF ORIGIN - No significant effect, with variation largely driven by PLUS rather than national origin.

RANDOM EFFECTS - Notable variation across individual speakers, reflected in the random effects and intraclass correlation coefficients (ICC).

MODEL FIT - The models vary in explanatory power:

PRONOUN PRESENCE - *marginal r^2* = .06, *conditional r^2* = .17

PREVERBAL SUBJECTS - *marginal r^2* = .09, *conditional r^2* = .18

CENTRALIZED FILLED PAUSES - *marginal r^2* = .44, *conditional r^2* = .73

These values indicate that while speaker-level variation (captured by *conditional r^2*) accounts for a substantial portion of the variance, fixed effects alone (*marginal r^2*) explain a smaller but meaningful proportion, particularly for centralized filled pauses.

These results provide further evidence that, among the speakers analyzed, features at the lower end of the salience continuum exhibit parallel patterns of change in the same direction—namely, structural convergence with English—highlighting a shared trajectory of adaptation in this contact setting.

6.3 Salient Features as Tools for Speaker Identity

Speakers' treatment of coda /s/ is analyzed across the full sample of 80 speakers and the subset of 22 speakers who are of primary interest in the present study. Figure 6-3a highlights that while there is variation in coda /s/ realization across both the overall and subset groups, its treatment is more consistent and less related to PLUS compared to low-salience features. Among the 22 speakers (Figure 6-3b), patterns in coda /s/ realization

remain tightly linked to national dialect norms, underscoring its salience and availability for speakers to use in their construction of identity. The analysis of coda /s/ mismatch

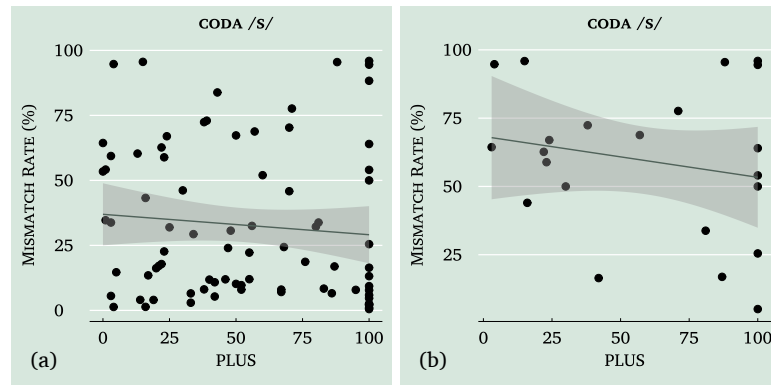


Figure 6.3: Comparison of the relationship between coda /s/ mismatch and PLUS in 80 and 22 Spanish-speaking Bostonians

among both the 80 and 22 speakers highlights the influence of national origin and the stability of this feature against contact-induced change. However, while coda /s/ provides valuable insights into dialectal maintenance and resistance to English structural convergence, it represents only one dimension of the broader picture of linguistic mismatch in these speakers. To more fully understand how these speakers use salient features in this context of language contact, it is crucial to consider all three features—coda /s/, tap-lateral mismatch, and trill-fricative trill mismatch—together.

Unlike coda /s/, the liquids were only analyzed in detail among the 22 speakers, representing one of the main contributions of this study. This focus enables a more comprehensive examination of how these features are associated with predictors like *PLUS* and *Country of Origin*, shedding light on the unique patterns of variation and stability that emerge within the group. The results for all three liquids are presented in Figure 6.4 and Table 6.6.

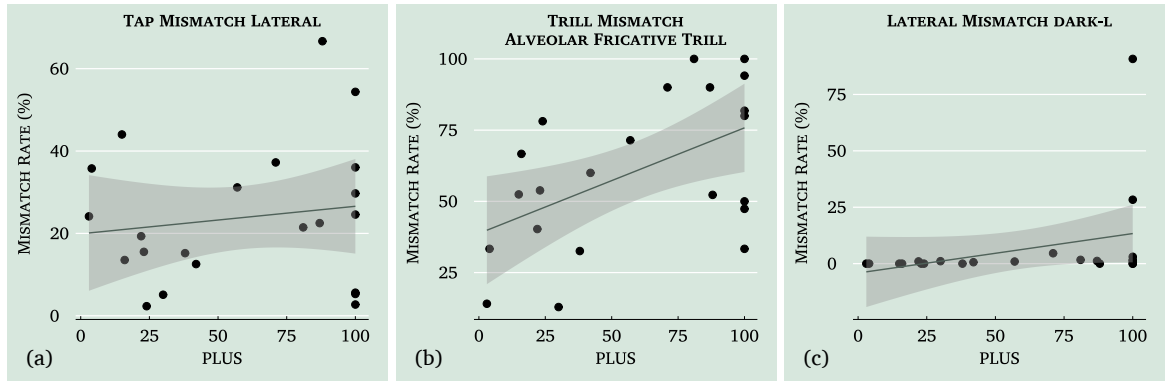


Figure 6-4: Relationship of tap mismatch, trill mismatch, and lateral mismatch to PLUS among 22 Spanish-speaking Bostonians

Table 6.6: Summary of Model Results Corresponding to Coda /s/ Mismatch, Tap-Lateral Mismatch, Trill-Fricative Trill, and Lateral Velarization Mismatch among 22 Spanish-speaking Bostonians

PREDICTORS	STAT	DEPENDENT VARIABLE: MISMATCH			
		CODA /s/	/r/ - [l]	/r/ - [r]	/l/ - [ɭ]
Intercept	β	-.89	-3.27***	.07	-9.57***
	SE	(1.30)	(.75)	(1.01)	(2.68)
Country of Origin (<i>ref:DR</i>)					
Puerto Rico	β	1.64**	1.43***	-.61	-.99
	SE	(.64)	(.38)	(.50)	(1.17)
PLUS	β	-.01	.00	.02***	.06***
	SE	(.01)	(.01)	(.01)	(.02)
Age	β	.03	.02	-.00	.02
	SE	(.03)	(.02)	(.02)	(.05)
Sex (<i>ref:female</i>)					
Male	β	.03	.43	-.69	.82
	SE	(.63)	(.37)	(.48)	(1.18)
RANDOM EFFECTS					
SD _{SPEAKERS}		1.41	.79	.94	2.08
ICC		.67	.38	.47	.81
Observations		3806	3244	735	2509
Marginal R ² / Conditional R ²		.13/.46	.12/.26	.17/.34	.37/.73

* $p < .1$; ** $p < .05$; *** $p < .01$

MODEL SPECIFICATIONS:

`glmer(MISMATCH ~ COUNTRY OF ORIGIN + PLUS + AGE + SEX + (1|SPEAKER))`

The results presented in Table 6.6 shed light on the relationship between linguistic mismatch patterns and key predictors among 22 Spanish-speaking Bostonians. The findings suggest distinct influences for each dependent variable:

CODA /s/ MISMATCH (6·3B) - *Country of origin* is a significant predictor ($\beta = 1.64, p < 0.1$), indicating that speakers from Puerto Rico are more likely to exhibit mismatch. This suggests a maintenance of linguistic patterns tied to nation-state varieties rather than convergence with English structures. *PLUS*, however, does not significantly predict mismatch in this feature ($p > 0.1$), suggesting that coda /s/ does not exhibit the same pattern of change observed in lower-salience features. Model fit statistics show a moderate contribution of fixed effects (*marginal* $r^2 = .13$), while speaker-level variation accounts for a substantial portion of the variance (*conditional* $r^2 = .46$), highlighting the role of individual differences in this feature.

TAP-LATERAL MISMATCH (6·4A) - *Country of origin* also plays a significant role ($\beta = 1.43, p < 0.01$), with speakers from Puerto Rico showing greater mismatch. This further supports the idea of dialectal maintenance linked to national origin. *PLUS* does not emerge as a significant predictor ($p > 0.1$) for this feature, suggesting that contact with other dialects of Spanish—whether through interaction with non-Caribbean speakers or exposure to standard norms—has not led to reconfiguration of this feature. The model explains a moderate amount of variance through fixed effects (*marginal* $r^2 = .12$), with speaker-level variation contributing substantially (*conditional* $r^2 = .26$), suggesting that individual speaker tendencies shape patterns of variation in this feature.

TRILL-FRICATIVE TRILL MISMATCH (6·4B) - *PLUS* has a significant positive effect ($\beta = 0.02, p < 0.01$). While this might initially be interpreted as a sign of structural alignment with English or possibly dialectal leveling, a closer analysis of

the specific variant used by speakers reveals an alternative explanation. Speakers with higher PLUS values are more likely to use a trill variant that is not found in English but is attested within the broader Spanish linguistic repertoire. Rather than signaling convergence, this pattern may represent a nuanced form of linguistic resistance—one in which adaptation to mainland U.S. residency is accompanied by the reinforcement of variation internal to Spanish. In this way, speakers may maintain aspects of their linguistic identity even as they navigate life in a bilingual environment. The model captures a portion of the variance through fixed effects (*marginal* $r^2 = .17$), while speaker-level variation remains influential (*conditional* $r^2 = .34$), indicating that both social factors and individual differences contribute to this pattern.

LATERAL-VELARIZED LATERAL MISMATCH (6.4B) - PLUS has a significant positive effect ($\beta = 0.06, p < 0.01$). As discussed in the chapter dedicated to the analysis of liquids, the [ɭ] variant is absent in monolingual Spanish contexts and emerges primarily in the context of contact with English. This sets it apart from the subsets of the other two liquids studied. Nonetheless, speakers recognize /l/ as a variable feature. It appears that /l/ lacks significant social indexing of national or regional identity, leading speakers to invest less effort in preserving a distinctly ‘Puerto Rican’ or ‘Dominican’ pronunciation of the lateral, while placing greater emphasis on rhotic sounds. The model demonstrates a strong ability to account for variation in this feature, with fixed effects explaining a substantial portion of the variance (*marginal* $r^2 = .37$) and speaker-level variation playing an even greater role (*conditional* $r^2 = .73$), suggesting

that individual differences strongly shape patterns of lateral variation.

6.4 Intermediate Conclusion: Saliency and Covariation

The results presented in this chapter demonstrate that features at the higher end of the saliency continuum do not exhibit the same patterns of covariation observed among lower-saliency variables, which show more consistent intergenerational trends. Among the low-saliency features—*pronoun presence*, *pre-verbal subjects*, and *centralized filled pauses*—there is clear evidence of parallel change: each shows a significant positive relationship with *PLUS*, suggesting that these features are undergoing rate-level change in comparable directions. These patterns reflect systematic shifts in usage that align with structures found in English, likely facilitated by their low social visibility and reduced indexical potential.

In contrast, *coda /s/* shows no evidence of change with respect to *PLUS*, and remains closely tied to national origin. Among the liquid variables, the results are more mixed: while *lateral velarization*([ɭ]) shows signs of contact-induced restructuring, *trill-fricative trill* alternation appears to reflect a more complex form of adaptation—one that resists convergence with English and draws instead from the broader Spanish linguistic repertoire. Notably, *tap-lateral alternation*, the highest-profile of the liquid variables, shows no evidence of change induced by either language or dialect contact, and is best predicted by national origin.

Taken together, these findings suggest that features at the higher end of the saliency continuum do not exhibit consistent patterns of covariation in response to contact. Rather than shifting uniformly, these features appear to offer speakers greater flexibility in how they are used. Speakers seem to selectively navigate their linguistic choices, balancing the

pressures of contact with a commitment to Spanish-specific variants. Some salient variables, such as *coda* /s/ and *rhotics*, carry greater social weight and are more likely to be actively maintained or strategically deployed to signal identity. In contrast, /l/, which may lack the same indexical significance, appears more susceptible to contact-driven restructuring—perhaps because speakers invest less effort in maintaining national or dialectal norms for this feature. These contrasts underscore the importance of salience not only in shaping patterns of change, but also in enabling speakers to navigate and negotiate linguistic variation in bilingual settings.

7 Discussion & Conclusions:

Linguistic Salience and Contact-Induced Change

Linguistic variation in contact settings presents an especially interesting opportunity to examine the interplay between salience, agency, and social meaning. The present study investigates this dynamic by analyzing patterns of variation in and potential relationships between multiple linguistic features in Spanish-speaking Bostonians. Two overarching research questions guided this study:

1. How do linguistic and social factors shape variation in the production of liquids in Spanish?
2. Do the data support the hypothesis that features at the lower end of the salience continuum and those at the higher end pattern differently in their relation to various metrics of contact experience?

The findings provide strong evidence that lower-salience features (e.g., *subject pronouns*, *subject placement*, *filled pauses*) exhibit structural convergence with English as an outcome of bilingual optimization strategies aimed at maximizing common ground across linguistic systems. Conversely, higher-salience features (e.g., *coda /s/* and *liquid variation*) tend to show greater stability—not because they are inherently resistant to change, but because they are governed by different processes in contact settings. Rather than reflecting cognitively motivated—and potentially subconscious, or at least non-socially moti-

vated—alignment across linguistic systems, these features function as socially charged resources. Speakers can—but do not always—strategically manipulate them to assert in-group membership, maintain dialectal identity, or resist external linguistic pressures. While this distinction between high- and low-salience features is well-documented in the literature (Erker 2017; Erker 2022; Guy et al. 2022), this study expands on previous work by incorporating liquid variation as a key sociolinguistic variable.

In what follows, I first summarize the findings on liquid variation and situate them within existing research. I then discuss patterns of covariation and coherence, focusing on the structural convergence of features on the lower end of the salience spectrum and the social significance of higher-salience features. Finally, I reflect on the broader implications of these findings, emphasizing how certain salient features afford greater indexical potential than others, and how speakers navigate this variation in bilingual contact settings.

7.1 Linguistic and Social Conditioning of Liquid Variation

The regression analyses presented in Chapter 5 reveal that the three Spanish liquids—taps, trills, and laterals—are not uniformly conditioned by linguistic and social factors. Instead, each liquid displays a distinct profile, shaped by a combination of articulatory, phonological, lexical, and sociolinguistic influences. These patterns not only echo previous findings in the literature but also add nuance by disaggregating effects across subsets of the data and exploring how these predictors interact in bilingual, urban contexts.

7.1.1 Taps: Strong Internal Conditioning, Dialect-Based Social Patterning

Tap variation is strongly constrained by internal linguistic factors. The most robust predictors of mismatch between presumed phonological form and observed phonetic realizations included syllable position, phonological context, lexical frequency, and speech rate. These findings align with prior studies showing that consonants in coda position are more susceptible to reduction or reanalysis (Poplack 1979; Alba 1990; Quilis 1993; Ramos-Pellicia 2004; Beaton 2016). The influence of following segment also mirrors classic accounts of coarticulatory influence on liquid realization, with taps more stable when flanked by vowels.

Notably, the data also reveal that taps are more likely to be realized canonically in prosodically weaker (i.e., unstressed) syllables—contrary to typological expectations, where reduction typically favors weaker positions. While—so far—limited to the present data, this finding suggests a potential divergence in Spanish tap behavior that warrants further investigation.

Importantly, social factors also play a role in shaping tap variation. Country of origin emerged as a significant predictor, with Puerto Rican speakers more likely than Dominicans to produce mismatches, especially lateralized taps—a hallmark feature of Puerto Rican Spanish. This is consistent with previous descriptions of lateralization in the Caribbean, particularly in Puerto Rico (Lipski 2012; Shenk 2021). Gender was also significant, with male speakers more prone to mismatch, echoing past findings that associate men with less standard or more locally marked speech forms (Beaton 2005).

Interestingly, contact-related measures, such as PLUS, did not significantly predict vari-

ation for taps, suggesting that tap lateralization persists primarily as a dialect-internal feature, resistant to contact-driven restructuring. These results reinforce the idea that taps are maintained as a regionally specific variant, particularly among Puerto Rican speakers, and are less affected by English influence or inter-dialectal interaction.

7.1.2 Trills: Bilingual Contact as a Primary Driver

Unlike taps, trill variation appears more sensitive to bilingual exposure. The most robust social predictor in the overall trill model was Percentage of Life in the U.S. (PLUS), which positively predicted mismatch. This suggests that speakers with more extensive U.S. residency are more likely to deviate from the canonical trill, often producing the variant [r̥], a fricative-like realization previously attested in Caribbean Spanish varieties (Delgado-Díaz & Galarza 2015).

However, a closer look at the mismatch forms complicates this interpretation. The most frequent non-canonical realizations are not borrowed from English, but rather emerge from within the broader inventory of Spanish phonetic variation—especially those common in Caribbean varieties. This suggests that the observed patterns may not reflect straightforward convergence with English phonology, but instead result from contact with other Spanish dialects. As such, the correlation between trill variation and U.S. residency likely indexes increased exposure not only to English, but also to Spanish varieties beyond a speaker’s own dialect background—especially in an urban environment like Boston.

This interpretation is supported by the trill subset model, where both PLUS and lexical frequency showed significant effects, and speaker gender and national origin emerged as marginally significant. Taken together, these findings suggest that bilingual contact, in the

broadest sense, shapes trill realization—not solely through structural influence from English, but through the cumulative effects of multilingual, multi-dialectal interaction. The fricative trill variant may thus be reanalyzed or reweighted in its salience or acceptability through this contact-driven exposure, further underscoring the complex dynamics of variation in diaspora communities.

7.1.3 Laterals: Contact-Induced Restructuring and Dialect Mixing

Among the three liquid phonemes, the lateral /l/ shows the clearest evidence of contact-induced restructuring. In contrast to both rhotic liquids, lateral variation (/l/) showed minimal alignment with national origin and was shaped primarily by contact with English and increased exposure to diverse Spanish dialects. In the lateral models, social predictors such as % English-only interaction with interlocutors and PLUS (Percentage of Life in the U.S.) emerged as significant, while national origin had no effect. This suggests that variation in /l/ is less tied to national identity and more reflective of bilingual and bidialectal interaction in U.S. urban environments.

The treatment of the lateral reflects broader linguistic experiences shaped by life in the United States—including not only increased contact with English, but also greater interaction among Spanish speakers from diverse dialect backgrounds. This interpretation aligns with prior work on L2 Spanish, such as Solon (2015), which shows that English-speaking learners often produce velarized [ɫ] in coda position due to transfer from English. Solon also notes that this cue significantly contributes to the perception of a foreign accent. While her study focuses on L2 learners, the current findings suggest that a similar phonetic influence may emerge in heritage and bilingual speakers as a result of regular use of

English.

While [ɫ], a variant not associated with Caribbean Spanish but common in English, was the most frequent mismatch form in the dataset (6.7%), the overwhelming majority of lateral tokens (88.9%) were produced as canonical [l]—a variant widely preferred across all Spanish varieties. The presence of velarized /ɫ/ appears to correlate with increased mainland U.S. life experience, likely reflecting cross-linguistic influence. However, it remains a minority realization, suggesting that even in contexts of sustained bilingualism, canonical /l/ continues to dominate production patterns. Unlike some L2 contexts where [ɫ] becomes a dominant or default realization, the speakers in this study do not exhibit wholesale convergence with English norms. Instead, their productions suggest that even in contexts of intense bilingualism, speakers continue to produce /l/ in ways consistent with Spanish phonological patterns. This, in turn, supports the idea that /l/, while arguably less socially charged than rhotics, is shaped more by language-internal phonological structure than by the conscious stylistic or social choices of speakers.

Moreover, the broader distribution of /l/ variants points to the possibility that dialect contact is also shaping variation. While processes such as deletion, aspiration, and vocalization—widely attested in Caribbean Spanish (Lipski 2008; Quilis 1993; Alba 1990; Alba 2004)—are present in the dataset, they are relatively rare, accounting for just 4.5% of tokens. This imbalance suggests a reconfiguration of how /l/ is socially interpreted in diaspora contexts, where contact with English and non-Caribbean Spanish varieties may be introducing new phonetic alternatives. As such, /l/ mismatch in this dataset appears to stem not from the preservation of inherited dialect norms, but from contact-induced restructuring in an urban, multilingual environment.

7.2 Covariation in Contact Settings: Structural Convergence vs. Sociolinguistic Indexing

This study reveals a striking asymmetry in the behavior of sociolinguistic variables in a contact setting: lower-salience features tend to shift together in systematic ways—showing shared alignment with contact-related metrics like PLUS—while higher-salience features show more differentiated patterning, often uncorrelated with those same metrics. This aligns with and extends existing models of covariation, coherence, and salience in bilingual communities (Erker & Otheguy 2016; Guy & Hinskens 2016; Erker 2022). Lower-salience features exhibit strong covariation and structural convergence with English—albeit modest in effect size—consistent with bilingual optimization strategies that prioritize communicative efficiency and cross-linguistic alignment. Rates of subject pronoun expression and preverbal subject placement increase significantly with U.S. mainland residency, reflecting alignment with English syntactic norms. Similarly, filled pauses shift toward centralized variants (e.g., [ə] or [a]), further indicating a systematic push toward structures that are more congruent across the speakers’ two languages. Higher-salience features, by contrast, show more differentiated patterning, shaped less by structural convergence and more by social meaning. Coda /s/ demonstrates robust dialectal maintenance, with no significant correlation to length of U.S. mainland residency, suggesting its continued role as an index of ethnonational identity. Liquid variation reveals a more complex picture. While rhotics function as sociolinguistic boundary markers closely tied to national origin, laterals appear more stable overall. The presence of velarized /l/—a form not typical of Caribbean Spanish but common in English—does suggest the possibility of contact-induced restructuring; however, this remains a minority pattern in the dataset. There is also tentative evidence

that intra-Spanish dialect contact may be contributing to this variation, though further research is needed to confirm this interpretation.

These findings underscore a core theoretical insight: salience shapes the coherence of linguistic behavior. Building on Labov's (1972) taxonomy of indicators, markers, and stereotypes, and Trudgill's (1986) work on awareness in contact settings, this study provides further support for the idea that features at the lower end of the salience continuum behave more "coherently"—i.e., they pattern together under bilingual pressure. In contrast, higher-salience features show a different kind of coherence—rather than shifting in tandem with contact-related pressures, they remain stable across metrics like PLUS, reflecting strong language and dialectal loyalty. This shared pattern suggests that, in this dataset, speakers treat salient variables not as markers of idiosyncratic expression but as stable components of group-linked identity—decoupling them from the shaping pressures of cognitive realignment in favor of maintaining their social significance.

This pattern highlights the central role of social meaning in shaping contact outcomes. Lower-salience variables tend to shift in similar ways, guided by bilingual optimization strategies. In contrast, salient variables such as coda /s/ and rhotics exhibit patterned stability—not because they are inherently resistant to change, but because speakers orient to them as key stylistic resources. As developed further below, it is through these socially prominent features that speakers engage in both accommodation and differentiation, mobilizing them to align with or distinguish themselves from interlocutors depending on context, stance, and social goals.

This interpretation aligns with and extends previous work by Erker & Otheguy (2016), Erker (2022), and Erker & Vidal-Covas (2024), which argues that non-salient features in

Spanish are more susceptible to structural convergence in contact settings, while salient features are more likely to persist. The current study corroborates this framework while adding nuance through the analysis of liquid variation. By showing that rhotics and laterals play distinct sociolinguistic roles—with the former indexing national origin and the latter reflecting linguistic experience and possible dialect contact—this study demonstrates that not all salient variables are treated identically. Their indexical meanings and stylistic affordances vary across features, groups, and contexts. In short, salience is not merely a cognitive or perceptual property—it is socially constructed and dynamically deployed. It plays a crucial role in mediating whether linguistic forms converge or persist, revealing the layered and agentic nature of variation in multilingual communities like Spanish-speaking Boston.

7.2.1 The Social Power of Salient Variables: In-Group Membership and Indexicality

The fact that speakers maintain dialectally particular patterns of liquid variation and coda /s/ while shifting lower-salience features suggests that salient variables provide speakers with sociolinguistic agency. This section explores how that agency manifests—specifically, how speakers use these features to index identity, assert dialectal distinctiveness, and position themselves within contact-induced linguistic change. While the previous section highlighted broader patterns of alignment and divergence, the focus here shifts to the social meanings that drive those patterns. Speakers can—and sometimes do—maintain highly salient variables in ways that reflect in-group alignment and ideological stance. But just as importantly, the socially charged nature of these forms makes their treatment more variable and context-dependent. The data suggest that such features are mobilized

to signal belonging and distinction—often in nuanced and ideologically charged ways.

Liquid variation offers a clear window into this process. Among the two liquids examined, rhotics serve as powerful boundary markers between Dominican and Puerto Rican speakers. Because rhotic mismatch patterns align more strongly with national origin than with U.S. residency, they function as symbolic resources—akin to linguistic shibboleths—that reinforce dialectal distinctiveness and signal in-group membership (Valentín-Márquez 2007; Shenk 2021; Mendoza-Denton 2008). Their persistence, even in contexts of intense language and dialect contact, reflects the depth of their embeddedness in identity work.

Laterals, however, follow a different trajectory. Although also perceptually salient and frequently referenced by speakers, /l/ variation does not appear to index national identity in the same way. Instead, it reflects more diffuse linguistic experiences, including prolonged exposure to English and contact with speakers from diverse Spanish dialect backgrounds. In this way, lateral variation seems to index broader bilingual and pan-dialectal orientations, rather than ethnonational affiliation.

The contrast between rhotics and laterals underscores a key point: salience alone does not determine indexical function. Not all salient features carry the same social weight—or are interpreted in the same way by all speakers. While both coda /s/ and liquids are perceptually salient and frequently identified in metalinguistic commentary, the data show they operate differently in terms of indexical function and speaker control. Chapter 4 established that these two variables were explicitly referenced by speakers. Participants offered detailed descriptions of rhotic and lateral variation across regional varieties, as well as widespread commentary on /s/-weakening. These features were often invoked

spontaneously and with considerable phonetic detail, underscoring their elevated status in community language ideologies. These patterns of metalinguistic awareness reinforce the idea that these features are socially meaningful and salient, not just to researchers but to speakers themselves. In this respect, the patterns observed among speakers in this study echo foundational insights from early sociolinguistic work—particularly Labov’s studies of Martha’s Vineyard (Labov 1963) and New York City (Labov 1966; Labov 1972)—which demonstrated that linguistic variation can carry social meaning and align with group identity, even when speakers are not consciously aware of their own behavior. Crucially, those studies took place outside of language contact settings, underscoring that this kind of socially embedded variation is not unique to bilingual contexts. The present findings thus instantiate a dynamic that spans both contact and non-contact settings, while also showing how contact adds further layers of social meaning and complexity to the variation. In cases of heightened salience, such features may also become available for stylistic manipulation, particularly when they intersect with salient social categories.

These ideological stances are not simply abstract—they have statistical correlates in the production data. To further explore the relationship between social meaning and the use of salient features, I conducted follow-up regression analyses focusing on coda /s/ and tap lateralization—the two variables that showed strong effects for national origin in the co-variation chapter. Building on that finding, and given the prominence of these features in both production patterns and metalinguistic commentary, I ran separate models for Puerto Rican and Dominican speakers to examine how their linguistic behavior related to their attitudes about national identification. Specifically, I analyzed how Puerto Rican speakers responded to being identified as Puerto Rican, and how Dominican speakers responded to

being identified as Dominican—whether they liked it, disliked it, or felt indifferent—and whether these stances were associated with differing patterns of linguistic behavior. This approach allowed for a more nuanced understanding of how stance and identity positioning influence the maintenance or shifting of salient dialect features.

Among Puerto Rican speakers, those who reported liking being identified as Puerto Rican were significantly more likely to exhibit mismatch for coda /s/—that is, to aspirate or delete /s/ in ways associated with Puerto Rican Spanish. The regression analysis reveals a positive association between favorable national alignment and coda /s/ weakening ($\beta = 6.00, p = 0.01$), while the “indifferent” category also trended in the same direction, though less robustly ($\beta = 2.42, p = 0.10$). This suggests that speakers who express alignment—or at least lack resistance—to Puerto Rican identity are more likely to reproduce canonical dialectal features. A similar pattern emerged among Dominican speakers. While the “like” category was not significant on its own ($\beta = 0.71, p = 0.21$), indifference was strongly predictive of mismatch ($\beta = 1.22, p = 0.00$), suggesting that even a neutral stance toward national identification correlates with dialect maintenance. Taken together, these findings confirm that coda /s/ is a socially powerful resource. It is not simply preserved due to habit or exposure—it is mobilized in ways that correlate with national pride and in-group alignment. This reinforces earlier qualitative findings (see Chapter 4), where speakers like Pacífica, Danilo, and Pilar articulated clear associations between /s/ retention or loss and the distinctiveness of different varieties.

In contrast, tap mismatch—especially lateralization—functions as a more context-dependent and speaker-specific resource. For Puerto Ricans, liking PR identification was a strong predictor of tap lateralization ($\beta = 1.42, p < 0.01$), whereas indifference

corresponded to lower rates of tap lateralization ($\beta = -0.33$, $p = 0.20$). This suggests that for some Puerto Rican speakers, lateralization is consciously embraced as a marker of identity, especially when positively aligned with their national group. Among Dominicans, both the “like” and “indifferent” categories were significantly associated with higher rates of mismatch—that is, producing the tap as [l]. Speakers who liked being identified as Dominican were strongly predictive of mismatch ($\beta = 4.80$, $p < 0.01$), as were those who expressed indifference ($\beta = 3.24$, $p < 0.01$). This pattern suggests that Dominican speakers actively mobilize tap lateralization as an identity marker, with strong effects for both positive and neutral stances toward national identification. This indicates that /r/ lateralization is a socially recognized feature of Dominican Spanish—one that is acceptable, even expected, across a range of ideological positions. This interpretation is supported by qualitative data from speakers like Pilar and Danilo, who describe liquid variation as a hallmark of regional and national speech patterns.

These results reinforce the idea that salience plays a crucial role in shaping how linguistic features behave in contact settings—not by inherently preserving them, but by making them more available for social work. Highly salient features give speakers the option to make ideologically motivated choices: to maintain, reject, or strategically manipulate them depending on the context. This is evident in the treatment of rhotics, which many speakers preserve as dialectal markers despite long-term exposure to English and contact with other non-Caribbean varieties of Spanish. In contrast, features that are less perceptually salient (such as pronouns, subject position, and filled pauses) or carry weaker social meaning (such as lateral variation) are more prone to shift, often aligning more closely with English norms. These patterns support the view that salience is not an inherent prop-

erty of linguistic forms but a socially mediated one, shaped by ideologies, experience, and identity work (Eckert 2008; Mendoza-Denton 2008). Ultimately, the data suggest that speakers make strategic linguistic decisions—sometimes preserving forms to assert in-group belonging, other times converging toward more neutral variants—navigating the dual pressures of bilingual optimization and social positioning.

7.2.2 Contextualizing the Findings: A Non-Generalizable Pattern

An important conclusion of this study is that these patterns are not universally applicable. Different contact groups, shaped by distinct linguistic loyalties, sociolinguistic ideologies, and historical experiences, will exhibit varying degrees of salience and may treat even highly salient features differently. For instance, communities prioritizing integration may demonstrate leveling in high-salience features, while those emphasizing differentiation, may preserve dialectal distinctiveness. This variability highlights the importance of contextualizing salience within the sociolinguistic and cultural dynamics of individual communities.

The findings align closely with Erker & Vidal-Covas's (2024) analysis of Spanish in Boston, which demonstrates that contact-induced change is not simply about structural convergence or attrition, but about the social meaning of linguistic features. In that study, the authors observed that socially weak variables undergo systematic convergence with English norms across generations. In contrast, socially strong variables remain stable or even become more entrenched over time. Crucially, this pattern was not framed as a linguistic inevitability but as an outcome of the sociolinguistic orientation of the community.

Indeed, the Boston case represents a context in which speakers show strong dialectal

loyalty and a preference for maintaining nationally recognizable speech patterns. As Erker & Vidal-Covas argue, contact does not simply flatten difference—it can also amplify the salience of already prominent features. In multilingual urban centers like Boston, where speakers of multiple dialects coexist, socially strong features may become even more powerful stylistic resources, enabling speakers to assert in-group belonging and resist dialectal leveling.

However, this is not a universal pattern. In other contexts—such as Salvadoran and Honduran communities in the Pacific Northwest—speakers have been shown to reduce the use of highly salient features (such as *voseo*) as a strategy of integration into predominantly Mexican-American speech communities (Woods & Rivera-Mills 2012; Sorenson 2013). In such cases, the same types of features that remain stable in Boston (e.g., /s/ weakening or liquid variation) are sites of change elsewhere. The difference lies not in the linguistic features themselves, but in the social value attributed to them by the speakers.

This underscores a critical point: salience is not an inherent or static property of a linguistic form. Instead, it is a contextually situated phenomenon—shaped by language ideologies, patterns of intergroup contact, and the social goals of speakers. As such, the outcomes of contact cannot be generalized across communities. Rather, they must be interpreted through the lens of local sociolinguistic dynamics, including how speakers perceive and mobilize particular forms as resources for identity, inclusion, or differentiation.

In this light, the findings presented here should be seen not as a blueprint for contact-induced change, but as one instantiation of how sociolinguistic salience, speaker agency, and language contact interact in a specific community. Future work would benefit from further comparative analyses across contact settings, with close attention to the social

meanings that linguistic forms carry—and how those meanings are shaped by local histories, ideologies, and interactional practices.

7.3 Conclusion and Future Directions

This dissertation has argued that variation in bilingual contact settings cannot be fully understood without attending to how speakers use linguistic features to navigate social meaning. Salient forms do not behave differently because they are intrinsically more stable or resistant to change. Rather, they are treated differently by speakers—mobilized strategically, preserved ideologically, or reconfigured socially—depending on local histories, community orientations, and the speaker’s position within them.

This study highlights the complex interplay between linguistic salience, contact-induced change, and sociolinguistic agency. Among the variables analyzed here, lower-salience features pattern in ways that reflect bilingual optimization strategies, converging toward English norms in predictable and systematic ways. It is worth noting, however, that while the statistical associations between contact experience and the use of lower-salience features are significant, their effect sizes are modest. These patterns point to a meaningful trend toward convergence, but they should not be interpreted as wholesale structural shift. A more fine-grained, token-level analysis of the linguistic factors conditioning these features would be necessary to fully assess the extent and nature of convergence. This underscores the importance of interpreting statistical results in light of broader linguistic patterns and interactional practices.

By contrast, highly salient features are subject to social pressures that vary across speakers and settings. They are sometimes maintained, sometimes reconfigured, and sometimes

attenuated, depending on how they are ideologically positioned by speakers. In this sense, the key insight is not that salience leads to stability, but that salience decouples linguistic features from the cognitive pressures that drive convergence. As a result, salient features are more available for stylistic manipulation—but also more contingent, more context-sensitive, and less predictable.

The patterns observed in Boston, while informative, are shaped by the specific sociolinguistic orientations of this community—particularly the value placed on dialectal distinctiveness and linguistic maintenance. As discussed earlier, other communities may exhibit different orientations, leading to markedly different outcomes. This reinforces the broader point that salience alone does not determine contact-induced change. Rather, it is speakers' social goals, ideological positions, and community dynamics that guide how linguistic features are interpreted, evaluated, and deployed.

This study contributes to sociolinguistic theory by centering speaker agency in models of contact-induced change, emphasizing that variation is shaped not only by structural pressures but by socially meaningful choices. It demonstrates that salience is not a fixed or purely perceptual property, but one that emerges through folk commentary, identity work, and patterns of interaction. In doing so, it challenges models that treat convergence as an inevitable outcome of contact, showing instead that contact can also amplify socially meaningful forms—especially when speakers orient toward them as resources for differentiation or belonging.

These findings also open several paths for future research. Comparative work across communities could help reveal how the same linguistic features are treated differently by language users depending on local ideologies, histories, and patterns of interaction.

Perception-based studies, including matched-guise experiments or salience ranking tasks, could shed light on how speakers evaluate forms and whether their perceptions align with actual usage. Finally, social network analysis and ethnographic methods offer promising tools for understanding how speakers' affiliations, relationships, and lived experiences shape their linguistic behavior over time.

In addition to these broader directions, future research could also expand the methodological approach to phonological analysis. While the match/mismatch framework¹ used here may simplify some of the finer-grained distinctions between variants, it enables a more inclusive analysis by capturing any surface realization that diverges from the phonological form predicted by distributional context. Some prior studies have grouped variants into broader categories—such as “weakening” or “lateralization”—allowing for focused analyses across specific subsets of variation. However, the approach adopted here offers a unified framework that accommodates a wider range of phonetic outcomes, including less frequent or less-categorically analyzed realizations, providing a more comprehensive view of the variation that occurs in naturalistic speech. Together, these approaches can deepen our understanding of how language variation operates not just as a system of rules, but as a terrain of social meaning navigated by real people in real communities.

7.3.1 Final Thoughts

Linguistic features do not behave—*speakers* do. This dissertation has shown that when it comes to outcomes of contact, change and/or stability are not random, nor are they in-

¹The use of terms like “predicted” or “mismatch” in this study is not intended to imply that certain variants are incorrect. Rather, these terms are used descriptively to reflect surface realizations that differ from the phonological form associated with a given lexical or distributional context. This framework is analytic, not prescriptive, and is motivated by a commitment to documenting and valuing the full range of linguistic variation as it naturally occurs.

evitable. They reflect speaker decisions—conscious or not—about how to sound, who to align with, and what social meanings to project. Salience, then, is not just about perception or frequency. It is about value. And that value is shaped by context: by the histories speakers inherit, the ideologies they reproduce or resist, and the social terrain they move through. Understanding contact-induced change means understanding how speakers *navigate that terrain*—sometimes adapting, sometimes differentiating, always making choices. The case of Spanish in Boston provides one map. It shows how bilingual speakers manage competing pressures: structural convergence, identity expression, inter-dialectal contact, and community expectations. But it is only one map. Other communities, other contexts, will draw their linguistic landscapes differently. And that, too, is the point.

A Speaker Demographics

This appendix provides an overview of the demographic characteristics and linguistic background of the participants in this study. It includes summary statistics, individual usage rates for key linguistic variables, and detailed demographic profiles for each speaker. It is organized into four sections:

SUMMARY STATISTICS OF KEY DEMOGRAPHIC VARIABLES (A.1)– This section presents descriptive statistics for age, age of arrival (AOA), percent life in the U.S. (PLUS), and self-reported language use, providing an overview of the participant sample.

RATES OF LINGUISTIC FEATURES ACROSS SPEAKERS (A.2) – This section details individual usage rates for the linguistic variables examined in the study, highlighting patterns of variation in phonology, morphosyntax, and language contact effects.

PARTICIPANT DEMOGRAPHICS (A.3)– This section provides a breakdown of participant characteristics by place of origin, summarizing demographic and language background information.

SPEAKER DEMOGRAPHIC PROFILES (A.4)– This section presents detailed individual profiles, describing each participant's linguistic history, social background, and self-reported language practices. These profiles help contextualize the qualitative and quantitative findings discussed in the dissertation.

Together, these sections provide an in-depth perspective on the study’s participants, illustrating the diversity of their linguistic experiences and the factors influencing their speech patterns.

A.1 Summary Statistics of Key Demographic Variables

Table A.1 presents summary statistics for age, age of arrival (AOA), and language use variables across all participants. The average age of participants in this study is 33.41 years, with ages ranging from 18 to 73 years. The average AOA is 14.82 years, though some participants were born in the mainland U.S. and thus have an AOA of 0. Percent

Table A.1: Summary Statistics for Age, AOA, and Language Use Variables

VARIABLE	MEAN	SD	MEDIAN	MAX	MIN
Age	33.41	13.44	33	73	18
AOA	14.82	14.26	17	36	0
PLUS	59.14	37.44	64	100	3
% English & Spanish	36.14	25.20	33	89	0
% Spanish Only	30.55	30.62	25	100	0
% English Only	33.09	28.60	31	83	0

¹ AOA = Age of Arrival

² PLUS = Percent Life in the US

³ % values indicate the percentage of time participants reported speaking each language with their interlocutors.

Life in the U.S. (PLUS) varies significantly among participants, with an average of 59.14%. This variation reflects different migration patterns, with some participants having spent the majority of their lives in the U.S., while others arrived more recently. Regarding language use, participants reported using both Spanish and English approximately 36.14% of the time on average, while Spanish-only use was reported at 30.55%, and English-only use at 33.09%. These values indicate a broad range of bilingual experiences, with some speakers maintaining a strong preference for one language, while others regularly switch between both.

A.2 Rates in Linguistic Features Across Speakers

Table A.2 provides an overview of the individual usage rates for each linguistic variable examined in the dissertation. These rates reflect the proportion of instances in which speakers exhibit particular linguistic features in their speech.

Table A.2: Rates of Usage for Linguistic Variables of Participants

PSEUDONYM	TAP MISMATCH	/ɾ/-[ɫ] MISMATCH	TRILL MISMATCH	/r/-[r̄] MISMATCH	LATERAL MISMATCH	/l/-[ɫ] MISMATCH	CODA /s/ MISMATCH	CENTRALIZED FPS	PRONOUNS PRESENT	PREVERBAL SUBJECTS
Dominican Republic										
Daniel	44	25	15	15	2	0	65	9	35	89
Daniela	29	14	67	67	2	0	44	43	22	84
David	27	16	59	54	3	0	59	8	33	91
Damaris	10	3	79	79	6	0	67	4	39	92
Diana	13	6	13	13	9	2	50	38	30	67
Danilo	53	32	75	72	5	1	69	85	31	78
Diego	55	38	90	90	10	5	78	56	27	94
Donaldo	29	22	100	100	4	2	34	93	64	96
Ernesto	25	3	93	50	91	91	26	86	6	82
Dafne	13	6	95	95	2	0	6	100	50	96
Delia	27	6	89	80	7	4	50	14	57	87
Puerto Rico										
Pascal	40	36	37	34	3	0	95	2	28	87
Paloma	52	45	54	53	5	0	96	61	23	70
Pilar	34	20	48	41	2	1	63	2	31	90
Pablo	27	16	33	33	3	0	73	6	21	86
Patricio	25	13	60	60	2	1	17	89	10	64
Paola	37	23	95	90	14	2	17	96	62	95
Pacifica	70	67	54	53	2	0	96	98	32	93
Priscila	60	25	92	82	39	29	96	100	60	98
Ponce	64	55	80	34	12	2	95	94	49	91
Pamela	37	30	100	100	11	0	55	98	57	95
Prospero	44	37	48	48	4	2	64	55	35	90

A.3 Participant Demographics

The study includes speakers from diverse backgrounds, representing different national origins and varying levels of exposure to English and Spanish. Table A.3 provides an overview of participant demographics by place of origin, showing the distribution of age, AOA, PLUS, and language use (percentage of time spent speaking Spanish, English, or a

mix of both). These variables help frame the linguistic experiences of participants, offering insight into their exposure to different languages in their daily lives.

Table A.3: Demographic Characteristics and Language Contact Measures of Participants

PSEUDONYM	SEX	AGE	AOA	PLUS	IMMIG CAT	% ENGL & SPAN	% SPAN ONLY	% ENGL ONLY
Dominican Republic								
Daniel	M	33	32	3	recent arrival	50	50	0
Daniela	F	38	32	16	recent arrival	60	30	10
David	M	47	36	23	established	13	63	25
Damaris	F	45	34	24	established	0	100	0
Diana	F	43	31	30	established	0	100	0
Danilo	M	46	20	57	established	34	34	34
Diego	M	73	18	71	established	50	25	25
Donaldo	M	21	4	81	established	0	17	84
Ernesto	M	21	0	100	US born	15	15	72
Dafne	F	21	0	100	US born	80	0	20
Delia	F	23	0	100	US born	29	0	72
Puerto Rico								
Pascal	M	37	36	4	recent arrival	29	72	0
Paloma	F	20	17	15	recent arrival	50	17	34
Pilar	F	45	24	22	established	50	50	0
Pablo	M	21	17	38	established	25	25	50
Patricio	M	36	21	42	established	34	34	34
Paola	F	23	3	87	US born	50	13	38
Pacifica	F	33	1	88	US born	89	0	12
Priscila	F	24	0	100	US born	20	0	80
Ponce	M	25	0	100	US born	34	34	34
Pamela	F	18	0	100	US born	17	0	84
Prospero	M	42	0	100	US born	72	0	29

¹ AOA = Age of Arrival

² PLUS = Percent Life in the US

³ Immig Cat = Immigration Category

⁴ % values indicate the percentage of time participants reported speaking each language with their interlocutors.

A.4 Speaker Demographic Profiles

This section provides individual demographic profiles for each speaker who participated in this study. These profiles are based on responses to the language background questionnaire (see Appendices D and C) and offer key details about each participant’s linguistic history, social background, and self-reported language use. The profiles are organized in order of increasing PLUS by speaker name and labeled according to their national origin.

A.4.1 Dominican Republic

DANIEL 33M – Daniel is a 33-year-old recently arrived Dominican male. They arrived in the U.S. at the age of 32 and have spent 3% of their life in the U.S. At the time of the interview, they worked as an engineer. Previously, they worked as an engineer in their place of origin. They were educated outside the US. The highest level of education completed was university. They learned Spanish as their first language. They acquired their second language through school. They report that they know Spanish better. Their English proficiency is very good. Their Spanish proficiency is excellent. They report using only Spanish with interlocutors 50% of the time and a mix of both English and Spanish 50% of the time. They believe that it is possible to be Dominican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should not be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identified them as Colombian, Cuban, Dominican, Guatemalan, Mexican, Puerto Rican, and Salvadoran. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

DANIELA 38F – Daniela is a 38-year-old recently arrived Dominican female. They arrived in the U.S. at the age of 32 and have spent 16% of their life in the U.S. Previously, they worked in marketing in their place of origin. They were educated outside the US. The highest level of education completed was postgraduate. They consider themselves working class. They learned Spanish as their first language. They acquired their second language at the age of 18 through school. They report that they know Spanish better. They prefer speaking Spanish. Their English proficiency is passable. Their Spanish proficiency is excellent. They report using only English with interlocutors 10% of the time, only Spanish 30% of the time, and a mix of both languages 60% of the time. They believe that it is not possible to be Dominican and only speak English. They believe that it is not possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would like being identified as Colombian, Cuban, Dominican, Guatemalan, Mexican, Puerto Rican, and Salvadoran. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

DAVID 47M – David is a 47-year-old established Dominican male. They arrived in the U.S. at the age of 36 and have spent 23% of their life in the U.S. At the time of the interview, they worked as a salesperson. Previously, they worked as a salesperson in their place of origin. They were educated outside the US. The highest level of education completed was university. They consider themselves working class. They learned Spanish as their first language. They acquired their second language at the age of 35 through school. They report that they know Spanish better. They like speaking both English and Spanish equally. Their English proficiency is passable. Their Spanish proficiency is excellent. They report using only English with interlocutors 25% of the time, only Spanish 62.5% of the time, and a mix of both languages 12.5% of the time. They believe that it is possible to be Dominican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in voting booths and medical services and are against bilingual services in driver's license exams. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identified them as Colombian, Cuban, Dominican, Guatemalan, Mexican, Puerto Rican, and Salvadoran. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

DAMARIS 45F – Damaris is a 45-year-old established Dominican female. They arrived in the U.S. at the age of 34 and have spent 24% of their life in the U.S. At the time of the interview, they worked as a supervisor. Previously, they worked as an accountant in their place of origin. They were educated outside the US. The highest level of education completed was university. They learned Spanish as their first language. They report that they know Spanish better. They prefer speaking Spanish. Their English proficiency is passable. Their Spanish proficiency is excellent. They report using only Spanish with interlocutors 100% of the time. They believe that it is not possible to be Dominican and only speak English. They believe that it is not possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should not maintain their national group's Spanish. They believe that Spanish speakers in Boston should avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should not be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identified them as Colombian, Cuban, Dominican, Guatemalan, Mexican, Puerto Rican, and Salvadoran.

DIANA 43F – Diana is a 43-year-old established Dominican female. They arrived in the U.S. at the age of 31 and have spent 30% of their life in the U.S. At the time of the interview, they worked in the food service industry. They were educated outside the US.

The highest level of education completed was primary. They consider themselves middle class. They learned Spanish as their first language. They report that they know Spanish better. They prefer speaking English. Their English proficiency is poor. Their Spanish proficiency is passable. They report using only Spanish with interlocutors 100% of the time. They believe that it is not possible to be Dominican and only speak English. They believe that it is not possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should learn Castilian Spanish. They believe that their Spanish should be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identified them as Colombian, Cuban, Dominican, Guatemalan, Mexican, Puerto Rican, and Salvadoran. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

DANILO 46M – Danilo is a 46-year-old established Dominican male. They arrived in the U.S. at the age of 20 and have spent 57% of their life in the U.S. They were educated outside the US. The highest level of education completed was university. They learned Spanish as their first language. They acquired their second language through school. They report that they know Spanish better. Their English proficiency is very good. Their Spanish pro-

iciency is excellent. They report using only English with interlocutors 33.33% of the time, only Spanish 33.33% of the time, and a mix of both languages 33.33% of the time. They believe that it is possible to be Dominican and only speak English. They believe that it is not possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams and medical services and are against bilingual services in voting booths. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should not be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They said they would be indifferent if someone identified them as Colombian, Cuban, Dominican, Guatemalan, Mexican, Puerto Rican, and Salvadoran.

DIEGO 73M – Diego is a 73-year-old established Dominican male. They arrived in the U.S. at the age of 18 and have spent 71% of their life in the U.S. At the time of the interview, they worked in construction. They were educated in both the US and outside. The highest level of education completed was university. They learned both English and Spanish at the same time through family. They prefer speaking English. Their English proficiency is excellent. Their Spanish proficiency is excellent. They report using only English with interlocutors 25% of the time, only Spanish 25% of the time, and a mix of both languages 50% of the time. They believe that it is not possible to be Dominican and only speak English. They believe that it is not possible to be American and only speak Spanish. They support bilingual services in driver's license exams, voting booths, and medical ser-

vices. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that US Spanish speakers should maintain their Spanish. They believe that it is not important that US Spanish speakers maintain their Spanish. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

DONALDO 21M – Donaldo is a 21-year-old established Dominican male. They arrived in the U.S. at the age of 4 and have spent 81% of their life in the U.S. At the time of the interview, they were a student. They were educated in the US. The highest level of education completed was university. They consider themselves working class. They learned Spanish as their first language. They acquired their second language at the age of 3 through school and TV. They report that they know English better. They prefer speaking English. Their English proficiency is excellent. Their Spanish proficiency is excellent. They report using only English with interlocutors 83.33% of the time and only Spanish 16.67% of the time. They believe that it is not possible to be Dominican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish

should not be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is important that US Spanish speakers maintain their Spanish. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

ERNESTO 21M – Ernesto is a 21-year-old US Born Dominican-Salvadoran male. They have spent 100% of their life in the U.S. At the time of the interview, they were a student. They were educated in the US. The highest level of education completed was university. They consider themselves middle class. They learned both English and Spanish at the same time through family. They acquired their second language at the age of 2. They report that they know English better. They prefer speaking Spanish. Their English proficiency is excellent. Their Spanish proficiency is excellent. They report using only English with interlocutors 71.43% of the time, only Spanish 14.29% of the time, and a mix of both languages 14.29% of the time. They believe that it is possible to be Dominican-Salvadoran and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identi-

fied them as Colombian and dislike being identified as Cuban, Dominican, Guatemalan, Mexican, and Puerto Rican and like being identified as Salvadoran. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

DAFNE 21F – Dafne is a 21-year-old US Born Dominican female. They have spent 100% of their life in the U.S. At the time of the interview, they were a student. They were educated in the US. The highest level of education completed was university. They consider themselves middle class. They learned both English and Spanish at the same time through family. They report that they know English better. They prefer speaking English. Their English proficiency is excellent. Their Spanish proficiency is excellent. They report using only English with interlocutors 20% of the time and a mix of both English and Spanish 80% of the time. They believe that it is possible to be Dominican and only speak English. They believe that it is possible to be American and only speak Spanish. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

DELIA 23F – Delia is a 23-year-old US Born Dominican female. They have spent 100% of their life in the U.S. At the time of the interview, they worked as an account manager. They were educated in the US. The highest level of education completed was postgraduate. They consider themselves middle class. They learned both English and Spanish at the same time through school and family. They report that they know English better. They prefer speaking Spanish. Their English proficiency is excellent. Their Spanish proficiency is very good. They report using only English with interlocutors 71.43% of the time and a mix of both English and Spanish 28.57% of the time. They believe that it is possible to be Dominican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should not be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identified them as Dominican, Guatemalan, Mexican, Puerto Rican, Salvadoran, and Peruvian and like being identified as Colombian. They think that the Spanish spoken by Colombians, Dominicans, Mexicans, and Puerto Ricans is different.

A.4.2 Puerto Rico

PASCAL 37M – Pascal is a 37-year-old recently arrived Puerto Rican male. They arrived in the U.S. at the age of 36 and have spent 4% of their life in the U.S. At the time of the interview, they worked as a painter and driver. Previously, they worked as a government employee in their place of origin. They were educated outside the US. The highest level of education completed was university. They consider themselves working class. They learned Spanish as their first language. They acquired their second language through TV. Their Spanish proficiency is excellent. They report using only Spanish with interlocutors 71.43% of the time and a mix of both English and Spanish 28.57% of the time. They believe that it is not possible to be Puerto Rican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is unknown. They believe that Spanish speakers in Boston should not maintain their national group's Spanish. They believe that their Spanish should be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identified them as Colombian, Cuban, Dominican, Guatemalan, Mexican, Puerto Rican, and Salvadoran. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

Paloma 20F – Paloma is a 20-year-old recently arrived Puerto Rican female. They arrived in the U.S. at the age of 17 and have spent 15% of their life in the U.S. At the time of the interview, they were a student. They were educated outside the US. The highest level of education completed was university. They consider themselves upper class. They

learned Spanish as their first language. They acquired their second language at the age of 4 through school, TV, and family. They report that they know both English and Spanish equally. They like speaking both English and Spanish equally. Their English proficiency is very good. Their Spanish proficiency is excellent. They report using only English with interlocutors 33.33% of the time, only Spanish 16.67% of the time, and a mix of both languages 50% of the time. They believe that it is possible to be Puerto Rican and only speak English. They believe that it is not possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identified them as Colombian, Cuban, Guatemalan, Mexican, Puerto Rican, and Salvadoran and dislike being identified as Dominican. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

PILAR 45F – Pilar is a 45-year-old established Puerto Rican female. They arrived in the U.S. at the age of 24 and have spent 22% of their life in the U.S. At the time of the interview, they worked as a missionary. Previously, they worked as a missionary in their place of origin. They were educated outside the US. The highest level of education completed

was university. They learned Spanish as their first language. They acquired their second language at the age of 5 through school. They report that they know Spanish better. Their English proficiency is excellent. Their Spanish proficiency is excellent. They report using only Spanish with interlocutors 50% of the time and a mix of both English and Spanish 50% of the time. They believe that it is possible to be Puerto Rican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

PABLO 21M – Pablo is a 21-year-old established Puerto Rican male. They arrived in the U.S. at the age of 17 and have spent 38% of their life in the U.S. At the time of the interview, they were a student. They were educated outside the US. The highest level of education completed was university. They consider themselves middle class. They learned Spanish as their first language. They acquired their second language through school. They report that they know both English and Spanish equally. They prefer speaking Spanish. Their English proficiency is excellent. Their Spanish proficiency is excellent. They report

using only English with interlocutors 50% of the time, only Spanish 25% of the time, and a mix of both languages 25% of the time. They believe that it is not possible to be Puerto Rican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identified them as Cuban, Mexican, and Salvadoran and dislike being identified as Dominican and Guatemalan and like being identified as Colombian and Puerto Rican. They think that the Spanish spoken by Colombians, Dominicans, Mexicans, and Puerto Ricans is different.

PATRICIO 36M – Patricio is a 36-year-old established Puerto Rican male. They arrived in the U.S. at the age of 21 and have spent 42% of their life in the U.S. They were educated in both the US and outside. The highest level of education completed was university. They learned both English and Spanish at the same time through school. They report that they know English better. They like speaking both English and Spanish equally. Their English proficiency is excellent. Their Spanish proficiency is excellent. They report using only English with interlocutors 33.33% of the time, only Spanish 33.33% of the time, and a

mix of both languages 33.33% of the time. They believe that it is possible to be Puerto Rican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identified them as Colombian, Dominican, Guatemalan, Mexican, Puerto Rican, and Salvadoran. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

PAOLA 23F – Paola is a 23-year-old US Born Puerto Rican female. They arrived in the U.S. at the age of 3 and have spent 87% of their life in the U.S. At the time of the interview, they worked as a teacher. They were educated in the US. The highest level of education completed was postgraduate. They consider themselves upper class. They learned Spanish as their first language. They acquired their second language at the age of 5 through school. They report that they know English better. They prefer speaking English. Their English proficiency is excellent. Their Spanish proficiency is very good. They report using only English with interlocutors 37.5% of the time, only Spanish 12.5% of the time, and a mix of both languages 50% of the time. They believe that it is not possible to be Puerto Ri-

can and only speak English. They believe that it is possible to be American and only speak Spanish. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would dislike being identified as Colombian, Cuban, Dominican, Guatemalan, Mexican, Puerto Rican, and Salvadoran. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

PACIFICA 33F – Pacifica is a 33-year-old US Born Puerto Rican female. They arrived in the U.S. at the age of 1 and have spent 88% of their life in the U.S. At the time of the interview, they worked as a nonprofit employee. They were educated in the US. The highest level of education completed was university. They consider themselves working class. They learned English as their first language. They acquired their second language at the age of 4 through family. They report that they know English better. They like speaking both English and Spanish equally. Their English proficiency is very good. Their Spanish proficiency is very good. They report using only English with interlocutors 11.11% of the time and a mix of both English and Spanish 88.89% of the time. They believe that it is possible to be Puerto Rican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They

support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should not be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identified them as Colombian, Dominican, Guatemalan, Mexican, Salvadoran, and Peruvian and dislike being identified as Puerto Rican. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, Salvadorans, and Peruvians is different.

PRISCILA 24F – Priscila is a 24-year-old US Born Puerto Rican female. They have spent 100% of their life in the U.S. At the time of the interview, they were a student. They were educated in the US. The highest level of education completed was university. They learned English as their first language. They acquired their second language at the age of 5 through family. They report that they know both English and Spanish equally. They like speaking both English and Spanish equally. Their English proficiency is excellent. Their Spanish proficiency is passable. They report using only English with interlocutors 80% of the time and a mix of both English and Spanish 20% of the time. They believe that it is possible to be Puerto Rican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in voting booths and medical services and are against bilingual

services in driver's license exams. They believe that Spanish speakers in Boston should not maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should be taught in schools. They said they would be indifferent if someone identified them as Colombian, Cuban, Dominican, Guatemalan, Puerto Rican, and Salvadoran and dislike being identified as Mexican. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

PONCE 25M – Ponce is a 25-year-old US Born Puerto Rican male. They have spent 100% of their life in the U.S. At the time of the interview, they worked in the food service industry. They were educated in the US. The highest level of education completed was university. They learned both English and Spanish at the same time. They report that they know English better. They like speaking both English and Spanish equally. They report using only English with interlocutors 33.33% of the time, only Spanish 33.33% of the time, and a mix of both languages 33.33% of the time. They believe that it is possible to be Puerto Rican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should not be taught in schools. They believe that

US Spanish speakers should maintain their Spanish. They said they would be indifferent if someone identified them as Dominican, Guatemalan, Mexican, and Salvadoran and like being identified as Colombian.

PAMELA 18F – Pamela is a 18-year-old US Born Puerto Rican female. They have spent 100% of their life in the U.S. At the time of the interview, they were a student. They were educated in the US. The highest level of education completed was secondary. They learned Spanish as their first language. They acquired their second language at the age of 5 through school. They report that they know English better. They prefer speaking Spanish. They report using only English with interlocutors 83.33% of the time and a mix of both English and Spanish 16.67% of the time. They believe that it is possible to be Puerto Rican and only speak English. Their opinion on bilingual education is positive. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should not avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should not learn Castilian Spanish. They believe that their Spanish should be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would dislike being identified as Mexican. They think that the Spanish spoken by Colombians, Dominicans, Mexicans, Puerto Ricans, and Salvadorans is different.

PROSPERO 42M – Prospero is a 42-year-old US Born Puerto Rican male. They have spent 100% of their life in the U.S. At the time of the interview, they worked as a carpenter. They

were educated in the US. The highest level of education completed was university. They consider themselves middle class. They learned Spanish as their first language. They acquired their second language through family. They report that they know English better. They prefer speaking Spanish. Their English proficiency is very good. Their Spanish proficiency is very good. They report using only English with interlocutors 28.57% of the time and a mix of both English and Spanish 71.43% of the time. They believe that it is possible to be Puerto Rican and only speak English. They believe that it is possible to be American and only speak Spanish. Their opinion on bilingual education is positive. They support bilingual services in driver's license exams, voting booths, and medical services. They believe that Spanish speakers in Boston should maintain their national group's Spanish. They believe that Spanish speakers in Boston should avoid unfamiliar words for other Spanish speakers. They believe that Spanish speakers in Boston should learn Castilian Spanish. They believe that their Spanish should not be taught in schools. They believe that US Spanish speakers should maintain their Spanish. They believe that it is very important that US Spanish speakers maintain their Spanish. They said they would be indifferent if someone identified them as Colombian and Dominican and dislike being identified as Guatemalan, Mexican, Salvadoran, and Peruvian and like being identified as Puerto Rican. They think that the Spanish spoken by Colombians, Dominicans, Guatemalan, Mexicans, Puerto Ricans, and Salvadorans is different.

B Coding Manual

This coding manual is designed for analyzing liquid variation in Spanish, specifically the variable production of /ɾ/, /r/, and /l/, in words from interviews in the **Boston Spanish Corpus (BSC)**. Originally developed for Lee-Ann Vidal-Covas' dissertation within the *Spanish in Boston Project*, the manual provides detailed guidelines for data collection and analysis of the **BSC** interviews.

Data Preparation

To ensure systematic and reproducible analysis, follow the process detailed below. The study uses audio recordings of sociolinguistic interviews, which are transcribed orthographically in *Praat*. These transcriptions are paired with audio to identify and annotate lexical items containing liquid segments.

1. IDENTIFYING AND SEGMENTING LIQUID TOKENS

Lexical items with liquid tokens are identified and manually segmented in Praat text grids. Each host word containing a liquid segment should be segmented on a designated tier. A minimum of 250 tokens must be collected per speaker. While identifying tokens for taps and laterals is relatively straightforward, trill tokens require special attention due to their lower frequency.

2. ANNOTATING LIQUID SEGMENTS

After segmenting the host words, create an additional text grid tier for annotating

the liquid segments within each host word. The annotation process follows the segmentation protocol detailed in the [Segmenting liquids](#) Section.

3. ADDING PREDICTOR VARIABLE TIERS

Additional tiers, as described in the [Tier Descriptions](#) section, are added to systematically code predictor variables for all tokens. This structured annotation ensures the dataset's integrity and supports robust acoustic and variationist analysis in subsequent stages.

Textgrid Tiers

Below is a list and description of the interval tiers to be added to the **Praat** (Boersma & Weenink 2020) TextGrid for data collection. These tiers represent the dependent and independent variables used to study liquid variation. While most variables are coded manually during TextGrid preparation, some independent variables are automatically extracted into a CSV file using a *Praat Script* after the coding process is complete. Detailed instructions for identifying sites of variation, segmenting liquid sounds, and applying the coding schema are provided in the sections that follow.

List of Tiers

The following tiers are added to the TextGrid during annotation:

- INTERVIEWER
- ORTHOGRAPHIC
- HOST WORD
- NUMBER OF TOKENS PER HOST WORD

- PHONOLOGICAL FORM
- SURFACE FORM
- NUMBER OF SYLLABLES
- SYLLABLE STRESS
- SYLLABLE TYPE
- POSITION IN SYLLABLE
- POSITION IN WORD
- WORD CLASS
- PREVIOUS SOUND
- FOLLOWING SOUND

Added with Praat extraction script After the TextGrids are completed, the following measurements are generated automatically using the *Praat Script* `Liquids_Data_Extraction.praat`, which compiles data from all TextGrids into a clean CSV file:

- DURATION OF HOST-WORD
- ACOUSTIC ANALYSIS (FORMANTS AT 11 POINTS)

Added in R Following the data extraction, the following variables are added during post-processing in R. The script `Create_liquids_master_df.Rmd` automates this step by calculating and incorporating these variables:

- LEXICAL FREQUENCY
- SPEECH RATE

$$\text{SPEECH RATE (IN MS PER SYLLABLE)} = \frac{\text{DUR(MS) OF HOST-WORD}}{\# \text{ OF SYLLABLES}}$$

Tier Descriptions

This section provides detailed descriptions of each tier included in the TextGrid annotation process. These tiers are critical for ensuring consistent and accurate data coding and represent the dependent and independent variables used in the analysis. Each tier description outlines its purpose, how it should be coded, and any relevant considerations or notes. By following these guidelines, the annotation process becomes more systematic, reducing the likelihood of errors and enabling efficient post-processing.

Orthographic Orthographic transcription of the consultant’s speech, which is used when segmenting the host words that include liquids. This tier will not need to be added, as the conversations have been transcribed by the time the liquids are to be extracted. It will also not appear in the data frame.

Interviewer Orthographic transcription of the interviewer’s speech. This tier will also be completed previous to liquid data collection and will not appear in the data frame.

Host Word Interval of speech which includes the lexical item that hosts the liquid tokens. Interval boundaries are essential as the duration of *Host Word* will be used to compute the variable *Speech Rate*.

Number of Tokens per host word Total number of liquid tokens in the host word. Examples are provided Table B.1.

Table B.1: Tokens per Word Examples

HOST WORD	TRANSLATION	# OF TOKENS
puerto	port	1
latinoamérica	Latin America	2
otorrinolaringólogo	otolaryngologist	4

Phonological Form This variable codes the presumed phonological identity of the token as one of three categories: /r/, /r/, or /l/. We use the orthographic spelling of the host word, its presumed phonological form, and an expanded version of Hualde’s (2014: 183) guide to Spanish rhotic distribution to differentiate between the two rhotics: /r/ and /r/. The expanded version introduces specific codes (e.g., a1 – d3) to facilitate working in

Table B.2: Hualde’s (2014) Spanish Rhotic Distribution Expanded

ENVIRONMENT	CODE	CONTEXT	EXAMPLE	TRANSLATION
Contrastive (<i>word-dependent</i>)				
V__V	a1	Intervocalic &	/ka.ro/ vs.	expensive vs.
	a2	Word-internal	/ka.ro/	car
/r/				
#__	b1	Word-initial (or root-initial)	/ro.ka/	rock
C.__	b2	After heterosyllabic consonant	/al.re.de.dor/	around
/r/				
#C__V	c1	After tautosyllabic consonant; syllable onset; word-initial	/bro.ma/	joke
C.C__	c2	After tautosyllabic consonant; syllable onset; word-internal; consonant in previous syllable coda	/en.tre/	between
V.C__	c3	After tautosyllabic consonant; syllable onset; word-internal; vowel in previous syllable coda	/a.bru.mar/	to overwhelm
V__#V	c4	Word final followed by vowel	/ser#a.mi.gos/	to be friends
Variable (<i>usually /r/</i>)				
V__C	d1	After vowel & Before consonant	/par.te/	part
V__#C	d2	Word final followed by consonant	/ser#po.e.ta/	to be a poet
V__##	d3	Word final followed by pause	/ser/	to be

¹ # represents word boundary

² ## indicates a pause or phonological phrase boundary

³ . represents a syllable boundary

Praat, streamlining the process of identifying and classifying instances of rhotic variation. Additionally, new contexts (*i.e.*, a1 – a2 & c1 – c3) and corresponding examples have been incorporated. These additions enable simultaneous coding of multiple independent variables when coding for Phonological Form, thereby saving time and enhancing accuracy.

See Table B.2 for the full coding schema.

Important Notes for Coding Phonological Form

- Please use the codes in the **Code** column when coding in Praat.
- When coding for the lateral /l/:
 - Add an “l” at the beginning of codes b–d
 - For the **Environment** V__V (codes a1 & a2), just write “la” and exclude the number.

Surface Form The *perceived* segmental identity of the liquid token. For this tier, the researcher will code what they hear, even if different from the reading in the spectrogram and waveform. This variable has the values:

[ɾ], [r], [ɹ], [ɻ], [ɽ], [l], [ɭ], [x], [v], [ð], [i], [h], [ʔ], [∅]

Number of Syllables The syllable count of each host word need to be coded to calculate speech rate; see Table B.3 for examples.

Table B.3: Number of Syllables in Word

HOST WORD	TRANSLATION	# OF SYLLABLES
las	the (fem. plural)	1
puer.to	port	2
la.ti.no.a.mé.ri.ca	Latin America	7
o.to.rrí.no.la.rin.gó.lo.go	otolaryngologist	9

Syllable Stress Whether the syllable to which the token belongs bears primary stress. The values are *stressed* or *unstressed*. Examples are provided in Table B.4.

Table B.4: Syllable Stress Example

HOST WORD	SYLLABIFICATION	STRESS
diferente	di.fe.'ren.te	stressed
escuela	es.'cue.la	unstressed
gustaría	gus.ta.'rí.a	stressed
otras	'o.tras	unstressed
problema	pro.'ble.ma	stressed
quieres	'quie.res	unstressed

- Primary stress is marked with a stress diacritic (') placed before the stressed syllable.
- Syllables containing the liquid token are **bolded**.
- Monosyllabic words should be coded as *stressed*.
- Please use the numbers that belong to each value when coding in Praat. This will avoid misspellings and cut down on coding time.

Syllable Type Whether the token belongs to an *open* or *closed* syllable. Open syllables end in a vowel sound, whereas closed syllables end with a consonant sound. Examples are provided in Table B.5.

1. open

Note: Please use the numbers that belong to each value when coding in Praat. This will avoid misspellings and cut down on coding time.

2. closed

Table B.5: Syllable Type Examples

HOST WORD	TRANSLATION	FIRST SYLLABLE	SYLLABLE TYPE	SECOND SYLLABLE	SYLLABLE TYPE
clase	class	cla	open	se	open
parte	part	par	closed	te	open

Position in Syllable Position of the liquid token within the syllable. Examples are listed in Table B.6. The values are:

1. onset

Note: Please use the numbers that belong to each value when coding in Praat. This will avoid misspellings and cut down on coding time.

2. coda

Table B.6: Position in Syllable Examples

HOST WORD	TRANSLATION	FIRST SYLLABLE	POSITION IN SYLLABLE	SECOND SYLLABLE	POSITION IN SYLLABLE
real	royal / real	re	onset	al	coda
partir	to leave / to split	par	coda	tir	coda

Position in Word Position of the liquid token within the within the host word. Refer to Table B.7 for examples. The values are:

1. initial

Note: Please use the numbers that belong to each value when coding in Praat. This will avoid misspellings and cut down on coding time.

2. internal

3. final

Table B.7: Position in Word Examples

HOST WORD	TRANSLATION	FIRST SYLLABLE	POSITION IN WORD	SECOND SYLLABLE	POSITION IN WORD
real	royal / real	re	initial	al	final
partir	to leave / to split	par	internal	tir	final

Word Class Word class to which the host word belongs. The same tier will also code for specific features in some classes to allow for post-hoc research, as seen in Table B.8.

Table B.8: Word Class Coding Schema

CLASS	CODE	FUNCTION	TYPE	PRONOUN +	EXAMPLES
Noun	1		proper (p) / common (c)		Puerto Rico, carro
Verb	2		infinitive (i) conjugated(c) past participle (t) gerund (g)	P	celebramos, celebrar, celebrarle Review of Spanish Participles
Determiner	3	modify nouns			el la los las un una unos unas todos algunos estos aquellos su sus numerals (primero, segundo)
Adjective	4	describe nouns			grande, largo, corto, hermoso, azul
Adverb	5	describe verbs			lentamente, rapidamente
Pronoun	6				yo, tú, él, ella
Preposition	7	explain relationships b/w nouns			a, ante, bajo, cabe, con, contra, de, desde, en, entre, hacia, hasta, para, por, según, so, sobre, tras
Conjunction	8	connect nouns, verbs, phrases, & sentences			y, o, pero, sin embargo, por lo tanto, así que, aunque, porque, para que
Other	9	catch-all			Use this for anything that doesn't fall into any of the previous categories

Note: PRONOUN+ will be coded for in the case of verbs with a suffixed pronoun, i.e. the verb *celebramos* would be coded as 2c while *celebrarle* would be coded as 2iP

Previous Sound The identity of the phonetic segment immediately preceding the liquid token.

- [a], [b], [β], [d], [ð], [e], [f], [g], [ɣ], [h], [i], [j], [k], [l], [m], [n], [ŋ], [o], [p],
[k], [r], [ɾ], [ʀ], [ɹ], [ɻ], [s], [t], [u], [v], [x], [z]
- “CT” for when the sounds are unrecognizable

- “Pause” for when there is no sound (i.e. there is a pause)
- These will be recoded to create new variables in R:
 - if *vowel*: Vowel Height, Vowel Displacement
 - if *consonant*, Place & Manner of Articulation

Following Sound The identity of the phonetic segment immediately following the liquid token. (This variable has the same possible values as the variable PREVIOUS SOUND, above.)

Identifying the Tokens

Inclusions

When identifying tokens, it is important to include all liquid segments present in a word. The extraction script is designed to accommodate multiple tokens from a single word, ensuring that no relevant data is overlooked. This comprehensive approach allows for the systematic analysis of liquid variation within and across words.

Segmenting liquids

Following the data preparation process, the segmentation of host words and individual liquid tokens is to be carried out in Praat. Each liquid token should be carefully segmented by closely examining both the waveform and the spectrogram to distinguish it from the surrounding sounds. In the segmentation process, two key objectives guide this protocol: (1) distinguishing the liquid from the surrounding sounds, and (2) classifying its identity based on its acoustic properties. These steps are essential for accurately analyzing the tokens, particularly in cases where the realization of liquids diverges from canonical ex-

pectations. The *Surface Form* variable, central to this study, captures such variation by providing a structured framework for classifying these sounds, even when they exhibit atypical acoustic characteristics.

The waveform, which represents frequency and amplitude over time, was analyzed to assess periodicity. Periodic speech sounds exhibit complex repeating waves, indicating regular vocal fold vibration, while aperiodic sounds display random, non-repetitive patterns. It should be noted that periodic and aperiodic components can co-occur within the same sound, as is often the case in speech. Spectrograms provide more detailed visual cues for identifying boundaries between sounds by displaying formant structure. For all liquids, changes in formant frequency—such as shifts in the first (F1), second (F2), or third (F3) formants—provide critical information for distinguishing between the preceding and following sounds. These changes often reflect the articulatory transitions between the liquid and adjacent sounds, such as the movement of the tongue or the opening and closing of the vocal tract, which can cause abrupt or gradual shifts in the formant structure. Additionally, intensity changes, such as a dip in amplitude when transitioning from a vowel to a liquid, are key cues for segmentation. The next section shows some examples to illustrate the segmentation and coding procedure for both rhotics and laterals.

Rhotics Consider the two rhotics in Figure B-1. Figure B-1a includes a flap in the word [para], while Figure B-1b includes a trill in the word [para].

- Some possible cues to segment rhotics are:
 - There is a-periodicity in the waveform.
 - A steep fall of the third formant is a defining acoustic signature of rhotic sounds,

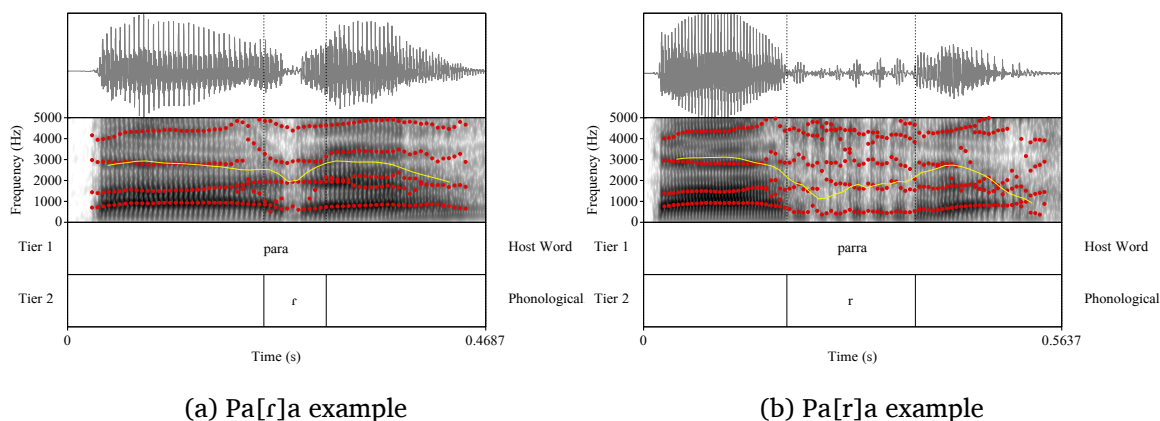


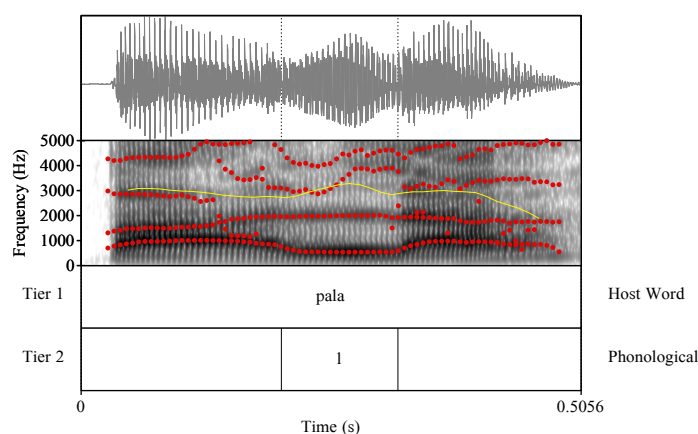
Figure B.1: Waveform, spectrogram, and annotation of pa[r]a and pa[r]a

particularly in retroflex or bunched articulations.

- Duration helps in distinguishing rhotics from adjacent vowels. Rhotics, especially flaps, tend to have shorter durations.
- To differentiate from adjacent vowels, a dip in intensity can help (yellow line in Praat picture).
- It should be noted that a trill is much easier to differentiate from adjacent vowels than a flap because trills typically consist of multiple rapid closures of the tongue against the alveolar ridge, creating distinct bursts of energy visible in both the waveform and spectrogram. In contrast, flaps are characterized by a single, brief contact of the tongue, resulting in a shorter and less acoustically distinct interruption, which makes them more challenging to segment from adjacent vowels.
- A radical drop in amplitude can be a cue to where rhotics begin, as vowels have a higher amplitude than rhotics.
- F4-F5 range: Subtle acoustic differences in F4 and F5 provide additional infor-

mation when coding the *Surface Form* of rhotics, particularly for distinguishing bunched and retroflex variants (Zhou et al. 2008: 4467).

Lateral Laterals are approximants, which means they are among the most sonorous consonants. As their classification hints, they are somewhere between a vowel and a consonant, a fact that can make it hard to differentiate between the sounds. Figure B·2 includes the lateral /l/ in the word [pala].



(a) Source: SIB 0049PR

Figure B·2: Waveform, spectrogram, and annotation of the word [pala]

- The following cues can be used to find and segment the lateral in order to differentiate between the previous and following sounds:
 - The lateral’s waveform is clearly periodic, a characteristic that can help differentiate it from flanking consonants. However, periodicity alone does not distinguish laterals from vowels, as both exhibit periodic waveforms. Additional acoustic cues, such as differences in intensity, formant transitions, and amplitude changes, are critical for segmentation.

- For a lateral, there is no major dip in amplitude or intensity (*yellow line in Praat picture*) in contrast to rhotics, which tend to have lower intensity relative to vowels. This distinction is particularly relevant in contexts where liquids may exhibit overlapping acoustic properties, such as when a purportedly phonological rhotic sound is realized with lateral-like qualities.
- When differentiating between a lateral and a vowel, the lateral typically shows slightly lower intensity due to the inherently higher sonority of vowels. This difference often appears as a subtle dip in intensity at the boundary between the two sounds. Additionally, while both vowels and laterals exhibit formant transitions, those of laterals tend to be more gradual and less pronounced.
- In post-vocalic positions, laterals often exhibit abrupt shifts in formant structure, providing a key cue for segmentation in these contexts.

It should be noted that the goal of the study is to examine variation in spontaneous speech. As such, “idealized” tokens, like the ones shown in Figures B·1 and B·2, will rarely be encountered. The ultimate goal is to make the best segmentation possible, within reason.

Data Extraction and Cleaning

The data processing workflow consists of two main steps: extraction and cleaning. These steps ensure a systematic approach to preparing the data for analysis, minimizing human error and automating repetitive tasks.

Step 1: Data Extraction with Praat Script

The first step involves extracting annotated data from Praat TextGrids using a custom *Praat Script*, `Liquids_Data_Extraction.praat`. This script extracts all relevant tiers and acoustic measurements, compiling them into a comprehensive CSV file. Additionally, it converts coded annotations from the coding manual into human-readable labels and performs the following tasks:

- Extracts tier labels, acoustic measurements (e.g., formants, segment durations), and contextual information.
- Converts Spanish diacritics (e.g., accents) and IPA codes for seamless compatibility with downstream analysis in **R**.
- Handles multiple tokens within a single word and calculates additional metrics, such as segment and word durations.
- Generates a structured CSV file containing fields for speaker metadata, phonological forms, surface forms, contextual sounds, and acoustic measurements at 11 equidistant time points (F1–F5).

The extraction script reduces manual errors, streamlines the data preparation process, and ensures consistency across all files. Detailed comments embedded in the script outline its functionality and modifications.

Step 2: Data Cleaning with R Script

Once the data has been extracted into a CSV file, the second step is data cleaning and transformation in **R**, using the script `Create_liquids_master_df.Rmd`. This script further

refines the data and performs the following tasks:

DATA TIDYING - Removes unnecessary whitespace, fills undefined values with NA, and corrects inconsistent labels.

TOKEN CATEGORIZATION - Separates word classes (e.g., verb, noun), creates phonological and surface form types (e.g., lateral, rhotic), and revalues contextual sound categories (e.g., vowel, consonant).

FEATURE ENGINEERING - Adds derived variables such as:

- SPEECH RATE - Calculates milliseconds per syllable based on word duration and syllable count.
- LEXICAL FREQUENCY - Counts occurrences of each word in the dataset.
- PHONETIC-PHONOLOGICAL AGREEMENT - Indicates whether surface and phonological forms align.
- CONTEXTUAL FEATURES - Extracts manner and place of articulation for preceding and following sounds, as well as vowel height and displacement.

DATA INTEGRATION - Merges linguistic data with speaker metadata (e.g., sociolinguistic background) for richer analysis.

EXPORT - Saves the cleaned and aggregated data to a new CSV file for downstream analysis.

This two-step process ensures that the final dataset is both comprehensive and ready for variationist and acoustic analysis.

c Spanish in Boston Questionnaire (Spanish)

BOSTON
UNIVERSITY

NÚMERO DEL INFORMANTE:

NOMBRE DEL INFORMANTE (SIN APELLIDO): _____

NOMBRE COMPLETO DEL ENTREVISTADOR: _____

Inglés y Español

1. Inglés

1a. ¿Habla inglés? Sí ___ No ___

1b. ¿Nivel? Excelente ___ Muy bien ___ Pasable ___ Pobre ___

2. Español

2a. ¿Habla español? Sí ___ No ___

2b. ¿Nivel? Excelente ___ Muy bien ___ Pasable ___ Pobre ___

3. ¿Cuál idioma aprendió primero? Inglés ___ Español ___ Ambos juntos ___

4. ¿A qué edad aprendió el otro idioma?

5. ¿Cómo/dónde lo aprendió? Escuela ___ TV ___ familia ___ otro ___

6. ¿Cuál idioma sabe más? Inglés ___ Español ___ Ambos ___

7. ¿Cuál le gusta más? Inglés ___ Español ___ Ambos ___

8. Lectura y escritura

8a. ¿Sabe leer ___ y escribir ___ en español?

8b. ¿Sabe leer ___ y escribir ___ en inglés?

Ideologías lingüísticas

9. Pertenecer al grupo nacional de uno

9a. ¿Se puede ser (de su grupo) y hablar solamente en inglés, (no saber español)? Sí ___ No ___

9b. ¿Se puede ser norteamericano y hablar solamente español, (no saber inglés)? Sí ___ No ___

10. ¿Cuál(es) idioma(s) habla [o hablaba] con su(s):

Ponga E, S, B en los espacios correspondientes (I = Inglés E = Español, A = Ambos).

___ Papá

___ Mamá

___ Hermanos/as

___ Hijos menores

___ Hijos mayores

___ Amigos

___ Jefe

___ Compañeros de trabajo

___ Compañeros de escuela

___ Pareja

11. ¿Cuánto español usa Ud. en :

Ponga P o M en los espacios correspondientes (P = Poco, M = Mucho).

- ___ casa
- ___ la escuela
- ___ el trabajo
- ___ actividades sociales
- ___ al leer
- ___ al escuchar la radio
- ___ al mirar la televisión

12. ¿Cree que el español que hablan los (mencione cada grupo) es diferente?

- | | | | | |
|-----------------|----|-----|----|-----|
| Colombianos | Sí | ___ | No | ___ |
| Dominicanos | Sí | ___ | No | ___ |
| Guatemaltecos | Sí | ___ | No | ___ |
| Mexicanos | Sí | ___ | No | ___ |
| Puertorriqueños | Sí | ___ | No | ___ |
| Salvadoreños | Sí | ___ | No | ___ |

13. ¿Cómo se distingue a un X por su forma de hablar? ¿Es decir, como se sabe que una persona es X al hablar?

- | | |
|-----------------|-------|
| Colombianos | _____ |
| Dominicanos | _____ |
| Guatemaltecos | _____ |
| Mexicanos | _____ |
| Puertorriqueños | _____ |
| Salvadoreños | _____ |

14. ¿Puede dar un ejemplo de cómo hablan en cada grupo? (trata si puede, e.j. una palabra? una pronunciación.

- | | |
|-----------------|-------|
| Colombianos | _____ |
| Dominicanos | _____ |
| Guatemaltecos | _____ |
| Mexicanos | _____ |
| Puertorriqueños | _____ |
| Salvadoreños | _____ |

15. ¿Cree Ud. que la gente de habla hispana en Boston:

- | | |
|--|---------------|
| 15a. ¿debe seguir hablando el español de sus compatriotas? | Sí ___ No ___ |
| 15b. ¿debe evitar palabras/dichos que los otros grupos no conocen? | Sí ___ No ___ |
| 15c. ¿deben aprender a hablar como los españoles? | Sí ___ No ___ |

16. ¿Cuánto contacto tiene Ud. con los siguientes grupos, y dónde [en la casa, el trabajo, el vecindario?]: Ponga una marca debajo de la columna correspondiente. Use las tres primeras columnas. Mucho, Algo, Poco. Si el informante ofrece información suplementaria, escríbala en una o dos palabras bajo la columna de Dónde.

	Mucho	Algo	Poco	¿Dónde?
Colombianos	_____	_____	_____	_____
Dominicanos	_____	_____	_____	_____
Guatemaltecos	_____	_____	_____	_____
Mexicanos	_____	_____	_____	_____
Puertorriqueños	_____	_____	_____	_____
Salvadoreños	_____	_____	_____	_____

17. Si alguien dijera que Ud. suena como (X), le gustaría (o se sentiría ofendida/o)?

Como un	Me gustaría	No me gustaría	¿Por qué?
Colombianos	_____	_____	_____
Dominicanos	_____	_____	_____
Guatemaltecos	_____	_____	_____
Mexicanos	_____	_____	_____
Puertorriqueños	_____	_____	_____
Salvadoreños	_____	_____	_____

18. ¿Se debe enseñar el español de su grupo en las escuelas? Sí ___ No ___

18a. ¿Por qué?

18b. ¿Si no el de su grupo, ¿cuál otro? ¿por qué?

19. ¿Alguna vez combina, Ud. los dos idiomas? Sí ___ No ___

19a. ¿Alguna razón? _____

19b. Conoce a otro(s) que lo hacen? Sí ___ No ___

19c. ¿Quiénes? _____

20. ¿Cree que la alternancia/mezcla ocurre porque los que lo hacen:

20a. saben los dos idiomas bien Sí ___ No ___

20b. no saben los dos idiomas bien Sí ___ No ___

20c. las dos posibilidades existen Sí ___ No ___

21. Mezclar idiomas

21a. ¿Qué opina sobre la mezcla de idiomas?

21b. Acepta la mezcla en situaciones:

Informal (por ejemplo, con amigos) Sí ___ No ___

Formal (por ejemplo, con el jefe) Sí ___ No ___

Ambas ___ Ninguna ___

22. ¿Cree que el mezclar daña el español? Sí ___ No ___

23. ¿Cree que el mezclar daña el inglés? Sí ___ No ___

24. ¿Cree que los hispanos en los EEUU deben mantener el español? Sí ___ No ___

25. ¿Qué tan importante es mantener el español? Muy importante ___ Importante ___ No importa ___

26. Le gustaría que sus hijos y nietos aprendieran Español __ Inglés __ Ambos __

26a. ¿Por qué? _____

26b. ¿Que hace/hará Ud. para asegurar que aprendan español?

La educación bilingüe

27. ¿Su opinión de la educación bilingüe es Positiva __ Negativa __ No sé __

27a. ¿Por qué? _____

27b. ¿Tienen/conoce a un niño en un programa bilingüe? Sí __ No __

28. Un programa de educación bilingüe debería:

28a. Continuar enseñando el español después que el niño aprenda el inglés Sí __ No __

28b. Dejar de enseñar el español y poner al niño en un salón donde solo se habla inglés:

Sí __ No __

29. Alguna gente cree que los jefes/patrones deberían tener el derecho de exigir que los empleados hablen solo inglés? ¿Que cree Ud.?

Datos Demográficos

30. Edad

31. Sexo _____

32. Profesión

32a. Profesión en los EE. UU. _____

32b. Profesión en tu país de origen _____

33. País de origen _____ Ciudad/Pueblo de origen _____

34. Educación

34a. Dónde se educó _____

34b. Nivel de estudios (hasta donde llegó):

primaria __ secundaria __ universidad __ posgrado __

34c. Años completados

35. Clase social a la que Ud. pertenece Alta __ Media __ Obrera __

36. Años en los EE.UU.

37. Años en Boston

38. Edad a la que llegó a EE.UU.

39. ¿Por que se vino a Boston?

40. Identificación nacional

40a. ¿Cómo se identifica Ud. (nacionalidad)? _____

40b. ¿Y sus hijos? _____

40c. Es Ud./tiene familia casado/a con otra nacionalidad?

Sí ___ No ___ ¿Cuál? _____

40d. ¿Cómo prefiere que se nos identifique: _____ como hispanos ___ latinos ___ ambos ___

41. ¿Cuál es el origen étnico de la mayoría de sus vecinos? _____

¿de sus amistades? _____

42. ¿Piensa regresar a su país de origen ___ o se queda aquí ___ ?

¿Por qué? _____

Inglés como lengua oficial

43a. ¿Ud. está a favor de ___ o en contra ___ el proyecto de ley para hacer el inglés el idioma oficial del estado de Massachusetts y de los Estados Unidos?

43b. ¿Ud. está a favor de ___ o en contra ___ de los servicios bilingües en:

Examen de licencia de conducir/manejo:

Sí ___ No ___

Las boletas electorales bilingües:

Sí ___ No ___

Servicios médicos bilingües:

Sí ___ No ___

D Spanish in Boston Questionnaire (English)

BOSTON
UNIVERSITY

INFORMANT NUMBER:

NAME OF INFORMANT (FIRST NAME ONLY): _____

FULL NAME OF INTERVIEWER: _____

English and Spanish

1. English

1a. Do you speak English? Yes ___ No ___

1b. Level? Excellent ___ Very well ___ Passable ___ Poor ___

2. Spanish

2a. Do you speak Spanish? Yes ___ No ___

2b. Level? Excellent ___ Very well ___ Passable ___ Poor ___

3. Which language did you learn first? English ___ Spanish ___ Both at the same time ___

4. When did you learn the other language?

5. How/Where did you learn it? School ___ TV ___ family ___ other ___

6. Which language do you know better? English ___ Spanish ___ The same ___

7. Do you prefer one? English ___ Spanish ___ The same ___

8. Reading and writing

8a. Do you read ___ and write ___ in Spanish?

8b. Do you read ___ and write ___ in English?

Language Ideologies

9. Regarding nationality

9a. Is it possible to be (informant's country of origin) and speak only English (and not know Spanish)? Yes ___ No ___

9b. Is it possible to be American and only speak Spanish (& not know English)? Yes ___ No ___

10. Which languages do/did you speak with your:

Answer E, S, B in the corresponding spaces (E = English S = Spanish, B = Both).

☐ Dad	☐ Friends
☐ Mom	☐ Boss
☐ Siblings	☐ Coworkers
☐ Younger children	☐ Schoolmates
☐ Older children	☐ Significant Other

11. How much Spanish do you use at:

Answer L or A in the corresponding spaces (L = Little, A = A lot).

☐ home
☐ school
☐ work
☐ social events
☐ when reading
☐ listening to the radio
☐ watching TV

12. Do you think that the Spanish spoken by (mention each national group in the study) is different?

Colombians	Yes ☐	No ☐
Dominicans	Yes ☐	No ☐
Guatemalans	Yes ☐	No ☐
Mexicans	Yes ☐	No ☐
Puerto Ricans	Yes ☐	No ☐
Salvadorans	Yes ☐	No ☐

13. What distinguishes the speech of X? That is, how do you know if someone is X when they talk?

Colombians	_____
Dominicans	_____
Guatemalans	_____
Mexicans	_____
Puerto Ricans	_____
Salvadorans	_____

14. Can you give an example – either a word or pronunciation that characterizes the speech of X:

Colombians	_____
Dominicans	_____
Guatemalans	_____
Mexicans	_____
Puerto Ricans	_____
Salvadorans	_____

15. Do you think that Spanish speakers in Boston:

15a. Should continue speaking Spanish with their countrymen?	Yes ☐ No ☐
15b. Should they avoid words that others might not know?	Yes ☐ No ☐
15c. Should they learn how to speak Castilian Spanish?	Yes ☐ No ☐

16. How much contact do you have with the following groups, and where (home, work, neighborhood, etc.)? Mark the corresponding column.

	A Lot	Some	Little	Where?
Colombians	_____	_____	_____	_____
Dominicans	_____	_____	_____	_____
Guatemalans	_____	_____	_____	_____
Mexicans	_____	_____	_____	_____
Puerto Ricans	_____	_____	_____	_____
Salvadorans	_____	_____	_____	_____

17. If someone told you that you sounded (X) would you like that?

Como un	I'd like it	I wouldn't like it	Why?
Colombians	_____	_____	_____
Dominicans	_____	_____	_____
Guatemalans	_____	_____	_____
Mexicans	_____	_____	_____
Puerto Ricans	_____	_____	_____
Salvadorans	_____	_____	_____

18. Should the kind of Spanish you speak be taught in schools? Yes ___ No ___

18a. If yes, explain why.

18b. If not yours, whose, and why?

19. Do you ever mix Spanish and English? Yes ___ No ___

19a. Can you explain why? _____

19b. Do you know other people who mix them? Yes ___ No ___

19c. Who? _____

20. Do you think this mixing happens because:

20a. Speakers know both languages Yes ___ No ___

20b. Speakers don't know both languages well Yes ___ No ___

20c. Both explanations are possible Yes ___ No ___

21. Language mixing

21a. Do you have an opinion on language mixing?

21b. Is it acceptable in the following situations:

Informal (e.g., with friends) Yes ___ No ___

Formal (e.g., with your boss) Yes ___ No ___

Both ___ Neither ___

22. Do you think language mixing corrupts Spanish? Yes ___ No ___

23. Do you think it corrupts English? Yes ___ No ___

24. Do you think Spanish speakers in the US should maintain their Spanish? Yes ___ No ___

25. How important is it to maintain Spanish? Very important ___ Important ___ Not important ___

26. Would you like for your kids and younger relatives to learn Spanish ___ English ___ Both ___

26a. Why? _____

26b. What would you recommend to make sure that they learn Spanish?

Bilingual Education

27. Your opinion of bilingual education is Positive ___ Negative ___ Not sure ___

27a. Why? _____

27b. Do you know any kids in such a program? Yes ___ No ___

28. A bilingual education program should:

28a. Continue teaching Spanish even after a child has learned English Yes ___ No ___

28b. Cease teaching Spanish quickly and put a child in an exclusively English environment as soon as possible

Yes ___ No ___

29. Some people believe that employers have the right to require that their workers speak exclusively in English. What do you think?

Demographics

30. Age

31. Sex _____

32. Profession

32a. Profession in the US _____

32b. Profession in your country of origin _____

33. Country of origin _____ City/Town of origin _____

34. Education

34a. Where educated _____

34b. Level of education (highest completed)

gradeschool ___ high school ___ college ___ graduate ___

34c. Years completed

35. Social class to which you belong Upper ___ Middle ___ Working ___

36. Years in the US

37. Years in Boston

38. Age when you came to the US

39. Why did you come to Boston?

40. National group

40a. How do you identify yourself (in terms of national group)? _____

40b. Your kids? _____

40c. Are you/anyone in your family married to someone of another nationality?
Yes ___ No ___ Which _____

40d. How do you/your family prefer to be identified: _____ as Hispanics ___ Latinos ___ both ___

41. What is the ethnic origin of most of your neighbors? _____
Your friends? _____

42. Do you plan to return to (country of origin) ___ or stay here ___ ?
Why? _____

English as an official language

43a. Are you in favor ___ or against ___ legislation that makes English the official language of the state of Massachusetts and the United States?

43b. Are you in favor ___ or against ___ bilingual services in:

driver's license examinations: Yes ___ No ___

voting booths: Yes ___ No ___

medical services: Yes ___ No ___

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